Informatica® Enterprise Data Catalog
10.5.11

MetaDex Scanner Configuration Guide

Informatica Enterprise Data Catalog MetaDex Scanner Configuration Guide
10.5.11
September 2026

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Publication Date: 2026-08-18

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Preface

Follow the instructions in the *MetaDex Scanner Configuration Guide* to create, configure, and manage metadata scanning jobs in the MetaDex tool. In addition to the configuration steps, the guide also indicates the types of metadata that MetaDex scanners can extract from various data sources.

This guide is written for the administrators of the MetaDex tool.

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CHAPTER 1

Configuration

This chapter includes the following topics:

- Overview of configurations, 9
- Where to configure the scanners, 9
- Metadata sources and conversion paths, 13
- Configuring MetaDex scanners, 16

Overview of configurations

A configuration is a set of parameters that defines a metadata scanning job. You can create configurations in MetaDex to extract metadata from supported data sources.

For data sources where Enterprise Data Catalog does not provide a resource, you can use the Advanced Custom Metadata Loader to automate the process of ingesting metadata into the catalog. For more information, see Appendix A, “Advanced Custom Metadata Loader” on page 136.

Where to configure the scanners

You can configure scanners in different tools. For example, you can configure the Salesforce scanner in the Catalog Administrator tool and the Salesforce V2 scanner in the MetaDex tool.

The following table classifies Informatica scanners by the tool in which you configure them:

Scanner	Configuration tool
Amazon Redshift database	Catalog Administrator
Amazon Redshift stored procedures and scripts	MetaDex
Amazon S3	Catalog Administrator
Amazon Web Services (AWS) Glue	Catalog Administrator
Apache Atlas	Catalog Administrator

Scanner	Configuration tool
Apache Cassandra	Catalog Administrator
Apache Hive	Catalog Administrator
Apache HiveQL	MetaDex
Apache Kafka	Catalog Administrator
Azure Microsoft SQL Data Warehouse	Catalog Administrator
Azure Microsoft SQL Server	Catalog Administrator
Business Glossary	Catalog Administrator
COBOL	MetaDex
Cloudera Navigator	Catalog Administrator
Data Integration Hub	Catalog Administrator
Databricks	MetaDex
Erwin	Catalog Administrator
Google BigQuery	Catalog Administrator
Google Cloud Storage	Catalog Administrator
Greenplum stored procedures and scripts	MetaDex
HDFS	Catalog Administrator
IBM Cognos	Catalog Administrator
IBM DB2 database	Catalog Administrator
IBM DB2 for z/OS	Catalog Administrator
IBM DB2 i5/OS3 through the JDBC connector	Catalog Administrator
IBM DB2 stored procedures and scripts	MetaDex
IBM InfoSphere DataStage	MetaDex
IBM Netezza database	Catalog Administrator
IBM Netezza stored procedures and scripts	MetaDex
Informatica Axon	Catalog Administrator
Informatica Data Quality	Catalog Administrator
Informatica Intelligent Cloud Services - Cloud Data Integration (IICS-CDI)	Catalog Administrator

Scanner	Configuration tool
Informatica Intelligent Cloud Services (IICS) V2	MetaDex
Informatica Master Data Management	Catalog Administrator
Informatica Platform	Catalog Administrator
JCL	MetaDex
JDBC	Catalog Administrator
MicroStrategy	Catalog Administrator
Microsoft Azure Blob Storage	Catalog Administrator
Microsoft Azure Data Factory	MetaDex
Microsoft Azure Data Lake Store	Catalog Administrator
Microsoft Azure Synapse Analytics stored procedures	MetaDex
Microsoft OneDrive, OneDrive for Business	Catalog Administrator
Microsoft Power BI V2	MetaDex
Microsoft SQL Server Analysis Services (SSAS)	MetaDex
Microsoft SQL Server Integration Services	MetaDex
Microsoft SQL Server Reporting Services (SSRS)	MetaDex
Microsoft SQL Server database	Catalog Administrator
Microsoft SQL Server stored procedures and scripts	MetaDex
Microsoft Sharepoint	Catalog Administrator
MongoDB	Catalog Administrator
MySQL through the JDBC connector	Catalog Administrator
Oracle Business Intelligence	Catalog Administrator
Oracle Data Integrator	MetaDex
Oracle database	Catalog Administrator
Oracle stored procedures and scripts	MetaDex
PowerCenter	Catalog Administrator
Qlik Sense	Catalog Administrator
QlikView	Catalog Administrator

Scanner	Configuration tool
SAP BW/4HANA	Catalog Administrator
SAP Business Object Data Services (BODS)	MetaDex
SAP Business Warehouse (BW)	Catalog Administrator
SAP BusinessObjects	Catalog Administrator
SAP PowerDesigner	Catalog Administrator
SAP R/3	Catalog Administrator
SAP S/4HANA	Catalog Administrator
SAS	MetaDex
Salesforce	Catalog Administrator
Salesforce V2	MetaDex
Snowflake database V2	MetaDex
Snowflake stored procedures and scripts V2	MetaDex
Sybase database	Catalog Administrator
Sybase stored procedures and scripts	MetaDex
TIBCO Spotfire	Catalog Administrator
Tableau	Catalog Administrator
Talend	MetaDex
Teradata database	Catalog Administrator
Teradata stored procedures and scripts	MetaDex
Workday	Catalog Administrator

For more information, see the *Enterprise Data Catalog Scanner Configuration Guide* or the *MetaDex Scanner Configuration Guide*, based on where you configure the scanner.

Metadata sources and conversion paths

You can divide the metadata sources of MetaDex scanners into types. Each source has a conversion path that you select in MetaDex to load the metadata into the catalog.

The MetaDex tool extracts metadata from the following source types:

- Database management. For example, Oracle.
- ETL. For example, Azure Data Factory.
- Reporting. For example, Microsoft Power BI.
- Mainframe. For example, COBOL.
- Applications. For example, Salesforce.

You use the conversion paths to extract specific assets. For example, the **Advanced Database Objects to EDC** path extracts stored procedures in the form of metadata to ingest into the catalog. For more details on the parameters used in each conversion path, see "Conversion paths reference" in the MetaDex Scanner Reference.

The following table lists conversion paths for database management metadata sources:

Database management metadata sources	Conversion paths	Reference information
Amazon Redshift stored procedures	Advanced Database Objects to EDC	"Amazon Redshift" on page 26 " Extracting metadata and lineage from database stored procedures" on page 24
Apache HiveQL scripts	Hive script to EDC	" Apache HiveQL" on page 29 " Extracting metadata and lineage from database scripts" on page 22
Greenplum stored procedures	Advanced Database Objects to EDC	"Greenplum" on page 32 " Extracting metadata and lineage from database stored procedures" on page 24
IBM DB2 scripts	Advanced PL/SQL to EDC	"IBM Db2" on page 35 " Extracting metadata and lineage from database scripts" on page 22
IBM DB2 stored procedures	Advanced Database Objects to EDC	"IBM Db2" on page 35 " Extracting metadata and lineage from database stored procedures" on page 24
IBM Netezza scripts	Advanced PL/SQL to EDC	"IBM Netezza" on page 37 " Extracting metadata and lineage from database scripts" on page 22
IBM Netezza stored procedures	Advanced Database Objects to EDC	"IBM Netezza" on page 37 " Extracting metadata and lineage from database stored procedures" on page 24

Database management metadata sources	Conversion paths	Reference information
Microsoft Azure Synapse Analytics	Advanced Database Objects to EDC	"Microsoft Azure Synapse Analytics" on page 40 " Extracting metadata and lineage from database stored procedures" on page 24
Microsoft SQL Server scripts	Advanced PL/SQL to EDC	"Microsoft SQL Server" on page 43 " Extracting metadata and lineage from database scripts" on page 22
Microsoft SQL Server stored procedures	Advanced Database Objects to EDC	"Microsoft SQL Server" on page 43 " Extracting metadata and lineage from database stored procedures" on page 24
Oracle scripts	- Advanced PL/SQL to EDC - SQL Loader to EDC	"Oracle" on page 45 " Extracting metadata and lineage from database scripts" on page 22 "Extracting metadata and lineage from Oracle SQL Loader scripts" on page 49
Oracle stored procedures	Advanced Database Objects to EDC	"Oracle" on page 45 " Extracting metadata and lineage from database stored procedures" on page 24
Snowflake **Note:** Including stored procedures.	Advanced Database Objects to EDC	"Snowflake" on page 51 " Extracting metadata and lineage from database stored procedures" on page 24
Sybase scripts	Advanced PL/SQL to EDC	"Sybase" on page 55 " Extracting metadata and lineage from database scripts" on page 22
Sybase stored procedures	Advanced Database Objects to EDC	"Sybase" on page 55 " Extracting metadata and lineage from database stored procedures" on page 24
Teradata scripts	- BTEQ to EDC - MultiLoad to EDC - FastLoad to EDC	"Teradata" on page 57 " Extracting metadata and lineage from database scripts" on page 22
Teradata stored procedures	Advanced Database Objects to EDC	"Teradata" on page 57 " Extracting metadata and lineage from database stored procedures" on page 24

The following table lists conversion paths for ETL metadata sources:

ETL metadata sources	Conversion paths	Reference information
Azure Data Factory	Azure Data Factory to EDC	"Microsoft Azure Data Factory" on page 82
Databricks	- Databricks to EDC - Advanced Database Objects to EDC. Use to extract metadata from Databricks Delta Lake assets.	"Databricks" on page 65
IBM InfoSphere DataStage	IBM DataStage to EDC	"IBM InfoSphere DataStage" on page 74
Informatica Intelligent Cloud Services	IICS to EDC	"Informatica Intelligent Cloud Services" on page 79
Microsoft SQL Server Integration Services	Microsoft SSIS to EDC	"Microsoft SQL Server Integration Services" on page 89
Oracle Data Integrator	ODI to EDC	"Oracle Data Integrator" on page 93
SAP Business Object Data Services (BODS)	SAP BODS to EDC	"SAP BODS" on page 103
Talend	Talend Data Integration to EDC	"Talend" on page 106

The following table lists conversion paths for reporting metadata sources:

Reporting metadata sources	Conversion paths	Reference information
Microsoft Power BI **Note:** Not related to Power BI Report Server. To successfully scan data from Microsoft Power BI, you need to enable Preview features for your organization in tenant settings. For more information, see the How to enable preview in Microsoft Power BI KB article.	Microsoft Power BI to EDC	"Microsoft Power BI" on page 108
Microsoft SQL Server Analysis Services (SSAS)	- Microsoft SSAS Multidimensional to EDC - Microsoft SSAS Tabular to EDC **Note:** Microsoft SQL Server Analysis Services applies to SSAS Tabular and SSAS Multidimensional.	"Microsoft SQL Server Analysis Services" on page 118
Microsoft SQL Server Reporting Services (SSRS)	Microsoft SSRS to EDC	"Microsoft SQL Server Reporting Services" on page 120
SAS	- SAS to EDC - SAS EGP to EDC	"SAS" on page 115

The following table lists conversion paths for mainframe metadata sources:

Mainframe metadata sources	Conversion paths	Reference information
COBOL	COBOL to EDC (for COBOL without JCL) JCL to EDC (for COBOL with JCL)	"COBOL and JCL" on page 123
JCL	JCL to EDC **Note:** JCL to EDC includes COBOL called from JCL.	"COBOL and JCL" on page 123

The following table lists conversion paths for application metadata sources:

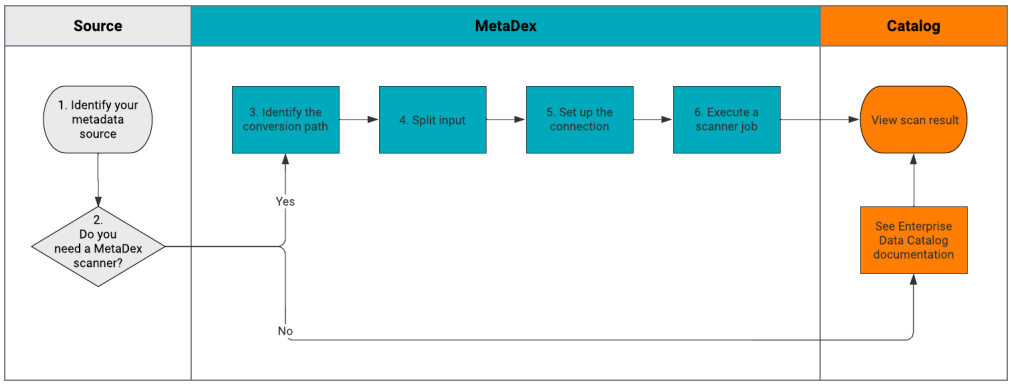
Application metadata sources	Conversion paths	Reference information
Salesforce	Salesforce to EDC	"Salesforce" on page 132

Note: With the recent acquisition, we are working to document the configuration of each MetaDex scanner. If you do not find a scanner in the documentation, send an email to infa_documentation@informatica.com and we will prioritize your request.

Configuring MetaDex scanners

Complete the following tasks to view the scan results from your data source using the MetaDex tool.

The following image shows the high-level workflow to configure MetaDex scanners:



Step 1. Identify the metadata source

You connect to and extract data from a metadata source. You then ingest the metadata into the catalog resources.

Step 2. Do you need a MetaDex scanner?

Identify the appropriate scanner based on the object that you want to scan:

- If you want to scan tables and columns, select a standard scanner. For more information, see the *Enterprise Data Catalog Scanner Configuration Guide*.
- If you want to scan stored procedures and scripts, select a MetaDex scanner.
  If you want to scan one or more external tables, stored procedures, views, materialized views, type definitions, table-valued functions, triggers, tasks, pipes, or tags, select the MetaDex scanner and navigate to **Advanced Database Objects to EDC Conversion Path Setting** > **Database object reader** with the applicable options selected.

  Note: For scanning any other objects, review the available conversion path settings in the MetaDex scanner.

Step 3. Identify the conversion path

Identify the appropriate conversion path for your metadata source. For more information, see "Metadata sources and conversion paths" on page 13.

Step 4. Split input

Plan to split input into logical applications or projects that a MetaDex configuration later loads into each catalog resource. Each metadata source has specific criteria that you need to identify. For example, if you are scanning IBM InfoSphere DataStage, identify projects. For other technologies, such as IBM Mainframe, identify logical applications.

Splitting input provides the following advantages:

- Reduces memory usage and saves time spent on troubleshooting errors.
- Allows you to represent each scan result as a separate catalog resource.
- Provides more flexibility in the security configuration for each resource.

Step 5. Set up the connection

In the connection registry, perform the following steps to configure the connection between your MetaDex scanner and the data source:

1. Log in to the MetaDex tool.
2. Access the connection registry.
3. Create a connection.
4. Test the connection.

Step 6. Execute a scanner job

Run a job to scan metadata:

1. Run a scanner job, analyze the logs.
   Optionally, adjust the configuration to run it again.
2. Optionally, configure the job scheduler. For more information, see "Customizing job schedules" in the MetaDex Scanner Administrator Guide.
3. Optionally, monitor statistics. For more information, see "MetaDex monitoring" in the MetaDex Scanner Administrator Guide.

You can now view and access assets extracted by MetaDex scanners from the list of resources in the Enterprise Data Catalog tool.

Logging in to the MetaDex tool

Use the MetaDex tool URL to log in to the MetaDex tool from a browser.

1. Open a browser.

   For a list of supported browsers, see the applicable Product Availability Matrix (PAM).

2. In the address field, enter the following URL: `http(s)://<MetaDex host>:<port number>`

   where:

   The <MetaDex host> is the server that hosts the MetaDex tool.

   8090 is the default port number. The administrator can set a different port number.

3. On the login page, perform the following actions:
   - If you are logging in for the first time, enter the default credentials:
     1. In the **Username** field, enter `admin`
     2. In the **Password** field, enter `admin`
   - Otherwise, enter your *username* and *password*.

4. If you configured authentication with the Informatica domain, select the applicable domain.
5. Click **Log In**.

   The **EDC MetaDex** tool home page appears.

If you are logging in for the first time, to maintain security, change the default password.

Changing the MetaDex user password

Change the MetaDex user password for security reasons.

1. In the top-right corner of the MetaDex tool page, click **admin** > **Change Password**.
2. In the **Change password** window, enter your credentials:
   a. In the **Old password** field, enter `admin`
   b. In the **New password** field, enter your new password.
   c. In the **Confirm new password** field, re-enter your new password.
3. Click **Submit**.

Creating a project

A project is a group of configurations and variables used in MetaDex. You can use a project to group multiple related configurations.

1. In the MetaDex tool, from the **Projects** menu, click **Create project**.

   The **New project** page appears.

2. Enter the name and an optional description for the project.
3. Optionally, you can add project variables for the configurations.

   You can also add variables that you want to use as specific parameter values across all the configurations within a project.

4. Click **Save**.

   The project page with your project name appears.

After you create a project, add the necessary project permissions and assign user roles. For more information see "Permission management" and "Assigning roles" in the *MetaDex Scanner Administrator Guide*.

Creating a configuration

A configuration is a set of parameters that defines a metadata scanning job. Use the MetaDex tool to create a configuration to scan metadata.

1. Open your project.

   The project page appears.
2. On the left pane, click **More** > **Create configuration** > **MetaDex**.

   If you have other types of configurations enabled in your instance, you can also create them from the **Create configuration** menu. For example, to create an ACML configuration, click **Create configuration** > **ACML**.

   The **New configuration** page appears.
3. Enter a name and a description for your configuration.
4. To scan multiple instances of the configuration simultaneously, select **Parallel Safe**.
5. From the **Conversion path** list, select the source and target of the metadata load for your scanner. For example, the conversion path for Oracle stored procedures is **Advanced Database Objects to EDC**.

   See "Metadata sources and conversion paths" on page 13.
6. From the **mode** list, based on your level of expertise, select the mode in which you want to create the configuration.

   You can select from the following modes:

Mode	Description
Basic mode	Default mode. Includes the minimum of data source connection properties required to scan metadata.
Advanced mode	Includes additional properties. You can use this mode to adjust processing metadata to your requirements.
Expert mode	Includes all available properties. If you are an experienced user, or if Informatica Global Customer Support suggests it, use this mode to adjust processing metadata to your requirements.

   A list of applicable configuration modules appears.
7. Configure properties required for your scanner in the selected mode:
   a. Expand each module to view the required and optional parameters.
   b. Provide the required parameter values.

      Note: The Connection registry path parameter is optional.

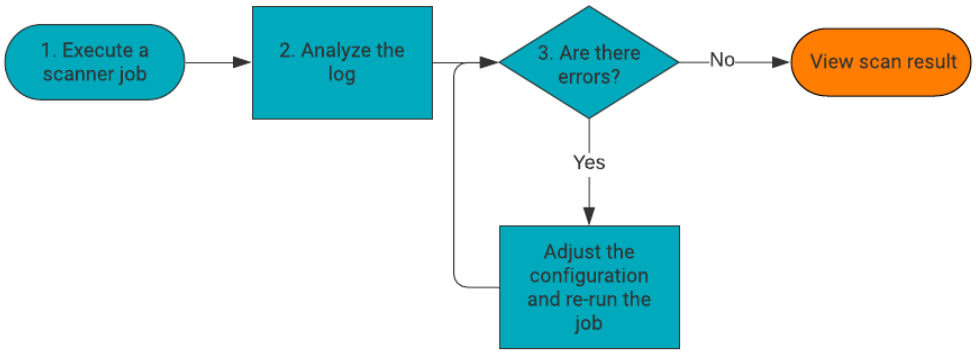
      For information about the parameter values, see "Modules Reference" in the MetaDex Scanner Reference.
8. Click **Save**.

For more information about MetaDex scanner projects and configurations, see "Projects page" and "Configuration page" in the MetaDex Scanner Administrator Guide.

Running a scanner job

Use the MetaDex tool to run a job to scan metadata.

The following image shows the workflow to complete a scanner job:



1. On the configuration page, click **Run**.

   A job is created.
2. Analyze the log.
3. If the log shows issues, perform the following steps:
   a. See "MetaDex logs and reports" in the MetaDex Scanner Administrator Guide.
   b. Provide advanced and expert mode parameters, if needed.
   c. Run the job again.
   d. Repeat step 3 until the log does not show any errors.

      Note: Input code might contain unsupported syntax, complex structures that a scanner can't parse, highly dynamic code or code written with non-standard tools. You can modify such code with the pattern replacer utility. To learn more, see the following How-to library article: *Replacing Complex Patterns in Source Code for MetaDex Scanners*

You can now view the scan result.

CHAPTER 2

Configuring database management scanners

This chapter includes the following topics:

- Overview of database management scanners, 21
- Extracting metadata and lineage from database scripts, 22
- Extracting metadata and lineage from database stored procedures, 24
- Amazon Redshift, 26
- Apache HiveQL, 29
- Google BigQuery, 30
- Greenplum, 32
- IBM Db2, 35
- IBM Netezza, 37
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- Microsoft SQL Server, 43
- Oracle, 45
- Snowflake, 51
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- Teradata, 57
- Troubleshooting database management scanners configuration issues, 59

Overview of database management scanners

The database management category of MetaDex scanners includes scanners that allow you to extract and migrate metadata from the associated database to the catalog.

Note: With the recent acquisition, we are working to document the configuration of each MetaDex scanner. If you do not find a scanner in the documentation, send an email to infa_documentation@informatica.com and we will prioritize your request.

Before you create any configuration, validate the prerequisites and necessary permissions, and familiarize yourself with the parameters, authentication methods, types of input and assets that a given scanner generates. If you haven't created a configuration with the MetaDex tool before, familiarize yourself with the general MetaDex scanners configuration workflow before you proceed.

Extracting metadata and lineage from database scripts

Extract metadata from SQL scripts with database management MetaDex scanners and load the metadata into the catalog.

You can create a sample basic configuration with the Advanced PL/SQL to EDC conversion path. For a detailed list of MetaDex scanners that you configure with the Advanced PL/SQL to EDC conversion path, see "Metadata sources and conversion paths" on page 13.

1. Create a database management project.
2. Assign roles to users.
3. In the project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.
4. Enter a name and an optional description for the configuration.
5. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
6. From the **Conversion path** list, select **Advanced PL/SQL to EDC**.

   A list of expandable modules appears.
7. From the **mode** list, select the **basic** mode.
8. Expand the **Advanced PL/SQL Reader** module.

   The parameter fields appear.
9. From the **Database vendor** list, select the vendor of the database management MetaDex scanner that you configure.
10. In the **Input scripts** field, enter the file path of scripts to include in the job or select it from the **File Manager**.

    You can edit the file path after you select it. Use an asterisk (*) to substitute multiple characters. For example, `/data/input/scripts/*.sql`
11. Optionally, in the **Input scripts to exclude** field, enter the path of scripts to exclude from the job.

    Use an asterisk (*) to substitute multiple characters. For example, `/data/input/scripts/*log.sql`
12. In the **Context** field, enter the SQL context in the following format: `<default host name>.<default database name>.<default schema name>`

    You can define the SQL context in the following ways:

    SQL context default value

    In an SQL script query, context is a default value that substitutes the unknown values in the source code. A context value cannot override a source code value.

    <default host name> and <default database name> must match the host and database entries in the connection registry.

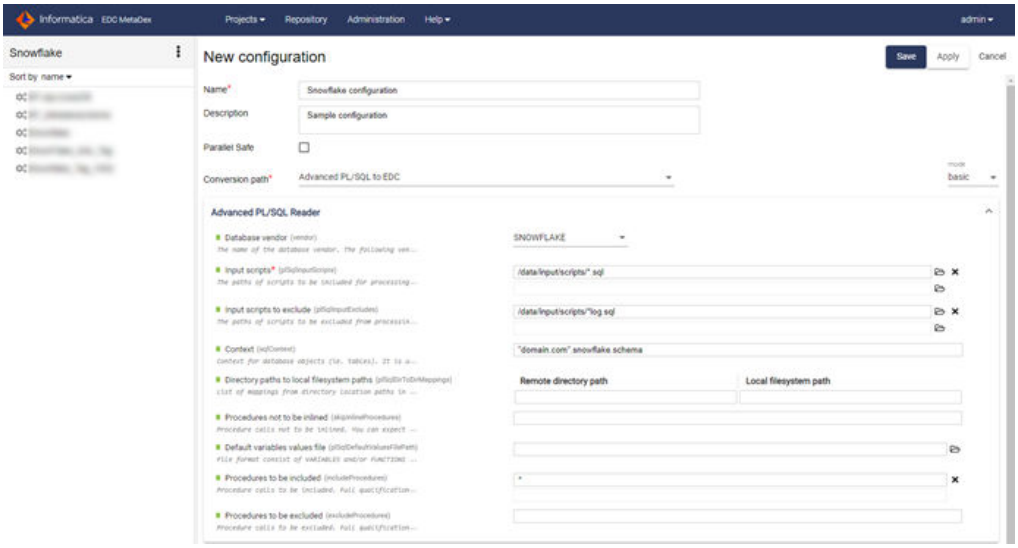
    Warning: If you specify more than three elements of the context syntax, the job fails with an error.

    If a name includes a period, enclose it within double quotation marks, for example, `"domain.com".oracledb.schema`

    SQL context database objects

    Database objects are tables that you can search by the suffix. You use SQL context to create and search the database objects.

The following image shows the **Advanced PL/SQL Reader** module parameters set to run a sample Snowflake configuration:



13. Expand the **Enterprise Data Catalog Writer** module.

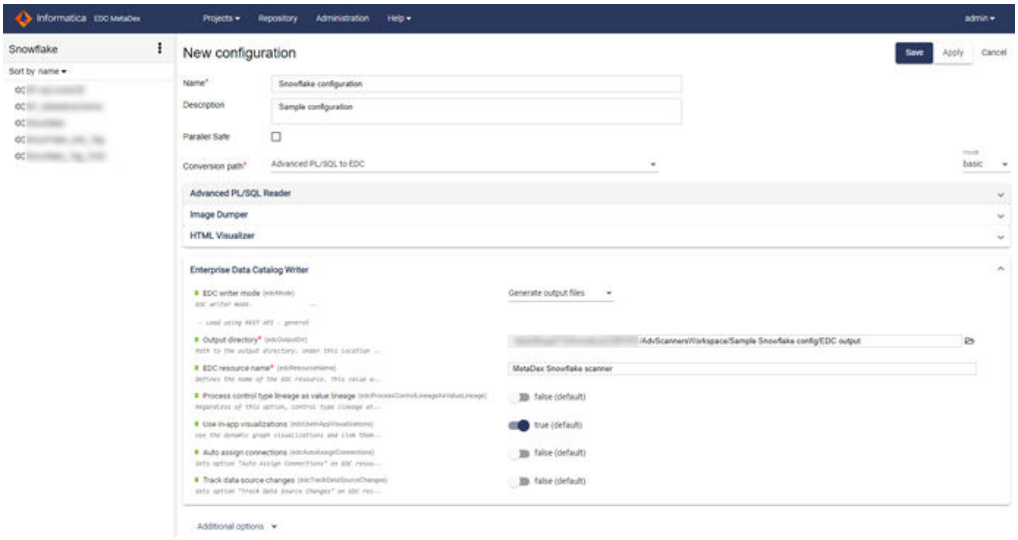
    The parameter fields appear.

14. In the **Output directory** field, enter the path to the output directory where the scanner stores extracted metadata or select the folder from the **File Manager**.

15. In the **EDC resource name** field, enter a unique name of the resource that the system creates in the Catalog tool.

    The resource contains metadata extracted by the configuration.

    The following image shows the **Enterprise Data Catalog Writer** module parameters set to run a sample Snowflake configuration:



16. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

17. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Extracting metadata and lineage from database stored procedures

Extract metadata from stored procedures with database management MetaDex scanners and load the metadata into the catalog.

You can create a sample basic configuration with the Advanced Database Objects to EDC conversion path. For a detailed list of MetaDex scanners that you configure with the Advanced Database Objects to EDC conversion path, see "Metadata sources and conversion paths" on page 13.

1. In the connection registry, configure the connection between the MetaDex scanner and the database.
2. Create a database management project.
3. Assign roles to users.
4. In the project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.
5. Enter a name and an optional description for the configuration.
6. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
7. From the **Conversion path** list, select **Advanced Database Objects to EDC**.

   A list of expandable modules appears.
8. From the **mode** list, select the **basic** mode.
9. Expand the **Database object reader** module.

   The parameter fields appear.
10. From the **Read from** list, select **Stored procedures**.
11. In the **Object include filters** field, enter the database objects to include in the SQL query in the following format: `<host name>.<database name>.<schema name>.<object name>`

    An object include filter, for example, `SNOWFLAKE.EDCDB.ADVSCANNER_QA.*`, consists of the following elements:

    <host name> is the connection registry database name.

    <database name> is the connection registry database name.

    <schema name> is a container for objects, such as tables. A schema name can contain an asterisk (*) to substitute multiple characters. For example, `SNOWFLAKE.EDCDB.*`

    <object name> is the name of an object in a schema. An object name can contain an asterisk (*) to substitute multiple characters. For example, `SNOWFLAKE.EDCDB.ADVSCANNER_QA.*`

    Note: The number of elements in a filter can vary, depending on the database vendor and object type. For example, Oracle database objects that are inside packages have the following format: `<schema name>.<package name>.<object name>`

    A MetaDex configuration processes database objects that match at least one include filter and don't match any of the exclude filters.
12. Optionally, in the **Object exclude filters** field, enter the database objects to exclude from the SQL query in the following format: `<host name>.<database name>.<schema name>.<object name>`

    An object exclude filter, for example, `SNOWFLAKE.EDCDB.ADVSCANNER_QA.log`, consists of the following elements:

    <host name> is the connection registry database name.

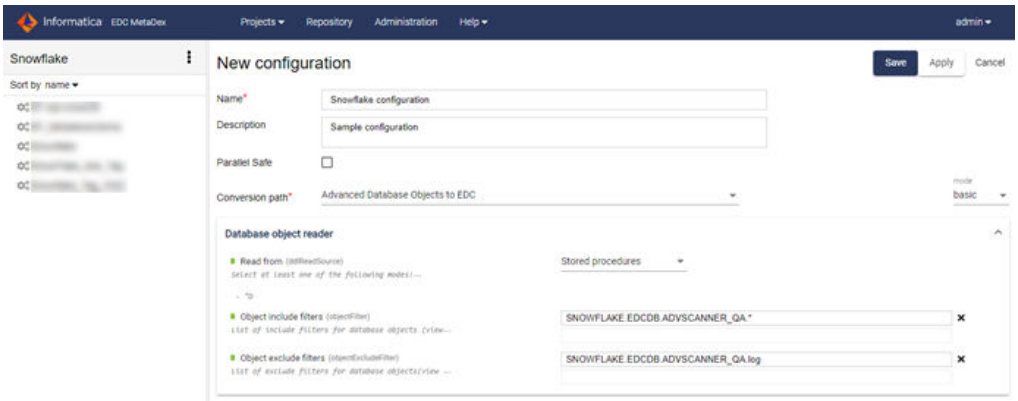
<database name> is the connection registry database name.

<schema name> is a container for objects, such as tables. A schema name can contain an asterisk (*) to substitute multiple characters. For example, `SNOWFLAKE.EDCDB.*.log`

<object name> is the name of an object in a schema. An object name can contain an asterisk (*) to substitute multiple characters. For example, `SNOWFLAKE.EDCDB.ADVSCANNER_QA.*`

Note: The number of elements in a filter can vary, depending on the database vendor and object type. For example, Oracle database objects that are inside packages have the following format: <schema name>.<package name>.<object name>

The following image shows the **Database object reader** module parameters set to run a sample Snowflake configuration:



13. Expand the **Enterprise Data Catalog Writer** module.

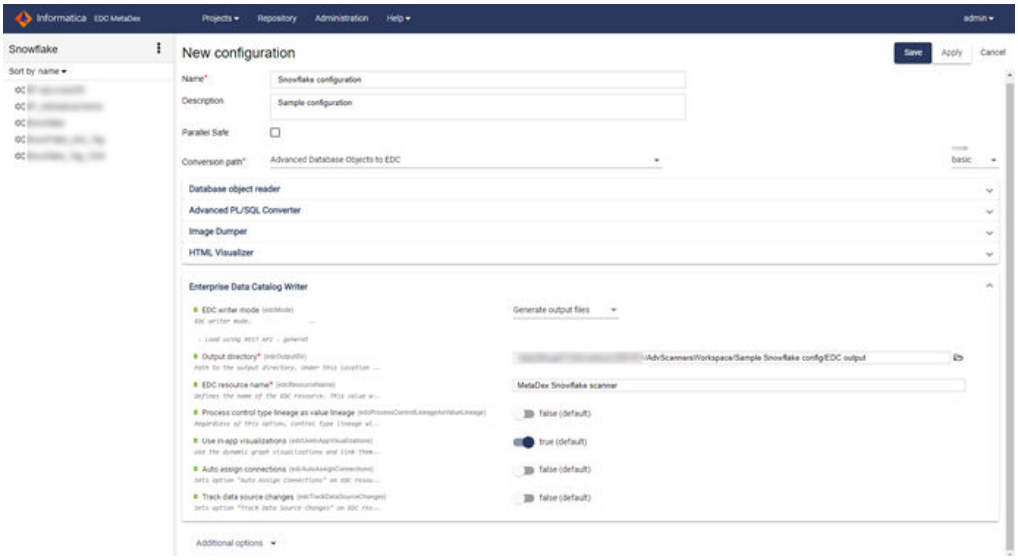
    The parameter fields appear.

14. In the **Output directory** field, enter the path to the output directory where the scanner stores extracted metadata or select the folder from the **File Manager**.

15. In the **EDC resource name** field, enter a unique name of the resource that the system creates in the Catalog tool.

    The resource contains metadata extracted by the configuration.

    The following image shows the **Enterprise Data Catalog Writer** module parameters set to run a sample Snowflake configuration:



16. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

17. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Amazon Redshift

Amazon Redshift is an Internet hosting service and data warehouse product. Amazon Redshift is part of Amazon Web Services, the cloud-computing platform offered by Amazon.

Prerequisites and permissions

To scan Amazon Redshift metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.

2. Get JDBC drivers from the website of your database vendor:

   a. Download JDBC driver files.
   Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM).

   b. Copy the JDBC driver files to the `${INFA_HOME}/AdvScannersWorkspace/jdbc` directory.

   For more information, see "JDBC connections" in the MetaDex Scanner Administrator Guide.

3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Amazon Redshift database. The default port is 5439.

4. Grant permissions that allow you to perform the following operations:

   - select on pg_catalog.PG_ATTRIBUTE
   - select on pg_catalog.PG_CLASS
   - select on pg_catalog.PG_CONSTRAINT
   - select on pg_catalog.PG_DESCRIPTION
   - select on pg_catalog.PG_LANGUAGE
   - select on pg_catalog.PG_NAMESPACE
   - select on pg_catalog.PG_PROC
   - select on pg_catalog.PG_TYPE
   - select on pg_catalog.PG_VIEWS
   - select on pg_catalog.SVV_EXTERNAL_TABLES
   - select on pg_catalog.SVV_EXTERNAL_COLUMNS
   - select on information_schema.COLUMNS
   - select on information_schema.TABLES

- select on pg_catalog.PG_TABLES
- select on pg_catalog.PG_CLASS_INFO
- select on pg_catalog.PG_PROC_INFO

5. Grant permissions to SHOW EXTERNAL TABLE on the tables that you want to process.
6. If you use the dynamic code that consists of solution building queries based on content of the database tables, grant read only access to all the tables that store queries or their components.
7. Configure the Amazon Redshift MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table lists properties that are required to create a connection to an Amazon Redshift database:

Property	Description
Host	Host name or IP address of the Amazon Redshift service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select Redshift as the required vendor from the list.
URL	Enter the JDBC URL. For example, `jdbc:redshift://example.com:5439/database`.
User	The user name used to access the database.
Password	The password associated with the user name.

For more information, see "Creating a connection" in the MetaDex Scanner Administrator Guide.

Types of input

In the MetaDex tool, you can extract Redshift metadata from the following types of input:

- SQL scripts (files)
- Stored procedures
- Functions
- Views, materialized views
- External tables

You can also use the following SQL functionalities:

- Static SQL
- PL/pgSQL
- Procedure calls with parameters tracing
- Dynamic SQL with metatables

  Note: For selected scenarios.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- External table
- Parameter
- Parameter container
- Procedure definition
- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence column
- Source column
- Statement
- Unresolved procedure definition
- Unresolved statement

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Amazon Redshift MetaDex scanner:

- You cannot extract metadata from overloaded procedures.
- You cannot extract metadata from Amazon Redshift functions that use the Python programming language.

Apache HiveQL

Apache Hive Query Language (HiveQL) is a query language in Apache Hive that is similar to SQL. After you define the data structure, you can use HiveQL to query and analyze the data.

Prerequisites and permissions

Before you configure the HiveQL MetaDex scanner to extract metadata from a Hive data source, complete the following prerequisite tasks:

1. Learn about the configuration workflow for MetaDex scanners. For more information, see "Configuring MetaDex scanners" on page 16.
2. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
3. Get the JDBC drivers from the database vendor website:
   a. Download the JDBC driver files.
   Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM). For example, the Hive database requires the hive-apache-jdbc-0.13.1-9.jar driver.
   b. Copy the JDBC driver files to the following directory: `${INFA_HOME}/AdvScannersWorkspace/jdbc`
   If the driver has dependencies, copy the dependency jars to the same location.
4. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Hive metastore database. The default port is 10000.
5. Grant permissions that allow you to perform the following operations:
   - select on sys.COLUMNS_V2
   - select on sys.DBS
   - select on sys.PARTITION_KEYS
   - select on sys.SDS
   - select on sys.TBLS

Connection properties

The following table describes properties that you enter to create a connection to a Hive metastore:

Property	Description
Host	Host name or IP address of the Hive service.
Database	The name of the Hive metastore. **Note:** In most cases, a Hive metastore is located in a Hive database. If the metastore is in a different database, provide the JDBC drivers that are specific to that database. For example, provide Oracle JDBC drivers for an Oracle database.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select HIVE as the required vendor from the list.

Property	Description
URL	Enter the JDBC URL. For example, enter the following JDBC URL for a Hive metastore located in the Hive database: `jdbc:hive2://<host>:<port>/<database>`
User	The user name used to access the Hive metastore.
Password	The password associated with the user name.

Types of input

You can extract Hive metadata from HiveQL script files.

Generated asset types

The MetaDex scanner creates the following types of folder objects in the catalog:

- Script
- Statement
- Unresolved statement

Lineage

You can view the following lineage:

- Statements.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the HiveQL MetaDex scanner:

- If you don't want to connect to a Hive database, you can use the Metadata Extractor to perform a one-time connection and prepare the extract. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.
- To enable Impala, enter the `Tool type` parameter in the configuration.

Google BigQuery

BigQuery is Google Cloud's analytics data warehouse that lets you run analytics over vast amounts of data in near real time. Use the GoogleBigQuery resource to extract metadata from assets in GoogleBigQuery. In the MetaDex tool, you can create and configure a metadata scanning job for Google BigQuery.

Note: Effective in version 10.5.5, Google BigQuery scanner is available for technical preview.

Technical preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Prerequisites and permissions

To scan Google BigQuery metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Google BigQuery database. The default port is 443 for Google BigQuery.
3. To extract metadata from a Google BigQuery source system, you can choose from one of the following options:
   - If you want to use the existing roles and avoid creation of custom roles, assign the BigQuery Data Viewer and BigQuery Metadata Viewer roles to the service account that you use to access the Google Cloud Platform project.
   - If you want to use minimal permissions, create a custom role with the following permissions and assign the custom role to the service account that you use to access the Google Cloud Platform project:
     - bigquery.datasets.get
     - bigquery.models.getMetadata
     - bigquery.routines.get
     - bigquery.tables.get
     - bigquery.tables.list
     - bigquery.jobs.create
     - bigquery.tables.getData
4. Configure the Google BigQuery MetaDex scanner based on information in the Administrator Guide.

Connection properties

The following table describes the properties that you configure to create a connection to a Google BigQuery database:

Property	Description
Host	Host name or IP address of the Google BigQuery service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select BIG_QUERY as the required vendor from the list.

Property	Description
URL	Enter the JDBC URL. For example: `jdbc:bigquery://https://www.googleapis.com/bigquery/v2:443` Default port is 443.
Project ID	Required with JDBC connection property. The ID of the Google Cloud Platform project in which the BigQuery dataset resides. For example: `jdbcConnectionProperty.ProjectId=ecosystemqainfa62360`
OAuth Private Key Path	Required with JDBC connection property. Path to the private key associated with the service account. Enter the location of the JSON file that stores the key with credentials. For example: `jdbcConnectionProperty.OAuthPvtKeyPath=/path/to/key.json` For information about encrypting the private key, see the following knowledge article: Encrypt a password in EDC Advanced Scanners WebUI
Region	Optional. Specifies the geographic location where your BigQuery data resides. For example: `bigquery.region=us`

For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

Lineage

You can view the following lineage:

- Data lineage at the data set level.
- Data lineage at the data element level.

Rules and guidelines

Consider the following rules and guidelines when you use the Google BigQuery MetaDex scanner:

- You cannot extract metadata from legacy SQL.

Greenplum

Greenplum is an open-source massively parallel processing SQL database.

Prerequisites and permissions

To scan Greenplum metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.

2. Get JDBC drivers from the website of your database vendor:
   a. Download JDBC driver files.
      Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM).
   b. Copy the JDBC driver files to the `${INFA_HOME}/AdvScannersWorkspace/jdbc` directory.

   For more information, see "JDBC connections" in the *MetaDex Scanner Administrator Guide*.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Greenplum database. The default port is 5432.
4. Grant the permissions that allow you to perform the following operations:
   - select on pg_catalog.PG_ATTRIBUTE
   - select on pg_catalog.PG_CLASS
   - select on pg_catalog.PG_CONSTRAINT
   - select on pg_catalog.PG_DESCRIPTION
   - select on pg_catalog.PG_LANGUAGE
   - select on pg_catalog.PG_NAMESPACE
   - select on pg_catalog.PG_PROC
   - select on pg_catalog.PG_TYPE
   - select on pg_catalog.PG_VIEWS
   - select on information_schema.COLUMNS
   - select on information_schema.TABLES
   - select on pg_catalog.PG_TABLES
   - select on pg_catalog.PG_MATVIEWS
5. If you use dynamic code that consists of solution building queries based on content of the database tables, grant read-only access to all the tables that store queries or their components.
6. Configure the Greenplum MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table lists properties that are required to create a connection to a Greenplum database:

Property	Description
Host	A folder in the connection registry. Host name or IP address of the Greenplum service.
Database	A folder in the connection registry. The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select GREENPLUM as the required vendor from the list.
URL	Enter the JDBC URL. For example, `jdbc:greenplum://example.com:5432/database`.
User	The user name to access the database.
Password	The password associated with the user name.

For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

Types of input

In the MetaDex tool, you can extract Greenplum metadata from the following types of input:

- SQL scripts (files)
- Stored procedures
- Functions
- Views, materialized views

You can also use the following SQL functionalities:

- Static SQL
- PL/pgSQL
- Procedure calls with parameters tracing
- Dynamic SQL with metatables for selected scenarios.

Generated asset types

TheMetaDex scanner creates the following types of objects in the catalog:

- Parameter
- Parameter container
- Procedure definition
- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence column
- Source column
- Statement
- Unresolved procedure definition
- Unresolved statement

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Greenplum MetaDex scanner:

- You cannot extract metadata from overloaded procedures.
- You cannot extract metadata from functions that use a non-SQL programming language.

IBM Db2

IBM Db2 is a database that enables low latency transactions and real-time analytics. In the MetaDex tool, you can create and configure a metadata scanning job to extract metadata from both IBM Db2 scripts and IBM Db2 stored procedures.

Prerequisites and permissions

To scan IBM Db2 metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. To connect to the IBM Db2 database, use the Type 4 JDBC driver included in the Informatica installation package.

   Note: If you want to use a different driver, you can download and use it.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the IBM Db2 database. The default port is 50000.
4. Grant the View Definition permissions that you need to perform the following operations on the IBM Db2 database:

   select on syscat.COLUMNS

   select on syscat.FUNCTIONS

   select on syscat.NICKNAMES

   select on syscat.PROCEDURES

   select on syscat.PROCPARMS

   select on syscat.TABLES

   select on syscat.USEROPTIONS

   select on syscat.VIEWS

   select on syscat.SCHEMATA

   select on syscat.TABCONST

   select on syscat.REFERENCES

   select on syscat.KEYCOLUSE
5. If you use dynamic code that consists of solution building queries based on content of the database tables, grant read-only access to all tables that store queries or their components.
6. Configure the IBM Db2 MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table describes properties that are required to create a connection to an IBM Db2 database:

Property	Description
Host	Host name or IP address of the IBM Db2 service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select IBM Db2 as the required vendor from the list.
URL	The JDBC URL. Enter the URL in the following format: `jdbc:db2://hostname:<port number>/<database name>` **Note:** The default pot number is 50000.
User	The user name used to access the database.
Password	The password associated with the user name.

For more information, see *Creating a connection* in the *MetaDex Scanner Administrator Guide*.

Types of input

In the MetaDex tool, you can extract IBM Db2 metadata from the following types of input:

- SQL scripts (files)
- Stored procedures
- Functions
- Views

Note: To extract metadata from SQL scripts, use the **Advanced PL/SQL to EDC** conversion path. To extract metadata from stored procedures, functions, and views, use the **Advanced Database Objects to EDC** conversion path.

You can also use the following SQL functionalities:

- Static SQL
- Procedure calls with parameter tracing
- Dynamic SQL with metatables
  Only for selected scenarios.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Package > Procedure Definition
- Package > Unresolved Procedure Definition
- Parameter
- Parameter container
- Pipe > Statement
- Pipe > Unresolved Statement

- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence > Sequence Column
- Source column

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the IBM Db2 MetaDex scanner:

- You cannot extract metadata from overloaded procedures.
- You cannot extract metadata from external tables.

IBM Netezza

IBM Netezza Performance Server is an advanced data warehouse that provides unified, scalable analytics and insights. In the MetaDex tool, you can create and configure a metadata scanning job for the IBM Netezza Performance Server database.

Prerequisites and permissions

To scan IBM Netezza metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. To connect to the IBM Netezza database, download the Netezza JDBC driver version from the IBM website.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the IBM Netezza database. The default port is 5480.

4. Grant the View Definition permissions that you need to perform the following operations on the IBM Netezza database:

   select on _V_DATABASE

   select on _V_JDBC_COLUMNS1

   select on _V_OBJECTS

   select on _V_PROCEDURE

   select on _V_RELATION_COLUMN

   select on _V_SCHEMA

   select on _V_SEQUENCE

   select on _V_SYNONYM

   select on _V_VIEW

5. If you use dynamic code that consists of solution building queries based on content of the database tables, grant read-only access to all tables that store queries or their components.
6. Configure the IBM Netezza MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table describes properties that are required to create a connection to an IBM Netezza database:

Property	Description
Host	Host name or IP address of the IBM Netezza service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select IBM Netezza as the required vendor from the list.
URL	The JDBC URL. Enter the URL in the following format: `jdbc:netezza://hostname:<port number>/<database name>` Where: <PortNumber> stands for the port number. The default port number is 5480.
User	The user name used to access the database.
Password	The password associated with the user name.
netezzaSchemasEnabled	netezzaSchemasEnabled=false|true Set to true to enable multiple schema support. Default is false.
netezzaEdition	netezzaEdition=NETEZZA|PDA Where PDA stands for PureData System for Analytics. Default is NETEZZA.

For more information, see *Creating a connection* in the *MetaDex Scanner Administrator Guide*.

Types of input

In the MetaDex tool, you can extract IBM Netezza metadata from the following types of input:

- SQL scripts (files)
- Stored procedures
- Functions
- Views

Note: To extract metadata from SQL scripts, use the **Advanced PL/SQL to EDC** conversion path. To extract metadata from stored procedures, functions, and views, use the **Advanced Database Objects to EDC** conversion path.

You can also use the following SQL functionalities:

- Static SQL
- Procedure calls with parameter tracing
- Dynamic SQL with metatables
  Only for selected scenarios.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Package > Procedure definition
- Package > Unresolved Procedure Definition
- Parameter
- Parameter container
- Pipe > Statement
- Pipe > Unresolved Statement
- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence > Sequence Column
- Source column

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the IBM Netezza MetaDex scanner:

- You cannot extract metadata from overloaded procedures.
- You cannot extract metadata from external tables.

Microsoft Azure Synapse Analytics

Microsoft Azure Synapse Analytics, which was previously called SQL Data Warehouse, is a metadata source for Azure Synapse stored procedures. In the MetaDex tool, you can create and configure a metadata scanning job to extract metadata from Azure Synapse stored procedures.

Note: SQL procedure is the only Synapse functionality this MetaDex scanner supports.

Prerequisites and permissions

To scan Microsoft Azure Synapse Analytics metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. Connect to the Azure SQL Pool with a Type 4 JDBC driver. Get JDBC drivers from the website of your database vendor:
   a. Download JDBC driver files.
      Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM).
   b. Copy the JDBC driver files to the `${INFA_HOME}/AdvScannersWorkspace/jdbc` directory.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Azure SQL Pool. The default port is 1433.
4. Grant the View Definition permissions and the permissions that you need to perform the following operations on the Azure SQL Pool:
   - select on sys.all_columns
   - select on sys.all_objects
   - select on sys.all_parameters
   - select on sys.database_principals
   - select on sys.databases
   - select on sys.foreign_key_columns
   - select on sys.indexes
   - select on sys.index_columns
   - select on sys.partitions
   - select on sys.schemas
   - select on sys.sql_modules

- select on sys.synonyms
- select on sys.tables
- select on sys.types

5. If you use the dynamic code that consists of solution building queries based on content of the database tables, grant read only access to all the tables that store queries or their components.
6. Configure the Azure SQL Server MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table describes properties that are required to create a connection to a Azure SQL Pool:

Property	Description
Host	Host name or IP address of the Azure SQL Server service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select SYNAPSE as the required vendor from the list.
URL	Enter the JDBC URL. For example: `jdbc:sqlserver://<poolname>.azuresynapse.net:1433;database=<DBNAME>;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.sql.azuresynapse.net`
User	The user name used to access the database.
Password	The password associated with the user name.

For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

Types of input

In the MetaDex tool, you can extract Azure Synapse Analytics metadata from the following types of input:

- Stored procedures
- Functions
- Views, materialized views

You can also use the following SQL functionalities:

- Static SQL
- T-SQL
- Linked servers
- Procedure calls with parameters tracing
- Dynamic SQL with metatables
  Only for selected scenarios.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Parameter
- Parameter container
- Procedure definition
- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence column
- Source column
- Statement
- Unresolved procedure definition
- Unresolved statement

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Azure SQL Server MetaDex scanner:

- You cannot extract metadata from overloaded procedures.
- You cannot extract metadata from external tables.
- SQL procedure is the only Synapse functionality this MetaDex scanner supports.

Microsoft SQL Server

Microsoft SQL Server is a metadata source for Microsoft SQL Server scripts and Microsoft SQL Server stored procedures. In the MetaDex tool, you can create and configure a metadata scanning job to extract metadata from both Microsoft SQL Server scripts and Microsoft SQL Server stored procedures.

Prerequisites and permissions

To scan Microsoft SQL Server metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. To connect to the Microsoft SQL database, use the Type 4 JDBC driver that comes with the installation.

   Note: The installation package provided by the installer contains the Microsoft SQL Server JDBC driver. You don't need to download it.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Microsoft SQL database. The default port is 1433.
4. Grant the View Definition permissions and the permissions that you need to perform the following operations on the Microsoft SQL database:

   SELECT ON sys.all_columns

   SELECT ON sys.all_objects

   SELECT ON sys.all_parameters

   SELECT ON sys.database_principals

   SELECT ON sys.databases

   SELECT ON sys.foreign_key_columns

   SELECT ON sys.indexes

   SELECT ON sys.index_columns

   SELECT ON sys.partitions

   SELECT ON sys.schemas

   SELECT ON sys.sql_modules

   SELECT ON sys.synonyms

   SELECT ON sys.tables

   SELECT ON sys.types

   SELECT ON sys.table_types, if your database uses table types.
5. If you use the dynamic code that consists of solution building queries based on content of the database tables, grant read only access to all the tables that store queries or their components.
6. Configure the Microsoft SQL Server MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table describes properties that are required to create a connection to a Microsoft SQL database:

Property	Description
Host	Host name or IP address of the Microsoft SQL Server service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select MSSQL as the required vendor from the list.
URL	The JDBC URL. Enter the following URL: `jdbc:sqlserver://localhost:1433;databaseName=<DatabaseName>`
User	The user name used to access the database.
Password	The password associated with the user name.

For more information, see *Creating a connection* in the MetaDex Scanner Administrator Guide.

Types of input

In the MetaDex tool, you can extract Microsoft SQL Server metadata from the following types of input:

- SQL scripts (files)
- Stored procedures
- Functions
- Views, materialized views
- Triggers

You can also use the following SQL functionalities:

- Static SQL
- T-SQL
- Linked servers
- Procedure calls with parameters tracing
- Dynamic SQL with metatables
  Only for selected scenarios.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Parameter
- Parameter container
- Procedure definition
- Procedure instance
- Procedure source
- Result set

- Result set column
- Sequence
- Sequence column
- Source column
- Statement
- Trigger
- Unresolved procedure definition
- Unresolved statement

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Microsoft SQL Server MetaDex scanner:

- You cannot extract metadata from overloaded procedures
- You cannot extract metadata from external tables

Oracle

Oracle is a metadata source for Oracle scripts, SQL Loader scripts, and stored procedures. Oracle Autonomous Database is a SaaS implementation of Oracle Database on the Oracle Cloud Infrastructure. In the MetaDex tool, you can create and configure a metadata scanning job to extract metadata from Oracle scripts, SQL Loader scripts, and database objects, using their respective conversion paths.

Prerequisites and permissions

To scan Oracle metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.

2. Get JDBC drivers from the website of your database vendor:
   a. Download JDBC driver files.
      Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM).
   b. Copy the JDBC driver files to the `${INFA_HOME}/AdvScannersWorkspace/jdbc` directory.

   For more information, see "JDBC connections" in the *MetaDex Scanner Administrator Guide*.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Oracle database or the Oracle Autonomous Database. The default port is 1521 for Oracle, and 1522 for the Oracle Autonomous Database.
4. Grant permissions that allow you to perform the following operations:
   - - select on SYS.DBA_ARGUMENTS
     - select on SYS.DBA_COL_COMMENTS
     - select on SYS.DBA_COLL_TYPES
     - select on SYS.DBA_CONS_COLUMNS
     - select on SYS.DBA_CONSTRAINTS
     - select on SYS.DBA_DB_LINKS
     - select on SYS.DBA_DIRECTORIES
     - select on SYS.DBA_EXTERNAL_LOCATIONS
     - select on SYS.DBA_EXTERNAL_TABLES
     - select on SYS.DBA_MVIEWS
     - select on SYS.DBA_OBJECTS
     - select on SYS.DBA_OPBINDINGS
     - select on SYS.DBA_SEQUENCES
     - select on SYS.DBA_SOURCE
     - select on SYS.DBA_SYNONYMS
     - select on SYS.DBA_TAB_COLUMNS
     - select on SYS.DBA_TAB_COMMENTS
     - select on SYS.DBA_TABLES
     - select on SYS.DBA_TRIGGERS
     - select on SYS.DBA_TYPE_ATTRS
     - select on SYS.DBA_USERS
     - select on SYS.DBA_VIEWS
     - select on SYS.NLS_SESSION_PARAMETERS
   - Grant a predefined SELECT_CATALOG_ROLE role.
5. If you use the dynamic code that consists of solution building queries based on content of the database tables, grant read only access to all the tables that store queries or their components.
6. Configure the Oracle MetaDex scanner. For more information, see one of the following procedures:
   - " Extracting metadata and lineage from database scripts" on page 22
   - " Extracting metadata and lineage from database stored procedures" on page 24
   - "Extracting metadata and lineage from Oracle SQL Loader scripts" on page 49

Connection properties

The following table describes properties that are required to create a connection to an Oracle database:

Property	Description
Host	Host name or IP address of the Oracle service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select ORACLE as the required vendor from the list.
URL	Enter the JDBC URL. For example: `jdbc:oracle:thin:@oracle01.exampleorg.local:1521:cspl`.
User	Required for Oracle database. The user name to access the database.
Password	Required for Oracle database. The password associated with the user name.
Truststore	Required for Oracle Autonomous Database. The file used to access the database. For example: jdbcConnectionProperty.TrustStore=/root/truststore/ewallet.p12
Truststore password	Required for Oracle Autonomous Database. The password to the file. For example: jdbcConnectionProperty.TrustStorePassword={PBE0}p17n9Xww1JJjejExr27jtE0agfmoZjfF6wjN6plWnww=

Note: You can use the connection properties separately. For Oracle Autonomous Database, you can use the following connection string details in the Informatica Developer tool:

```
jdbc:informatica:oracle://<hostname>:<portno>;
ServiceName=<servicename>;EncryptionMethod=SSL;CryptoProtocolVersion=TLSv1.2,TLSv
1.1,TLSv1;
KeyStore=<Absolute path to ewallet.p12>;KeyStorePassword=<keystorepassword>
```

where:

<hostname> is the host name or the IP address of the Oracle Autonomous Database service.

<portno> is the port number for the Oracle Autonomous Database.

<servicename> is the name of the service.

<Absolute path to ewallet.p12> is the path to the file used to access the database.

<keystorepassword> is the password to the file.

For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

Types of input

In the MetaDex tool, you can extract Oracle metadata from the following types of input:

- SQL scripts (files)
- SQL Loader scripts
- Stored procedures
- Functions
- Packages

- Views, materialized views
- External tables
- Triggers

You can also use the following SQL functionalities:

- Static SQL
- PL/SQL
- Database links
- Procedure calls with parameters tracing
- Dynamic SQL with metatables

  Note: Only for selected scenarios.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- External table
- Package
- Parameter
- Parameter container
- Procedure definition
- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence column
- Script
- Source column
- Statement
- Trigger
- Unresolved procedure definition
- Unresolved statement

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Lineage from files to Oracle tables generated with the Oracle SQL Loader.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Oracle MetaDex scanner:

- You cannot extract metadata from object-oriented PL/SQL
  Applies to object types with methods.
- You cannot extract metadata from overloaded procedures.
- You cannot extract metadata from custom operators binding.
- Oracle Continuous Query Language works in some cases, depending on the parsing capabilities available.

Extracting metadata and lineage from Oracle SQL Loader scripts

Extract metadata from Oracle SQL Loader scripts with the Oracle MetaDex scanner and load the metadata into the catalog.

You can create a sample basic configuration with the SQL Loader to EDC conversion path.

1. Create a database management project.
2. Assign roles to users.
3. In the project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.
4. Enter a name and an optional description for the configuration.
5. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
6. From the **Conversion path** list, select **SQL Loader to EDC**.

   A list of expandable modules appears.
7. From the **mode** list, select the **basic** mode.
8. Expand the **SQL Loader Reader** module.

   The parameter fields appear.
9. From the **Input mode** list, select one of the following options:
   - **Control and Loader file**. Default value. Applies to SQL Loader parameter files and control files.
   - **Control file only**

   For more information about SQL Loader parameter files and control files, see Oracle documentation.
10. In the **Input parameter files** field, enter the file path of files to include in the job or select it from the **File Manager**.

    You can edit the file path after you select it. Use an asterisk (*) to substitute multiple characters. For example, `/data/input/scripts/*.sql`
11. In the **CONTROL files directory** field, enter the path to the directory where you store control files. For example: `Y:\input\SQLLoader\CONTROL\`
12. In the **Context** field, enter the SQL context in the following format: `<default host name>.<default database name>.<default schema name>`

    You can define the SQL context in the following ways:

SQL context default value

In an SQL script query, context is a default value that substitutes the unknown values in the source code. A context value cannot override a source code value.

<default host name> and <default database name> must match the host and database entries in the connection registry.

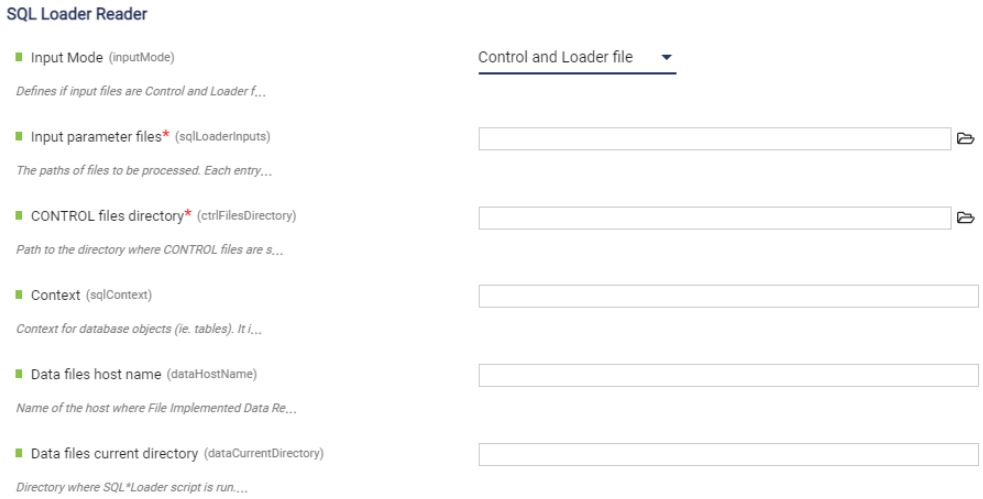
Warning: If you specify more than three elements of the context syntax, the job fails with an error.

If a name includes a period, enclose it within double quotation marks, for example, `"domain.com".oracledb.schema`

SQL context database objects

Database objects are tables that you can search by the suffix. You use SQL context to create and search the database objects.

The following image shows a sample **SQL Loader Reader** module with the input mode selected:



13. Expand the **Enterprise Data Catalog Writer** module.

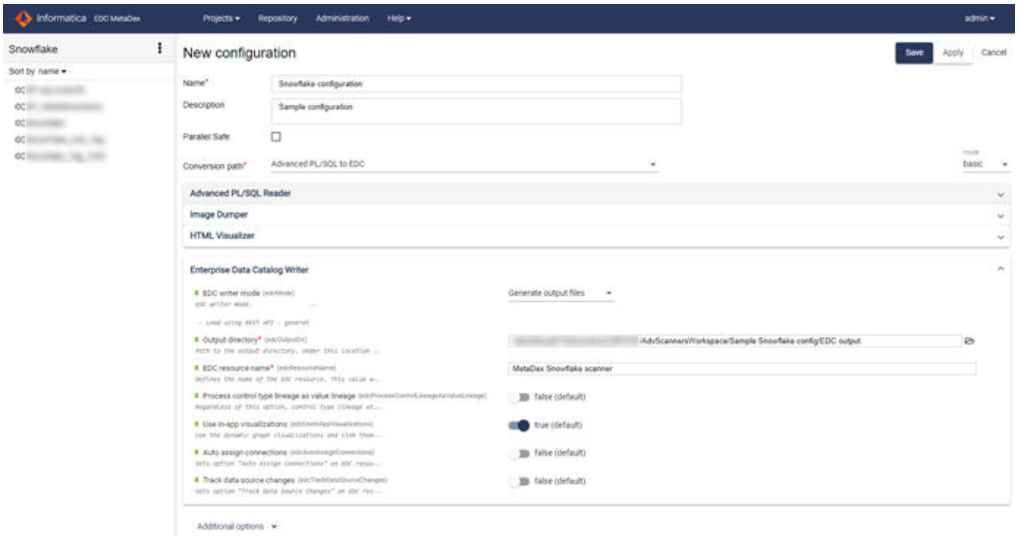
    The parameter fields appear.

14. In the **Output directory** field, enter the path to the output directory where the scanner stores extracted metadata or select the folder from the **File Manager**.

15. In the **EDC resource name** field, enter a unique name of the resource that the system creates in the Catalog tool.

    The resource contains metadata extracted by the configuration.

The following image shows the **Enterprise Data Catalog Writer** module parameters set to run a sample Snowflake configuration:



16. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

17. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Snowflake

Snowflake is an analytic data warehouse provided as Software-as-a-Service (SaaS). The Snowflake data warehouse uses a SQL database engine with a unique architecture designed for the Cloud.

Prerequisites and permissions

Before you configure the Snowflake MetaDex scanner, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.

2. Get JDBC drivers from the website of your database vendor:

   a. Download JDBC driver files.
   Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM).

   b. Copy the JDBC driver files into the following directory: `${INFA_HOME}/AdvScannersWorkspace/lib/jdbc`
   If the lib or lib/jdbc directory does not exist, create it.

   For more information, see "JDBC connections" in the *MetaDex Scanner Administrator Guide*.

3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Snowflake database. The default port is 443.
4. To perform the following operations on the Snowflake database, get the following permissions from the Snowflake administrator:
   - select on information_schema.EXTERNAL_TABLES
   - select on information_schema.FUNCTIONS
   - select on information_schema.PIPES
   - select on information_schema.PROCEDURES
   - select on information_schema.SCHEMATA
   - select on information_schema.SEQUENCES
   - select on information_schema.STAGES
   - select on SNOWFLAKE.ACCOUNT_USAGE.TAGS
   - select on SNOWFLAKE.ACCOUNT_USAGE.TAG_REFERENCES
   - show objects
   - show columns
   - show primary keys
   - show imported keys
   - show streams
   - show materialized views
   - show tasks
   - show databases
5. To scan procedures and functions, set the following roles depending on permissions:
   - With administrator permission. In the JDBC connect string, use: `role=<procedure owner>` or `role=SYSADMIN`

     Only the SYSADMIN can scan procedures or functions defined with the EXECUTE AS OWNER statement.
   - Without requiring administrator permission. Set the USAGE privileges for all procedures and functions. Create procedures with the EXECUTE AS CALLER statement.
   - To extract the definitions of secured views and generate view-to-table lineage, ensure that the role used by the MetaDex scanner role has one of the following privileges on the secured views: Ownership or References privilege. A role without these privileges cannot read the definition of a secured view, and the scanner skips the view during lineage extraction.

     Important: If you leave the Role field blank, Snowflake uses the user's default role. The default role may lack the necessary privileges required to perform required actions. It is recommended to explicitly assign an appropriate role with the required privileges.
6. If you use the dynamic code that consists of solution building queries based on content of the database tables, grant read only access to all the tables that store queries or their components.
7. Configure the Snowflake MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table lists properties that are required to create a connection to a Snowflake database:

Property	Description
Host	Host name or IP address of the Snowflake service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select Snowflake as the required vendor from the list.
URL	Enter the JDBC URL. For example: `jdbc:snowflake://account.snowflakecomputing.com?warehouse=<warehouse name>&db=<database name>`
User	The user name used to access the database.
Password	Optional. The password associated with the user name.

For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

Types of input

You can extract Snowflake metadata from the following types of input:

- Dynamic tables
- External tables
- Hybrid tables
- JavaScript functions
- JavaScript stored procedures
- Pipes
- SQL functions
- SQL scripts (files)
- Stages

  Note: Amazon S3 only.
- Views, materialized views

You can use the following SQL functionalities:

- Static SQL
- Procedure calls with parameters tracing
- Dynamic SQL and JavaScript with metatables
  Only for selected scenarios.

Generated asset types

MetaDex scanners create assets that you can view in the Catalog tool as Enterprise Data Catalog objects.

You can see the hierarchy of the following Snowflake objects:

- Package > Procedure definition > Procedure instance

- Package > Unresolved procedure definition
- Tag
- Unresolved procedure definition > Procedure instance > Result set > Result set column
- Unresolved procedure definition > Procedure instance > Parameter container > Parameter
- Unresolved procedure definition > Procedure instance > Procedure source > Source column
- Unresolved procedure definition > Procedure instance > Statement
- Unresolved procedure definition > Procedure instance > Unresolved statement
- Pipe > Statement
- Pipe > Unresolved statement
- Stage > Stage column
- Unresolved view
- Sequence > Sequence column
- Task > Statement
- Task > Unresolved statement
- External table > External column

To find the generated asset types in the Catalog Administrator tool, click: **Manage** > **Models** > **Advanced Scanners - Database Objects**.

Lineage

You can view the following lineage:

- Stored procedure calls
- Pipes and tasks
- External tables
- Views
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

You can view the foreign key relationship.

For more information about the **Relationships** tab in the Data Catalog tool, see "Relationships View" in the *Enterprise Data Catalog User Guide*.

Rules and guidelines

Consider the following rules and guidelines when you use the Snowflake MetaDex scanner:

- You can't extract column-level lineage for files and stages.
- You can extract lineage from procedures and functions in the JavaScript language.
- While extracting metadata from Snowflake tables, certain types of columns, such as DATE, do not display the column length in bytes. The length is not specified by the database.

Sybase

Sybase is an SQL database server that uses a relational model. In the MetaDex tool, you can create and configure a metadata scanning job for a Sybase database.

Prerequisites and permissions

To scan Sybase metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. To connect to the Sybase database, download the jConnect 4 JDBC driver from the Sybase website.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Sybase database. The default port is 5000.
4. Grant the View Definition permissions that you need to perform the following operations on the Sybase database:

   select on dbo.SYSCOLUMNS

   select on dbo.SYSCOMMENTS

   select on dbo.SYSOBJECTS

   select on dbo.SYSSERVERS

   select on dbo.SYSTYPES

   select on dbo.SYSUSERS
5. If you use dynamic code that consists of solution building queries based on content of the database tables, grant read-only access to all tables that store queries or their components.
6. Configure the Sybase MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table describes properties that are required to create a connection to an Sybase database:

Property	Description
Host	Host name or IP address of the Sybase service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select Sybase as the required vendor from the list.
URL	The JDBC URL. Enter the URL in the following format: `jdbc:sybase:Tds:<Host>:<PortNumber>?ServiceName=<DatabaseName>` The default port number is 5000.

Property	Description
User	The user name used to access the database.
Password	The password associated with the user name.

For more information, see *Creating a connection* in the *MetaDex Scanner Administrator Guide*.

Types of input

In the MetaDex tool, you can extract Sybase metadata from the following types of input:

- SQL scripts (files)
- Stored procedures

Note: To extract metadata from SQL scripts, use the **Advanced PL/SQL to EDC** conversion path. To extract metadata from stored procedures, use the **Advanced Database Objects to EDC** conversion path.

You can also use the following SQL functionalities:

- Static SQL
- Procedure calls with parameter tracing
- Dynamic SQL with metatables
  Only for selected scenarios.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Package > Procedure Definition
- Package > Unresolved Procedure Definition
- Parameter
- Parameter container
- Pipe > Statement
- Pipe > Unresolved Statement
- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence > Sequence Column
- Source column

Lineage

You can view the following lineage:

- Statements within functions and stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.

- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Sybase MetaDex scanner:

- You cannot extract metadata from overloaded procedures.
- You cannot extract metadata from external tables.
- You cannot extract metadata from views.

Teradata

Teradata is an information repository. You can use tools and utilities that make it a complete relational database management system.

Prerequisites and permissions

To scan Teradata metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. Get JDBC drivers from the website of your database vendor:
   a. Download the following JDBC driver files:
      - tdgssconfig.jar
      - terajdbc4.jar

      Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM).
   b. Copy the JDBC driver files to the `${INFA_HOME}/AdvScannersWorkspace/jdbc` directory.

   For more information, see "JDBC connections" in the *MetaDex Scanner Administrator Guide*.
3. Establish network connectivity between any port on the machine where you install Enterprise Data Catalog and the listening port of the Teradata database. The default port is 1025.
4. Grant the permissions that allow you to perform the following operations:
   - select on dbc.COLUMNSV
   - select on dbc.TABLESV
   - select on dbc.TABLETEXTV
   - select on dbc.INDICESV
   - select on dbc.ALL_RI_CHILDRENV
   - SHOW PROCEDURES

- SHOW MACROS
- HELP PROCEDURES
- HELP MACROS
- SHOW VIEWS
- HELP VIEWS

5. If you use dynamic code that consists of solution building queries based on content of the database tables, grant read only access to all the tables that store queries or their components.
6. Configure the Teradata MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

The following table lists properties required to create a connection to a Teradata database:

Property	Description
Host	Host name or IP address of the Teradata service.
Database	The name of the database instance.
Repository type	Select JDBC as the repository type from the list.
Vendor	Select Teradata as the required vendor from the list.
URL	Enter the JDBC URL. For example: `jdbc:teradata://10.65.42.221/1025/QA`
User	The user name used to access the database.
Password	The password associated with the user name.

For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

Types of input

You can extract Teradata metadata from the following types of input:

- BTEQ
- MultiLoad
- FastLoad
- FastExport
- Macros
- Stored procedures
- Views

You can also use the following SQL functionalities:

- Static SQL
- PL/SQL
- Procedure calls with parameters tracing
- Dynamic SQL with metatables only for selected scenarios

Generated asset types

The MetaDex scanner creates the following types of objects in the catalog:

- Parameter
- Parameter container
- Procedure definition
- Procedure instance
- Procedure source
- Result set
- Result set column
- Sequence
- Sequence column
- Source column
- Statement
- Unresolved procedure definition
- Unresolved statement

Lineage

You can view the following lineage:

- Statements within macros.
- Views.
- Stored procedures.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

You can't extract metadata from Teradata functions that use a non-SQL programming language.

Troubleshooting database management scanners configuration issues

This section contains the common configuration issues that a database management MetaDex scanner can report after you run a job for the first time.

"Failed to process PL/SQL call" error appears.

```
ERROR: Failed to process PL/SQL call. Cannot determine value for: @[variable].
```

This error appears when the value of a variable is unknown.

Perform the following actions to solve the issue:

1. In the MetaDex tool, select the applicable job.
2. Click **Download** > **Download debug log**.
3. In Notepad ++, open the full debug log and search for the following message: `Cannot determine value for`
   The following example shows a sample log message:

```
ERROR (DefaultPlSqlFunctionCallVisitor:143) - Failed to process PL/SQL call.
Cannot determine value for: @strQuery.
Visitors stack:
  + script
  + definition of function doc_proc
  + root function TEST.TEST.dbo.doc_proc
  + block @5.0:15.2
  + call sp_executesql(@strQuery) @13.1:13.28
com.compactsolutionsllc.cdimc.common.exceptions.CannotDetermineValueException:
 Cannot determine value for: @strQuery.
```

4. Search for other occurrences of the `Cannot determine value for` message.
   The following example shows a sample result of the search:

```
DEBUG (DefaultAssignmentStatementVisitor:80) - Assignment @strQuery := '
                select a, b
                from test
                where a in (' + @input_var +
')
                order by b'; @8.1:12.2 causes scalar variable @strQuery to be
set to <EXCEPTION_VALUE:
Cannot determine value for: @input_var.> of type scalar(VARCHAR(4000)).
```

   The message indicates that the <@input var> value is not provided in the default variables values file on the project **Details** tab.

5. Provide the missing value in the default_values.txt file.
   The following example shows a sample file:

```
[VARIABLES]
@input_var = 12345
[FUNCTIONS]
```

"Failed to process PL/SQL call (...) Function name is not deterministic" error appears.

```
ERROR: Failed to process PL/SQL call. Cannot determine value for: @[variable].
```

Value returned by a function cannot be determined. This error appears when a procedure uses a function that dynamically determines values. The following example shows dynamically determined current date as part of a table name::

```
CREATE PROCEDURE [dbo].[doc_proc5]
AS
BEGIN
      Declare @strQuery nvarchar(4000)
```

```
      Select @strQuery = '
      select a, b
      from test_' + convert(varchar(8), getdate(),112) + '
      order by b'
      exec sp_executesql @strQuery
END
GO
```

Perform the following actions to solve the issue:

1. In the MetaDex tool, select the applicable job.
2. Click **Download** > **Download debug log**.
3. In Notepad ++, open the full debug log and search for the following message: `Cannot determine value for`
   The following example shows a sample log message:

   ```
   DEBUG (DefaultAssignmentStatementVisitor:80) - Assignment @strQuery := '
                select a, b
                from test_' + CONVERT (VARCHAR(8), GETDATE, 112) +
   '
                order by b'; @7.1:10.1 causes scalar variable @strQuery to be
   set to
   <EXCEPTION_VALUE: GETDATE is not deterministic - add default value for
   function call.> of type scalar(VARCHAR(4000)).
   ```

4. Create a default values file with the following content:

   ```
   [VARIABLES]
   [FUNCTIONS]
   GetDate()='YYYYMMDD'
   ```

"Missing connection information for link" error appears.

```
ERROR (MSSQLRepository:244) - Missing connection information for link: linked_server_name
```

```
WARN (MSSQLRepository:246) - Add entry in MDREPO/host_name/db_name/db.properties: connection.
[X]=linked_server_name=[VENDOR]:[HOST].[DB].[SCHEMA]
```

This error appears when there is no information about the linked server. The following example shows a sample of the error message:

```
ERROR (MSSQLRepository:244) - Missing connection information for link:
linked_server_name
WARN  (MSSQLRepository:246) - Add entry in MDREPO/host_name/db_name/
db.properties: connection.[X]=linked_server_name=[VENDOR]:[HOST].[DB].[SCHEMA]
DEBUG (BaseRepository:712) - Searching MSSQL repository (EXTERNAL: Repository
created from databases, csvs or ddls) for sequence
(Context : sqlPart): [host_name].[db_name].[dbo] : table_name .
DEBUG (BaseRepository:746) - Doing MSSQL catalog search for name [host_name].
[db_name].[dbo].[remote_table_name]
ERROR (PlSqlQueryConverter:66) - Unknown table: [linked_server_name].
[linked_server_table_name].[dbo].[remote_table_name]
Statement at 26.0:39.52:
<SQL>
insert into db_name.dbo.tbl_name
```

```
select
*
from [linked_server_name].linked_server_table_name.dbo.remote_table_name
where 1=1
</SQL>
com.compactsolutionsllc.cdimc.sql.exceptions.UnknownTableException: Unknown
table:
[linked_server_name].[linked_server_table_name].[dbo].[remote_table_name]
```

Perform the following actions to solve the issue:

1. In the MetaDex tool, select **Administration** > **Connection Registry**.
2. For the applicable host, click **Edit**.
3. From the drop-down list, select **connection**.
4. Add the connection specified in the WARN message.

Note: Your database configuration can have more than one connection. Ensure that the connection number is unique.

"No viable alternative at input" error appears.

```
[main] INFO (LogSeparatorsWriterWritingToLogs:19) - --BEGIN: D:\MSSQL_DEMO\input
\CalculateTotalProfits.sql

main ERROR (AdvancedPlSqlReaderModule:311) - Failed to analyze PL/SQL script. Unable to parse
script.

com.compactsolutionsllc.cdimc.app.modules.exceptions.InvalidInputException: Unable to parse
script.

at
com.compactsolutionsllc.cdimc.plsql2.mssql.parser.MsSqlPlSqlParser.parseScript(MsSqlPlSqlPars
er.java:44) (...)

at com.compactsolutionsllc.cdimc.app.Application.main(Application.java:170)

Caused by: java.lang.RuntimeException: position 3:8: no viable alternative at input 'select'

at
com.compactsolutionsllc.cdimc.sql.mssql.MSSQLSQLParser.selectListExpr(MSSQLSQLParser.java:223
10)

at
com.compactsolutionsllc.cdimc.plsql2.mssql.parser.MsSqlPlSqlParser.parseScript(MsSqlPlSqlPars
er.java:42)

... 17 common frames omitted

Caused by: org.antlr.runtime.NoViableAltException: null

at org.antlr.runtime.DFA.noViableAlt(DFA.java:158)

at org.antlr.runtime.DFA.predict(DFA.java:116)

at
com.compactsolutionsllc.cdimc.sql.mssql.MSSQLSQLParser.selectListExpr(MSSQLSQLParser.java:218
54)

... 36 common frames omitted
```

This error suggests an issue in line 3 if the CalculateTotalProfits.sql file, around the word "select".

Perform the following actions to solve the issue:

1. Open the file and verify the indicated issue.
2. Fix the issue.

"Number of target column names does not match the number of query outputs" error appears.

```
ERROR Number of target column names does not match the number of query outputs

2 expected: 3 Statement at 4.0:4.48:

<SQL>

insert into test_table select * from test_table_2

</SQL>

com.compactsolutionsllc.cdimc.sql.exceptions.InvalidColumnListException: Number of target
column names does not match the number of query outputs. Found: 2 expected: 3

at

com.compactsolutionsllc.cdimc.sql.names.NameReslverHelper.checkColumnListSize(NameResolverHel
per.java:70)
```

This error appears when the structures of two tables do not match. The following example shows a sample of query that inserts data from test_table_2 to test_table and causes the error:

```
Use [test]
GO
CREATE PROCEDURE [dbo].[doc_proc3]
AS
BEGIN
insert into test_table select * from test_table_2
END
GO
```

test_table has two fields and test_table_2 has three, which causes the error.

Perform the following action to solve the issue:

1. Verify if the number of rows is identical for both tables.

"Unknown table" error appears.

```
ERROR Unknown table

<SQL>

insert into test_table select a, b from test_table_2

</SQL>

com.compactsolutionsllc.cdimc.sql.exceptions.UnknownTableException: Unknown tsble: [TEST].
[TEST].[dbo].[test_table_2]

at

com.compactsolutionsllc.cdimc.sql.mssql.MsSqlPlSqlNameResolver.process(MsSqlPlSqlNameResolver
.java:318)
```

This error appears when a table is missing. To generate a lineage, MetaDex server requires definitions of all tables used in the procedure. The following example shows a sample of query that causes the error:

```
Use [test]
GO
CREATE PROCEDURE [dbo].[doc_proc2]
AS
BEGIN
insert into test_table select a,b from test_table_2
END
GO
```

Initially, both tables were in the database but test_table_2 is no longer there, which causes the error.

Perform the following action to solve the issue:

1. Verify if the table mentioned in the error message exists and you can access it.

CHAPTER 3

Configuring ETL scanners

This chapter includes the following topics:

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- Databricks, 65
- IBM InfoSphere DataStage, 74
- Informatica Intelligent Cloud Services, 79
- Microsoft Azure Data Factory, 82
- Microsoft SQL Server Integration Services, 89
- Oracle Data Integrator, 93
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Overview of ETL scanners

The ETL category of MetaDex scanners includes sources that move and transform data between data stores, such as applications, databases or file systems.

Before you create any configuration, validate the prerequisites and necessary permissions, and familiarize yourself with the parameters, authentication methods, types of input and assets that a given scanner generates. If you haven't created a configuration with the MetaDex tool before, familiarize yourself with the general MetaDex scanners configuration workflow before you proceed.

Databricks

Databricks combines data warehouses and data lakes into an AI-driven Databricks Lakehouse platform. Delta Lake is an open format storage layer for a data lake. Databricks Notebooks is a web-based interface to a document that contains code that you can run, visualizations, and narrative text. You can use the MetaDex tool to extract lineage from Databricks Delta Lake and Databricks Notebooks.

Note: Effective in version 10.5.3.0.1, the Databricks Delta Lake scanner is available for technical preview.

Technical preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in

accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Prerequisites and permissions

To scan Databricks metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. To extract Databricks Notebooks files, select one of the following methods:
   - Export Databricks Notebooks files in a DBC format.
     Does not require connectivity to Databricks Notebooks. Set the **Databricks input read mode** parameter to **Files** and use the **Input files** parameter to indicate the paths to files that you want to process.
   - Connect to a live instance of Databricks Notebooks.

   For more information, see Databricks documentation.
3. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Setting up connectivity" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in Databricks Notebooks. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.
4. Configure the Databricks MetaDex scanner.

Databricks Notebooks connection properties

Perform the following actions to connect to a live instance of Databricks Notebooks through REST API:

1. In the MetaDex tool, set the **Databricks input read mode** parameter to **Workspace**.
2. Enter the following parameters:

Parameter	Description
Workspace host	Required. The host name of the Databricks Notebooks workspace.
Workspace authentication token	Required. Authentication token to access the Databricks Notebooks workspace.
Workspace path	Required if you download a workspace larger than 10 MB. An advanced parameter that you can set to narrow down the scope to a single path.

3. To download a workspace that is larger than 10 MB, in Databricks Notebooks, manually unload the workspace into a DBC file. Use the DBC file instead of live connection to the Databricks Notebooks instance.

For more information about Databricks Notebooks parameters, see "Databricks Reader" in the *MetaDex Scanner Reference*.

Databricks Delta Lake connection properties

Perform the following actions to connect to Databricks Delta Lake:

1. In the connection registry, create a connection to the Databricks database. For example: `MyHost/MyDatabricks`

   For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

2. In the **Databricks Reader** module, enter a value for the **Delta Lake SQL Context** parameter to define a connection to the Databricks database. For example: `MyHost.MyDatabricks.Schema`

Types of input

You can extract Databricks Notebooks metadata from the following types of input:

- Live connection to the Databricks Notebooks instance.
- Notebooks files, which are DBC files exported from Databricks Notebooks.

Input for both Notebooks live connection and Notebooks files must include all direct and transitive dependencies. For example, if you use "notebook a" which uses "notebook b," the input must include both "notebook a" and "notebook b."

You can use the following Python functionalities:

- Standard language constructions
- Standard built-in functions
- Supported modules:
  - pyspark
- Partially supported modules:
  - azure
  - datetime
  - gc
  - glob
  - json
  - matplotlib
  - numpy
  - os
  - pandas
  - re
  - requests
  - sys
  - time
  - urllib
  - warnings
- Custom libraries

You can also use DeltaLake SQL in SQL commands and PySpark calls.

For the detailed list of languages, see "Supported Languages" in the "Databricks Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

MetaDex scanners create assets that you can view in the Catalog tool as Enterprise Data Catalog objects.

You can see the hierarchy of the following Databricks objects:

- Folder > Notebook > Command
- Folder > Notebook definition > Notebook instance > Command
- Folder > Notebook definition > Notebook instance > Parameter
- Folder > Notebook definition > Notebook instance > Result

In Databricks Notebooks scans, Notebook definition represents Notebooks that you create in Databricks. A single Notebook definition can have multiple Notebook instances. You run a Notebook instance with specific parameter values.

In Databricks Delta Lake scans, you can use relational and file model objects.

Lineage

You can view the following lineage:

- Summary and command or transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Databricks Notebooks MetaDex scanner:

- You can extract metadata from Python and SQL Notebook command types only. You can't extract metadata from other command types, such as Scala or R.
- If you have incomplete lineage that is caused by wildcard characters, you can configure the scanner to access information about the file names and directory paths that include wildcards to resolve them. For more information, see the following How-to library article: *Configure ETL scanners to use information about files that match a pattern*

Creating a Databricks configuration

You can extract metadata from Databricks Notebooks and Databricks Delta Lake assets. When you create a Databricks configuration, you configure parameters for the Reader, Assets, and Writer modules.

1. In the MetaDex tool, open the Databricks project.
2. In the Databricks project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.
3. Enter a name and an optional description for the configuration.
4. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.

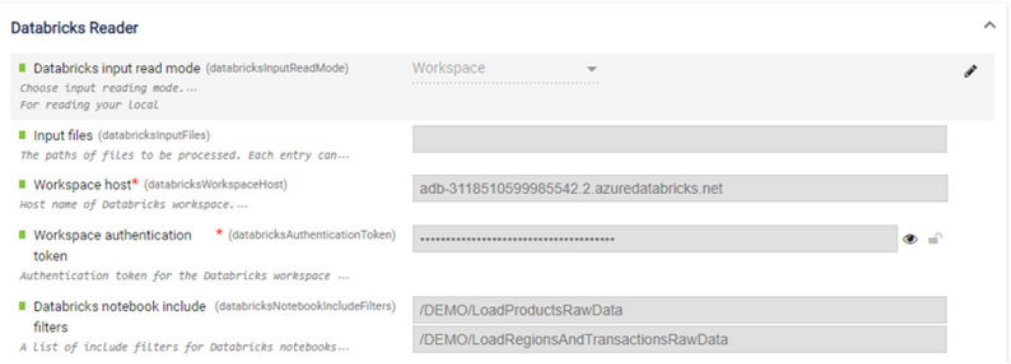
5. From the mode list, based on your level of expertise, select the mode in which you want to create the configuration. You can select from the following modes:

Mode	Description
Basic mode	Default mode. Includes the minimum of data source connection properties required to scan metadata.
Advanced Mode	Includes additional properties. You can use this mode to adjust processing metadata to your requirements.
Expert mode	Includes all available properties. If you are an experienced user, or if Informatica Global Customer Support suggests it, use this mode to adjust processing metadata to your requirements.

   A list of possible configuration modules appears.

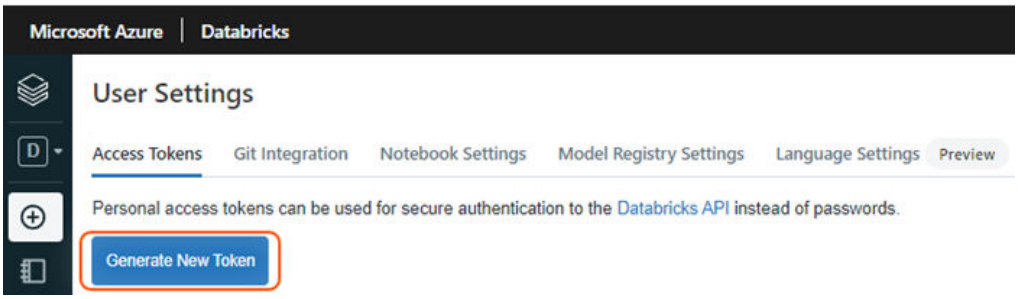
6. From the **Conversion path** list, select one of the following options:
   - To extract metadata from Notebooks, select: **Databricks to EDC**
   - To extract metadata from Notebooks and assets, select: **Databricks to EDC**
   - To extract metadata from assets, select: **Advanced Database Objects to EDC**
7. In the **Databricks Reader** module, navigate to the **Databricks input read mode** parameter and select one of the following options:
   - **Files**. Choose it to use DBC and .py files. You can manually export the files from a Databricks Notebooks instance.
   - **Workspace**. Recommended choice to connect to a Databricks Notebooks workspace end extract the notebooks automatically.
8. Depending on your choice in the previous step, perform one of the following actions:
   - Provide paths to the previously extracted files in the **Input files** parameter.
   - Provide the **Workspace host** and **Workspace authentication token**.

   The following image shows example values:



   For information about the parameter values, see "Modules Reference" in the MetaDex Scanner Reference.

You can generate the **Workspace authentication token** in the Databricks tool. Use the **Generate New Token** option available on the **User Settings** page, as shown in the following image:



9. Enter a value for the **Notebooks Preload Paths** parameter to specify the relative workspace path to scan and preload notebooks. For example, specify `/notebook directory` as the relative workspace path to preload all notebooks from the mentioned directory.
10. Enter a value for the **Databricks notebook include filters** parameter to filter notebooks which you want to extract.

    The **Workspace** list includes all Databricks workspace notebook paths. To ensure that you provide the correct path, you can copy a file path, available for every notebook in the Databricks tool. The name of the option is **Copy File Path**. To access the **Copy File Path** option, click the arrow next to a notebook name.
11. Connect to the Databricks database:
    a. In the connection registry, create a connection to the Databricks database. For example: `MyHost/MyDatabricks`

       Note: When you configure a connection to extract metadata from Databricks Unity Catalog, ensure that the database name in the connection is `hive_metastore` and the JDBC URL includes the `ConnCatalog=hive_metastore` parameter.

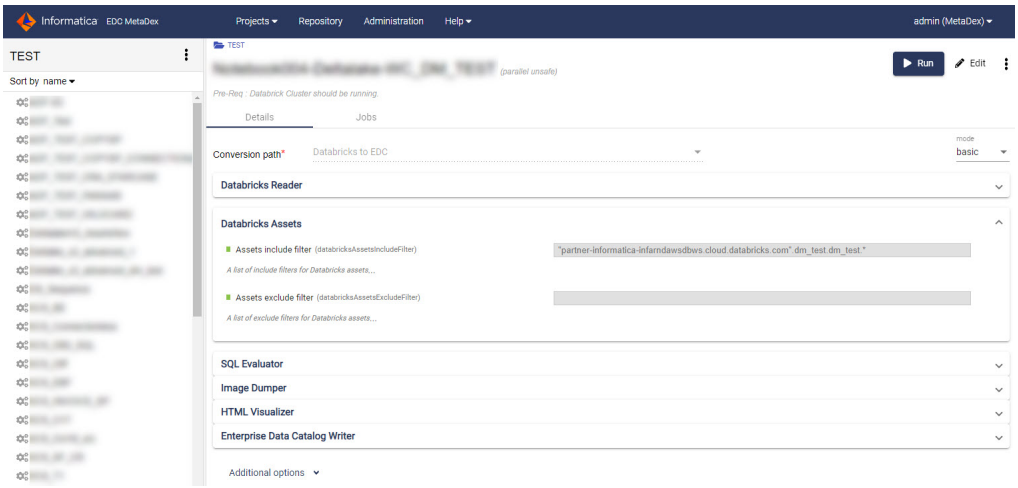
       For more information, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

    b. In the **Databricks Reader** module, enter a value for the **Delta Lake SQL Context** parameter to define a connection to the Databricks database. For example: `MyHost.MyDatabricks.Schema`

12. Expand the **Databricks Assets** module and provide values for the following parameters:

Parameter	Description
Assets include filter	Processes a limited set of Databricks Delta Lake assets, such as tables or views. Enter at least one filter in the following format: <host name>.<database name>.<schema name>.<asset name>. For example: `"InformaticaUser.cloud.databricks.com".test.test.*` An asset name can contain a question mark (?) to substitute any single character, or an asterisk (*) to substitute multiple characters. A MetaDex configuration processes assets that match at least one include filter and don't match any of the exclude filters.
Assets exclude filter	Optional. Processes a limited set of Databricks Delta Lake assets, such as tables or views. Enter at least one filter to exclude in the following format: <host name>.<database name>.<schema name>.<asset name>. For example: `"InformaticaUser.cloud.databricks.com".test.test.log` An asset name can contain a question mark (?) to substitute any single character, or an asterisk (*) to substitute multiple characters. A MetaDex configuration processes assets that match at least one include filter and don't match any of the exclude filters.

The following image shows the Databricks Assets module parameters set to run a sample Databricks configuration:



13. Optionally, expand the **Unity Lineage** module and provide values for the following parameters:

Parameter	Description
Lineage from Unity	Select one of the following metadata extraction options from Databricks Unity Catalog: - Disabled. Doesn't extract lineage from Unity Catalog. Default is disabled. - Enabled for filtered assets. Extracts lineage from assets selected in filter conditions. - Enabled for all assets. Extracts complete lineage including referenced resources.
Workspace host	Specify the host name of the Databricks workspace.

Parameter	Description
Workspace authentication	Specify the authentication token for the Databricks workspace access. It contains a random string of 38 symbols that Databricks generates. For example, `dapi8be46d02f290489a1defbad48bbbbf77-3`.
Delta Lake SQL Context	Define the connection to the Databricks database. For example, `MyHost.MyDatabricks.Schema`.

14. Expand the **Enterprise Data Catalog Writer** module and provide values for the following parameters:

Parameter	Description
EDC writer mode	Set the value to **Generate output files**. After a successful parsing, you can change it to **Load using REST API**. After the job is complete, the extracted metadata loads to your Enterprise Data Catalog instance.
Output directory	The path to the output directory of the extracted metadata files.
EDC address	The URL of Enterprise Data Catalog. Do not change the global variables if the target of the metadata is a local Enterprise Data Catalog instance.
EDC user	The Enterprise Data Catalog user name that you need to connect to the Enterprise Data Catalog instance.
EDC password	The Enterprise Data Catalog password that you need to log in to the Enterprise Data Catalog instance.
EDC resource name	The name of the resource that the system creates in the Enterprise Data Catalog instance. It contains metadata extracted by the configuration.

For a complete list of parameter descriptions, see the "Databricks Reader" chapter of *MetaDex Scanner Reference*.

15. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

16. Use the **WARN** filter field to filter and review the warning or error messages.

    When you select **WARN**, you can review the warnings and errors in one list.

17. Analyze the reported warning or error messages.
18. Adjust the configuration.
19. Run the job again.

    For more information, see "Running a scanner job" on page 20.

20. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Troubleshooting Databricks Notebooks configuration issues

This section describes the most common configuration issues that a Databricks Notebooks MetaDex scanner can report after you run a job for the first time.

"Connection was created using connectionless approach". warning appears.

```
Missing connection definition for: "jdbc:sqlserver://metadex-test-
sql.database.windows.net:1433;database=ace-
demo;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;l
oginTimeout=30;". Connection was created using connectionless approach.
```

This warning appears when the scanned Databricks Notebooks cannot establish a connection with a database.

When the connection is missing, the scanner operates in offline mode and tries to analyze SQL statements to establish source data structures. However, in some cases it requires a connection to the real source database, otherwise it might fail to scan some data. For more information, see "Connection requirements" in *MetaDex Scanner Administrator Guide*.

Perform the following actions to solve the issue:

1. Add the correct database to the **Connection Registry** in the MetaDex tool.
2. Add the connection to the configuration.

   Note: The connection mapping entry consists of the **Database Connection key**, which you can find in the warning message, and a **Resolved MDREPO entry**. The following example presents the correct **Resolved MDREPO entry** syntax: <VENDOR>:<USER>@HOST.DATABASE.<SCHEMA>

   where: <HOST> and <DATABASE> are values that you configured when you added the connection to the Connection Registry.

   For more information about how to add a connection, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.
3. Run the configuration again and verify the logs. For more information, see "Running a scanner job" on page 20.

"Failed to resolve file name (...) Cannot determine value (...)" warning appears.

```
Failed to resolve file name (<EXCEPTION_VALUE: Cannot determine value for: GET('metadex-key-
vault', 's3-bucket-metadex-dev-test-access-key').>), will use ''s3a://' + GET('metadex-key-
vault', 's3-bucket-metadex-dev-test-access-key') + ':' + REPLACE(GET('metadex-key-vault',
's3-bucket-metadex-dev-test-secret-key'), '/','%2F') + '@' + 'metadex-dev-test-1'' instead
```

This warning appears when there is a missing default variable value in the input code. Missing variable values can affect the accuracy of the metadata loaded to an Enterprise Data Catalog instance.

Perform the following actions to solve the issue:

1. In the input code, check the variables quoted in the warning message.
   For example, the following fragment of sample input code contains the values stated in the warning:



2. Create a file to define the missing variable values.

The following example shows the correct structure of the file:

```
[VARIABLES]
access_key='s3-bucket-metadex-dev-test-access-key'
secret_key='s3-bucket-metadex-dev-test-secret-key'
```

3. Navigate to **Enterprise Data Catalog Writer** > **Python default variables values file** and enter the path to the file.
4. Run the job again and verify the log file.

IBM InfoSphere DataStage

IBM InfoSphere DataStage is an ETL platform that is used to integrate data across multiple systems.

Prerequisites

To scan IBM InfoSphere DataStage metadata, perform the following prerequisite tasks:

1. Split input into projects, and then create a configuration for each project. For example, if you split the input into 10 projects based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into projects.
2. To extract IBM InfoSphere DataStage metadata, export all your project files in the ISX format. You can export the files with the istool utility provided as part of the DataStage installation or with the IBM InfoSphere Information Server Manager.
   For more information, see IBM documentation.
3. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in IBM InfoSphere DataStage. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.
4. Configure the IBM InfoSphere DataStage MetaDex scanner. For more information, see “Configuring MetaDex scanners” on page 16.

Permissions

Not applicable.

Connection properties

Not applicable.

Types of input

You can use the IBM InfoSphere DataStage MetaDex scanner to extract metadata from exported ISX files. From the ISX files, MetaDex scanners process the following types of jobs:

- Parallel jobs
- Sequence jobs
- Server jobs

For the list of stages that the DataStage Reader module uses, see "Supported Stages" in the "DataStage Reader" section of the *MetaDex Scanner Reference*.

Optionally, you can provide additional XML files that contain operational metadata. You can use the files to generate lineage for parametrized jobs. For more information, see IBM documentation.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Engine
- Project
- Folder
- Job

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- Pre/post SQL.
- Stored procedure calls.
- Jobs with parameters defined from sequences and from operational metadata.

Relationships

Not applicable.

Rules and guidelines

You can extract lineage from IBM InfoSphere DataStage jobs that use runtime column propagation (RCP). If the scanner cannot fully establish the list of propagated columns, lineage might not be complete and correct.

Creating an IBM DataStage configuration

When you create an IBM DataStage configuration, you can configure parameters for the Reader and Writer modules. Before you create a configuration, prepare your input.

To learn more about how to prepare your input, see the
HOW TO: Export IBM InfoSphere DataStage jobs to an ISX file Knowledge Base article.

1. From the **More** menu in the DataStage project, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.
2. Enter a name and an optional description for the configuration.
3. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.

4. From the mode list, based on your level of expertise, select the mode in which you want to create the configuration. You can select from the following modes:

Mode	Description
Basic mode	Default mode. Includes the minimum of data source connection properties required to scan metadata.
Advanced mode	Includes additional properties. You can use this mode to adjust processing metadata to your requirements.
Expert mode	Includes all available properties. If you are an experienced user, or if Informatica Global Customer Support suggests it, use this mode to adjust processing metadata to your requirements.

   A list of possible configuration modules appears.

5. From the **Conversion path list**, select **IBM DataStage to EDC**.
6. Expand the **DataStage Reader** module.
7. In the **Input files or folders** field, provide a path to the ISX files or folders that you want to process.
8. Optionally, in the **Data source connections** field, provide the **Database connection key** and the **Resolved MDREPO entry** values. For more information, see "Connection requirements" in the *MetaDex Scanner Administrator Guide*.
9. Expand the **Enterprise Data Catalog Writer** module and provide values for the following parameters:

Parameter	Description
EDC Writer mode	Determines how the Writer module operates. Set this parameter to **Generate output files**. After a successful parsing, you can change the parameter to **Load using REST API**. After the job is complete, the extracted metadata loads to the catalog.
Output directory	The path to your output directory. The scanner generates files containing extracted metadata and stores them in the output directory.
EDC resource name	The name of the resource that the system creates in the Enterprise Data Catalog tool. It contains metadata extracted by the parsing configuration.

   You can find descriptions of all the other parameters in the "Module Reference" chapter of the *MetaDex Scanner Reference*.

10. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

11. Use the **WARN** filter field to filter and review the warning or error messages.

    When you select **WARN**, you can review the warnings and errors in one list.

12. Analyze the reported warning or error messages.
13. Adjust the configuration.
14. Run the job again.

    For more information, see "Running a scanner job" on page 20.

15. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Troubleshooting IBM DataStage configuration issues

This section describes the most common configuration issues that an IBM DataStage MetaDex scanner can report after you run a job for the first time.

"Missing connection definition for: "ConnectionName" warning appears.

```
WARN Missing connection definition for: "ConnectionName". Processing module:
DataStageReaderModule. Connection was created using connectionless approach. WARN Analysis
performed in the connectionless repository: [host].[ConnectionName]
```

This warning appears when you don't define the ConnectionName property in **Data source connections** on the DataStage Reader page, and the MetaDex tool uses the connectionless approach.

Perform the following actions to resolve the issue:

1. In the **DataStage Reader** module, go to **Data source connections**.
2. Define the missing connection.
3. Run the job again.

Note: You can ignore the warning and use the connectionless approach, but this approach has certain limitations. For more information, see "Connection Requirements" in the *MetaDex Scanner Administrator Guide*

"Failed to analyze PL/SQL script" warning appears.

```
WARN Failed to analyze PL/SQL script.
com.compactsolutionsllc.cdimc.app.modules.exceptions.InvalidInputException: There is no
parser capable of parsing
```

This warning appears when the SQL query is incorrect and the default values of parameters in the SQL query are missing.

Perform the following actions to resolve the issue:

1. In the log message between `<SQL>` and `</SQL>`, find the SQL query that causes the warning.
2. Verify the SQL query.
3. In the **DataStage Reader** module, go to **Job parameter override**.
4. Add the missing default parameter values.
5. Run the job again.

"Failed to process deferred PL/SQL script" error appears.

```
ERROR Failed to process deferred PL/SQL script
com.compactsolutionsllc.cdimc.app.modules.exceptions.InvalidInputException: Unable to parse
script(...)Caused by: com.compactsolutionsllc.cdimc.common.exceptions.ParseException:
```

This warning appears when the SQL query is incorrect and the default values of parameters in the SQL query are missing.

Perform the following actions to resolve the issue:

1. In the log message, after `Caused by`, check the reason for the error.

2. Verify the SQL query.
3. In the **DataStage Reader** module, go to **Job parameter override**.
4. Add the missing default parameter values.
5. Run the job again.

"Cannot convert ISX job" error appears.

```
ERROR Cannot convert isx job: jobName.pjb: Cannot find shared container: containerName
```

This warning appears when the ISX file is incomplete and the "containerName" shared container is missing.

Perform the following actions to resolve the issue:

1. Export the jobs with dependent items.
   For instructions, see Export IBM InfoSphere DataStage jobs to an ISX file.
2. Run the job again.

"ISX file 'fullPath\fileName.isx' is incomplete" warning appears.

```
WARN ISX file 'fullPath\fileName.isx' is incomplete. The following Parameter Set files are
missing for the job jobName: parametrSetName
```

This warning appears when the ISX file is incomplete and the "parameterSetName" parameter set is missing.

Perform the following actions to resolve the issue:

1. Export the jobs with dependent items.
   For instructions, see Export IBM InfoSphere DataStage jobs to an ISX file.
2. Run the job again.

Note: If you ignore the warning, you must enter the default parameter values from the missing parameter sets in the **Job parameter override**.

"Cannot process expression of column columnName " warning appears.

```
WARN Cannot process expression of column columnName in stageName stage of type Transformer:
Cannot resolve variable: parameterJobName
```

This warning appears when the default value of the "parameterJobName" parameter is missing.

Perform the following actions to resolve the issue:

1. In the **DataStage Reader** module, go to **Job parameter override**.
2. Add the missing default parameter values.
3. Run the job again.

"Cannot analyze sql stage: stageName" error appears.

```
Cannot analyze sql stage: stageName: Expected single last component, but have: 0 for stage:
stageName. It's due to previously reported parsing errors
```

This warning appears when the MetaDex tool can't parse SQL queries from the "stageName" stage.

Perform the following actions to resolve the issue:

1. In the logs, verify that all parameters inside the SQL query have default values.
2. If values are missing, in the **DataStage Reader** module, go to **Job parameter override**.
3. Add the missing default parameter values.

4. Run the job again.

"Error in job jobName" error appears.

```
ERROR Error in job jobName, stage stageName of type STAGE_TYPE. Could not find sql file: /path/fileName
```

This warning appears when the MetaDex tool can't find the `/path/fileName` file.

Perform the following actions to resolve the issue:

1. On DataStage server, find the `/path/fileName` file.
2. Copy the file to the machine where the MetaDex tool is installed.
3. Use the **Filename Mapper** to create a mapping for the file path.
4. Run the job again.

Informatica Intelligent Cloud Services

Informatica Intelligent Cloud Services is an on-demand subscription service that provides access to applications, databases, platforms, and flat files hosted on premise or in the cloud. Informatica Intelligent Cloud Services runs at a hosting facility.

The Informatica Intelligent Cloud Services MetaDex scanner extracts metadata from the Cloud Data Integration and MDM SaaS assets.

Prerequisites and permissions

To scan metadata, perform the following prerequisite tasks:

1. Split input into projects, and then create a configuration for each project. For example, if you split the input into 10 projects based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into projects.
2. To access Informatica Intelligent Cloud Services assets, you must have read permissions on the assets and your organization must be licensed to use the Informatica Intelligent Cloud Services REST API.
3. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in IICS. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.
4. Configure the scanner. For more information, see “Configuring MetaDex scanners” on page 16.

You don't need to download any drivers. Informatica provides the required drivers in the installation package provided by the installer.

Connection properties

Enter the following parameters:

- IICS Cloud URL

- IICS Cloud User
- IICS Cloud Password

Types of input

You can extract metadata from the following types of input:

- Mapping tasks
- PowerCenter tasks
- Synchronization tasks
- Business 360 entities

Generated asset types

MetaDex scanners create assets that you can view in the Catalog tool as Enterprise Data Catalog objects.

You can see the hierarchy of the project objects extracted from Informatica Intelligent Cloud Services metadata:

- Folder > Business entity
- Folder > Mapping
- Folder > Task
- Business entity > Field
- Mapping
- Task > Task instance
- Mapping task instance
- Synchronization task instance
- PowerCenter task
- Mapping task
- Synchronization task

To find the generated asset types in the Catalog Administrator tool, click: **Manage** > **Models** > **Advanced Scanners - IICS V2**.

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- Parametrized mappings.

The Informatica Intelligent Cloud Services MetaDex scanner extracts B360 native connectors in mappings to generate lineage.

Relationships

You can view the following usage associations for IICS assets:

- Mapping
- Mapping Task

Parametrization

Informatica supports the following Informatica Intelligent Cloud Services - Cloud Data Integration mapping parametrization:

- Input parameters:
  - Source/Target objects parameterization for relational and flat files
  - Source/Target connections parameterization
  - Transformations that use the Sorter/Aggregator input parameters
  - Partial parametrization
- InOut parameters:
  - Different datatypes for parameters
  - Functions in expressions
- Parameter context

Informatica does not support the following cases of parametrization in version 10.5.2:

- SQL Override
- Pre/Post SQL
- SQL transformation connection parameterization
- PowerCenter tasks with parameters
- Advanced Session properties

Rules and guidelines

Consider the following rules and guidelines when you use the Informatica Intelligent Cloud Services - Cloud Data Integration MetaDex scanner:

- You cannot extract tasks that were never executed.
- If you executed the same task multiple times with different parameters, only the last task execution parameters are imported to the catalog.
- You can find the transformation-level details in the visualizations available in the MetaDex tool. You can find the URL in the system attributes.

Microsoft Azure Data Factory

Microsoft Azure Data Factory is a data integration service that you can use to visually integrate various data sources.

Prerequisites and permissions

To scan Azure Data Factory metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. Provide metadata in one of the following ways:
   - Export the Azure Resource Manager (ARM) template in the form of JSON files.
   - Authenticate as service principal.
   - Authenticate as admin user.

   For more information, see "Connection properties".
3. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in Azure Data Factory. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.
4. Get access to Azure Data Factory. At minimum, you need the Reader role on the Azure Data Factory resource that you want to scan.
   For more information, see Azure Data Factory documentation.
5. Configure the Azure Data Factory MetaDex scanner. For more information, see “Configuring MetaDex scanners” on page 16.

You don't need to download any drivers. Informatica provides the required drivers in the installation package provided by the installer.

Connection properties

To connect Azure Data Factory to Microsoft Azure through REST API, choose one of the following methods depending on your authentication method:

- For the service principal authentication method, in the MetaDex tool, provide the following parameters:
  1. REST API tenant ID
  2. REST API client secret
  3. REST API client ID
  4. REST API subscription ID

  For more information on Azure Data Factory REST API parameters, see "Azure Data Factory Reader" in the *MetaDex Scanner Reference*.

  Optionally, you can scan Azure Data Factory metadata directly from JSON files.
- For the admin user role method, in the MetaDex tool, provide the following parameters:
  1. REST API user

2. REST API password
3. REST API client ID
4. REST API subscription ID

For more information on Azure Data Factory REST API parameters, see "Azure Data Factory Reader" in the *MetaDex Scanner Reference*.

Optionally, you can scan Azure Data Factory metadata directly from JSON files.

Types of input

In the MetaDex tool, you can extract Azure Data Factory metadata from the following types of input:

- Pipelines with the following activities:
  - Copy
    Including embedded SQL.
  - Get metadata
  - Lookup
  - Stored procedure
  - Execute data flow

    Note: Effective in version 10.5.1.0.1, you can view generic lineage at the data set level with element or column-level transformations.
- Data sets:
  - Delimited text
  - JSON
  - Parquet
  - Salesforce objects
  - Amazon Redshift table
  - Azure SQL data warehouse table
  - SQL Server table
  - Oracle table
  - Azure SQL table
  - Snowflake table

For more details, see "Supported Pipeline Activities, Datasets and Data Flow components" in the "Azure Data Factory Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Factory
- Folder
- Pipeline
- Activity
- Data set
- Data flow

Lineage

You can view the following lineage:

- Summary lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- Stored procedure calls.
- Embedded SSIS package.

Relationships

Not applicable.

Rules and guidelines

If you have incomplete lineage that is caused by wildcard characters, you can configure the scanner to access information about the file names and directory paths that include wildcards to resolve them. For more information, see the following How-to library article: *Configure ETL scanners to use information about files that match a pattern*

Creating a Microsoft Azure Data Factory configuration

When you create a Microsoft Azure Data Factory configuration you can configure parameters for the Reader and Writer modules. You can also filter the metadata based on pipelines and activities that you want to extract.

1. In your Microsoft Azure Data Factory project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.
2. Enter a name and an optional description for the configuration.
3. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
4. From the mode list, based on your level of expertise, select the mode in which you want to create the configuration. You can select from the following modes:

Mode	Description
Basic mode	Default mode. Includes the minimum of data source connection properties required to scan metadata.
Advanced Mode	Includes additional properties. You can use this mode to adjust processing metadata to your requirements.
Expert mode	Includes all available properties. If you are an experienced user, or if Informatica Global Customer Support suggests it, use this mode to adjust processing metadata to your requirements.

   A list of possible configuration modules appears.

5. From the **Conversion path** list, select **Azure Data Factory to EDC**.
6. Expand the **Azure Data Factory Reader** module.
7. In the **Reader mode** field, select one of the following options:

Option	Description
Files	Uses JSON files as input. Extract the files from the Azure Data Factory environment beforehand.
Admin User	Connects to the Azure Data Factory environment as the administrator user and extracts the content.
Service Principal	Connects to the Azure Data Factory environment with the service principal authentication method and extracts the content.

8. Depending on your choice in the previous step, perform one of the following actions:
   - If you chose the **Files** option, provide paths to the extracted files in the **Input files** parameter.
   - If you chose any of the other options, provide additional credentials described in the "Connection properties" subsection of the "Microsoft Azure Data Factory" on page 82 section.
9. To filter the pipelines and activities that you want to extract, provide values for the **ADF input filters** parameter available in the **Azure Data Factory Reader** module.

   Use the following syntax: <Include> or <Exclude>:<Factory Pattern>:<Pipeline pattern>:<Activity Pattern>

   where:

   - <Include> is an include filter keyword.
   - <Exclude> is an exclude filter keyword.
   - <Pattern> is a filter syntax that supports wildcards.

     Note: Use a question mark for a single character. Use an asterisk for multiple characters.

   You can choose one of the following sample values to fill in the **ADF input filters** parameter:

Action	Description and Syntax
Process all activities in all pipelines and all factories.	This is the default option. Use the following syntax: `Include:*:*:*`
Exclude all activities from a designated pipeline.	Use the following syntax: `Exclude:ADF_PROD:DAILY_PROCESS|CUSTOMERS_DIM|LoadCustomers:*` In this example, the pipeline name is LoadCustomers. The pipeline exists in the ADF_PROD factory, in the CUSTOMERS_DIM subfolder of the DAILY_PROCESS folder.
Process all activities from all pipelines in a designated factory.	Use the following syntax: `Include:*prod:*:*` This processes all activities from factories whose name ends with "prod."
Process all activities from a designated factory and pipeline.	Use the following syntax: `Include:ADF-LNT:DEMO|LoadTransactions:*` In this example, the factory name is ADF-LNT. The pipeline name is LoadTransactions and it exists in the DEMO folder.

10. Expand the **Enterprise Data Catalog Writer** module and provide the following parameters:

Parameter	Description
EDC Writer mode	Determines how the Enterprise Data Catalog Writer module operates. Depending on the configuration stage you completed, you can set the parameter to one of the following values: - Generate output files. Select this option before you run the job. - Load using REST API. Select this option after the job succeeds and you want to load the results to the catalog.
Output directory	The path to the output directory where the scanner stores extracted metadata.
EDC address	The host name and port number of the Catalog Service in the following format: <host>:<port>
EDC user	The user name configured for the catalog administrator user.
EDC password	The password configured for the catalog administrator user. When you add the password, click the **Hide password** icon.
EDC resource name	The name of the resource that the system creates in the Enterprise Data Catalog tool. The resource contains metadata extracted by the parsing configuration.

For other parameter descriptions, see the "Module Reference" chapter of the *MetaDex Scanner Reference*.

11. Click **Save** and then click **Run**.
12. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Troubleshooting Microsoft Azure Data Factory configuration issues

The MetaDex scanner reads and parses the input. If it encounters any issues, it generates error or warning messages. Use the **WARN** filter field to filter and review the warnings or error messages. When you select **WARN**, you can review the errors and warnings in one list. Analyze the reported warnings or error messages, change the configuration as required and run the job again. This section describes the most common configuration issues that a Microsoft Azure Data Factory MetaDex scanner can report after you run a job for the first time.

"Connection was created using connectionless approach."

```
Missing connection definition for: "ace_demo_metadex_test_sql". Connection was created using connectionless approach.
```

This warning appears when the scanned Azure Data Factory cannot establish a connection with a database.

When the connection is missing, the scanner operates in offline mode and tries to analyze SQL statements to establish source data structures. However, in some cases, it requires a connection to the real source database, otherwise the scanner might fail again. For more information, see "Connection requirements" in the *MetaDex Scanner Administrator Guide*.

Perform the following actions to solve the issue:

1. In the MetaDex tool, add the correct database to the **Connection Registry** .
2. Add the connection to the configuration.

   Note: The connection mapping entry consists of the following elements: **Database Connection key**, which you can find in the warning message, and a **Resolved MDREPO entry**. The following example presents the correct **Resolved MDREPO entry** syntax: <VENDOR>:<USER>@HOST.DATABASE.<SCHEMA>

   where: <HOST> and <DATABASE> are values that you configured when you added the connection to the Connection Registry.

   For more information about how to add a connection, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.
3. Run the configuration again and verify the logs. For more information, see "Running a scanner job" on page 20.

"Cannot determine value for: ProductCategoriesFileName"

```
Could not evaluate ADF expression @dataset(). SourceFileName is context ADF-
LNT.LoadTransactions.Import Product Categories, due to: Cannot determine value for:
ProductCategoriesFileName.
```

This warning can appear when there is a missing default variable value in the input code. Missing variable values can affect the accuracy of the metadata loaded to Enterprise Data Catalog.

Perform the following actions to solve the issue:

1. Create a default values configuration file.
2. In the file, define the default value for the "@dataset().SourceFileName" parameter. This parameter is used in the Import Product Categories activity. The activity exists in the LoadTransactions pipeline created in the ADF-LNT factory.
   To define this value, create the following sample section in the value configuration file:

   ```
   ["ADF-LNT"."LoadTransactions"."Import Product Categories"]
   SourceFileName='ProductCategories.csv'
   ```
3. Navigate to **Enterprise Data Catalog Writer** > **Parameter default value file** and enter the path to the default values configuration file.
4. Run the job again and verify the log file.

Similar warning messages can occur when other variable values are missing. To solve such issues, define one or more of the following variables in the variable value file.

Pipeline parameters and the system variable.

To define the default values for pipeline parameters and the system variable, create the following sample section in the variable value file:

```
["My factory"."My pipeline"]
parameters.Pipeline_Param='X'
RunId='X'
```

where:

- <Pipeline_Param> is the name of the parameter in the "My pipeline" pipeline, in the "My factory" factory.
- <RunId> is a system variable.

Pipeline activity items.

To define the default values for pipeline activity items, create the following sample section in the variable value file:

```
["My factory"."My pipeline"."My Activity"]
Dataset_Param='X'
utcnow()='...'
```

where:

- <X> is a default value for the sample "@dataset().Dataset_Param" parameter used in the "My Activity" expression.
- <utcnow()> is a sample function used in an expression in "My activity" properties.

Expressions that are in the input and output sections of an activity.

To define default values for expressions that are in the input and output sections of an activity, create the following sample section in the variable value file:

```
["My factory"."My pipeline"."My Activity"."input"]
Dataset_Param='S'
["My factory"."My pipeline"."My Activity"."output"]
Dataset_Param='T'
```

where:

- <S> is a sample default value for "@dataset.Dataset_Param" that is used in the "My Activity" input expression.
- <T> is a sample default value for the "@dataset().Dataset_Param" parameter that is used in the "My Activity" output expression.

Pipeline variables referenced as "variables('variable_name')."

To define default values for pipeline variables referenced as "variables('variable_name')," create the following sample section in the variable value file:

```
["My factory"."My pipeline"]
variables('variable_name')='default_value'
```

where:

- <default_value> is a sample default value for the "variable_name" variable used in the "My pipeline" pipeline, created in the "My factory" factory.

Microsoft SQL Server Integration Services

Microsoft SQL Server Integration Services (SSIS) is a component of the Microsoft SQL Server database that can be used to perform data transformation and integration tasks.

Prerequisites

To extract SSIS metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. Choose one of the following options:
   - If you store the packages in a database, export the input files together with the corresponding DTSCONFIG, CONMGR, and PARAMS files.
     Note that the database can be the SSISDB database from the Project Deployment Model or the MSDB database from the Package Deployment Model.

     You export the input files to the server where you previously installed MetaDex with the dtutil utility. You get the utility during the Microsoft SQL Server installation.
   - If SSISDB has the packages deployed, MetaDex scanners can connect to it to extract DTSX, CONMGR, and PARAMS files.
   - If you store the packages as files in the DTSX format, export the input files from the files system. You must also export the corresponding DTSCONFIG and CONMGR files.
3. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in SSIS. To prepare the extract, use the Metadata Extractor.
4. SSISDB catalog operations require a user with Windows authentication. To use the SSISDB mode, create an SSISDB database connection registry entry by adding the sqlServer.ENABLE_SSISDB=ON property to the SSISDB database connection registry entry properties.
   For Windows authentication on a Linux machine, consider NTLM user authentication. The following example shows the connection string for a Microsoft SQL server database that uses NTLM authentication in a domain named informatica.com: `jdbc:sqlserver://host01:1433;DatabaseName=SSISDB;integratedSecurity=true;authenticationScheme=ntlm;domain=informatica.com`
5. Configure the SSIS MetaDex scanner. For more information, see “Configuring MetaDex scanners” on page 16.

Permissions

To extract the SSIS project files from an SSISDB database in the SSISDB mode, grant permissions to perform the following operations on the SSISDB database:

- select on catalog.folders
- select on catalog.projects
- select on catalog.packages
- select on catalog.environments

- select on catalog.environment_variables
- select on catalog.object_parameters
- execute on catalog.get_project procedure

To read specific folders, projects, or packages, assign one of the following roles or permissions:

- Membership to the ssis_admin database role
- Membership to the sysadmin server role
- READ permission on the folders and projects

Connection properties

In the MetaDex tool, enter the following parameters to extract the SSIS project files from the SSISDB database:

Parameter	Description
SSISDB database connection name	Name for the SSISDB database connection.
Data source connections	The Data source connections parameter consists of the database connection key and the resolved MDREPO entry, which is the MetaDex connection registry entry. In the **Database Connection key** field, enter the SSISDB database connection name. In the **Resolved MDREPO entry** field, map the SSISDB database connection name to the SSISDB database connection registry entry. The entry has the following syntax: <vendor name>@<host name>.<name of the database defined in the connection registry>

Types of input

You can extract SSIS metadata from the following types of input:

- Data flows
- Control flows

For the detailed list of components, see "Supported Components" in the "SSIS Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Folder
- Package
- Data task
- Script

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.

- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- SQL task.
- Stored procedure calls.
- Defined project-level parameters.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the SSIS MetaDex scanner:

- You can read variable values and expressions from SQL Server package configurations.
- You can extract the packages files, project parameter files, and connection manager files, if the packages are deployed to the SSISDB.

Troubleshooting Microsoft SQL Server Integration Services configuration issues

This section contains the common configuration issues that a Microsoft SQL Server Integration Services (SSIS) MetaDex scanner can report after the first time you run a job.

"Missing connection definition" error appears.

```
ERROR Missing connection definition for: "MSSQL02.AdventureWorksDW2012". Connection
"METADEX_UNRESOLVED_CONNECTION_MSSQL02" created for the following job: element "Product
Subcategory Procedure Load Task-0"
```

This error appears when connection definition for an SSIS connection is missing.

Perform the following actions to solve the issue:

1. In the MetaDex tool, go to **Administration** > **Connection Registry**.
2. Click **Add host**.
3. Add applicable host information and click **Save**.
4. Click **Add entry**.
5. Add applicable Database information and click **Save**.
6. Go to **Projects** and select applicable SSIS project.
7. In the **SSIS reader** module, enter required information in the Data source connections parameter.

"No connection manager" error appears.

```
ERROR No connection manager id={0f }: available for the OleDBConnection connection. Define
every project-level connection manager with the "SsisReaderModule"."External Connection
Manager Files" parameter.
```

This error appears when a .conmgr file associated with the SSIS package is missing. The MetaDex server can't extract the required files from the SSIS repository.

Perform the following actions to solve the issue:

1. In the MetaDex tool, go to **Projects** and select the applicable project.

2. Select the applicable configuration.
3. Expand the **SSIS Reader** module.
4. In the **External Connection Manager Files** parameter, enter the path to the .conmgr file.
5. Click **Save**.

"No files found under host" error appears.

```
ERROR No files found under the following host: SSIS_HOST, path: \\dir\ExcelReport, nor in the internal repository, and MDREPO.

Output to that directory might not be registered yet. Run the MetaDex job again or add missing column definitions for the following file: \\dir\ExcelReport.

Add column definitions edit or create Files.csv a file in the folllowing location: MDREPO/SSIS_HOST
```

This error appears when a file definition is missing.

Perform the following actions to solve the issue:

1. Go to the following location: ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app/MDREPO/SSIS_HOST
   The file location results from the **Application host name** parameter settings.
2. Add or edit the Files.csv file so that it contains the following content:

```
"FilePath"."ColumnName"
"\\dir\ExcelReport","field1"
"\\dir\ExcelReport","field2"
```

"No matching files found in Foreach File Enumerator file or MetaDex repository" warning appears.

```
WARN Found no matching files in the Foreach File Enumerator file nor the MetaDex repository.

This can lead to unexpected behavior during the Foreach File Enumerator processing.

Set the proper file path in the "SsisReaderModule"."
```

This warning appears when a file list for a Foreach Loop Component is missing.

Perform the following actions to solve the issue:

1. In the MetaDex tool, go to **Projects** and select the applicable project
2. Select the applicable configuration.
3. Expand the **SSIS Reader** module.
4. In the **File containing paths used by Foreach File Enumerator** parameter, enter the path to the files that match the SSIS Enumerator Configuration from the Foreach Loop Container Configuration.
5. Click **Save**.
6. Ensure that the files are defined in the Files.csv file.

"No default value for variable" error appears.

```
ERROR No default value for the following variable: [Package::SSIS_DTKEY].

The following default value will be used = [SSIS_PROPERTY_VALUE::$Package::SSIS_DTKEY].
```

This error appears when you did not set the default value for a variable.

Perform the following actions to solve the issue:

1. In the MetaDex tool, go to **Projects** and select the applicable project.
2. Select the applicable configuration.
3. Expand the **SSIS Reader** module.
4. In the **Default variables values file** parameter, enter the path to the file that contains the missing value.
5. Click **Save**.

Oracle Data Integrator

Oracle Data Integrator (ODI) is a comprehensive ELT data integration platform that allows you to extract, load, and then transform large amounts of data.

Prerequisites

Before you configure the ODI MetaDex scanner to extract metadata from an ODI data source, complete the following prerequisite tasks:

1. Familiarize yourself with the configuration workflow for MetaDex scanners. For more information, see "Configuring MetaDex scanners" on page 16.
2. Learn about ODI mapping executions and procedure executions in Enterprise Data Catalogs. For more information, see the mapping executions and procedure executions knowledge article.
3. Verify that you're using ODI version 12c. To verify the version, confirm that the exported files contain `RepositoryVersion="05.<AB.CD.EF>"`, where <AB.CD.EF> are numbers of the minor version.

   Note: You must install the Oracle Data Integrator 12c client on the same machine as the MetaDex server.
4. Split input into projects, and then create a configuration for each project. For example, if you split the input into 10 projects based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into projects.
5. Get the Oracle JDBC Thin driver from the Oracle website:
   a. Download the driver files.
      Use drivers that support the version of the database that you want to connect to. For database versions, see the applicable Product Availability Matrix (PAM).
   b. Copy the driver files to the `${INFA_HOME}/AdvScannersWorkspace/jdbc` directory.

   For more information, see "JDBC examples" in the *MetaDex Scanner Administrator Guide*.
6. Identify the metadata extraction method that you want to use:
   Automatic export with the ODI direct database connection

   The ODI direct database connection reduces the number of configuration errors. To use the feature, configure the ODI MetaDex scanner in the MetaDex tool with the **Exporter** option. For more information, see "Extracting Oracle Data Integrator metadata with the ODI Direct Database Connection" on page 99.

Manual export from ODI files

Use one of the following options only when Informatica Global Customer Support recommends it:

- Manual export of the entire repository that is easier to perform but can take a long time for large repositories.
- Manual selective export that you can use during tests.

For more information, see "Extracting Oracle Data Integrator metadata from manually exported ODI files" on page 96 and the how to export ODI data knowledge article.

7. If you selected the automatic export option, request the following types of users:
   - Oracle database user that you need for the JDBC connection.
   - ODI Studio user that you need to connect to ODI Studio and the ODI Studio client.
8. If you selected the manual export option, request the Oracle database user that you need for the JDBC connection.

Connection properties

In the automatic export scenario, to connect to the live ODI instance, in the **ODI Reader** module, enter the following parameters:

Parameter	Description
ODI repository Connection registry entry	The entry that points to the connection registry of the ODI master repository.
ODI Studio password	The user password to ODI Studio.
ODI Studio user	The name of the ODI Studio user.
WORK REPO name	The name of the work repository that contains the user projects.

For detailed descriptions of the parameters with example values, see "ODI Reader" in the *MetaDex Scanner Reference*.

Permissions

To use the ODI extractor tool, set the following minimum permissions for the ODI Studio user:

- A combination of Connect and Designer or Design_Reviewer to access projects
- Topology admin or Release manager to access the topology

If you integrate ODI with external Version Control Systems (VCS), you can also set the VCS Version admin permission.

For information about the permissions to the connection properties, see "Configuring Oracle Data Integrator Studio" in the Oracle document *Fusion Middleware Installing and Configuring Oracle Data Integrator*.

Types of input

You can extract ODI metadata from the following types of input:

Knowledge modules

Knowledge modules are code templates. Each template is dedicated to an individual task in the overall data integration process.

Package steps

The package is the largest unit of execution in Oracle Data Integrator. Each unit is made up of a sequence of steps organized into an execution diagram.

Technologies

The classification of the possible technology types.

For a comprehensive list, see "Supported Knowledge Modules, Package Steps and Technologies" in the "ODI Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

MetaDex scanners create assets that you can view in the Catalog tool as Enterprise Data Catalog objects.

You can see the hierarchy of the following project objects:

- Folder > Interface > Scenario
- Folder > Interface > Interface execution
- Folder > Mapping > Scenario
- Folder > Mapping > Mapping execution
- Folder > Procedure > Scenario
- Folder > Procedure > Procedure execution
- Folder > Package > Scenario

Lineage

You can view the following lineage:

- Lineage at the summary and transformation levels.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.

Relationships

You can view the relationship between scenarios and the mappings that the scenarios use.

Rules and guidelines

Consider the following rules and guidelines when you use the ODI MetaDex scanner:

- Some production instances of ODI use compiled scenarios without projects, mappings, and packages.
- To get lineage from ODI scenarios with additional information on the ODI mappings from which the scenarios were created from, you need to provide metadata from both scenarios and mappings.
- Install the Oracle Data Integrator 12c client on the same machine as the MetaDex server. To install the exporter tool on a different machine than the MetaDex server, see the Knowledge Base article.
- For information about ODI, see the following Oracle documentation:
  - *Fusion Middleware Developer's Guide for Oracle Data Integrator*
  - *Fusion Middleware Installation Guide for Oracle Data Integrator*

Extracting Oracle Data Integrator metadata from manually exported ODI files

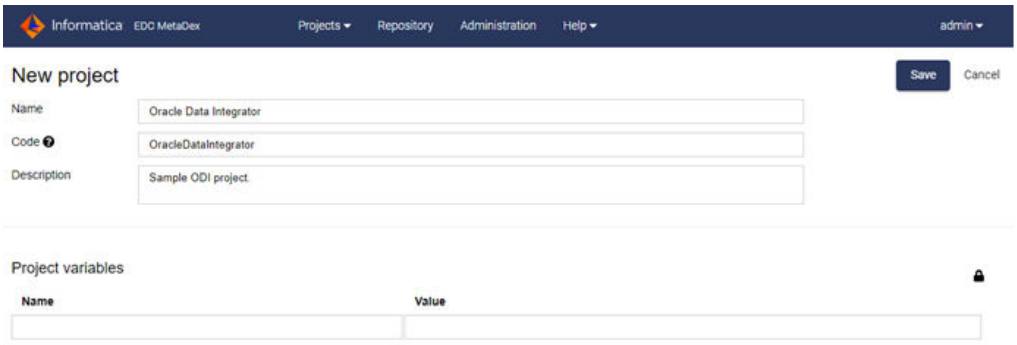
Create a sample Oracle Data Integrator configuration to extract metadata with the Oracle Data Integrator MetaDex scanner and load the metadata into the catalog.

Important: Use this option only when Informatica Global Support recommends it. Otherwise, extract metadata with the ODI exporter tool. For more information, see "Extracting Oracle Data Integrator metadata with the ODI Direct Database Connection" on page 99.

If the input does not include ODI scenarios, before you extract mappings, procedures, or packages, generate the scenarios to compile the code for the objects. For more information about generating a scenario, see "Using Scenarios" in the Oracle documentation.

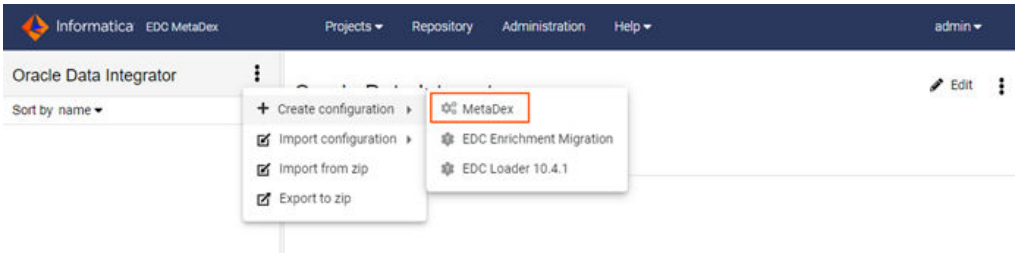
1. Create an Oracle Data Integrator project.

   The following image shows a **New project** page for a sample Oracle Data Integrator project:



2. In **Administration**, assign roles to users in the Oracle Data Integrator project.
3. In the project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The following image shows the **Create configuration** menu:



   The **New configuration** page appears.
4. Enter a name and an optional description for the configuration.
5. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
6. From the **Conversion path** list, select **ODI to EDC**.

   A list of expandable modules appears.
7. From the **mode** list, select the **basic** mode.
8. Expand the **ODI Reader** module.

   The parameter fields appear.
9. From the **Method for providing input files** list, select **Files**.

You can choose one of the following options:

- Manual export of the entire repository that is easier to perform but can take a long time for large repositories.
- Manual selective export that you can use during tests.

For more information about the manual export, see the Knowledge Base article.

10. In the **Input files** field, enter the file paths of XML files or ZIP files that contain XML files to include in the job or select the files from the **File Manager**.

    You can edit a file path after you select it. Use an asterisk (*) to substitute multiple characters. For example: `Y:\RootDir\input\*.zip`

11. In the **Odi context name** field, enter the name of a physical location, such as a database schema, for example, `Production`

    In Oracle Data Integrator, you use contexts to group physical resources together in the same environment.

12. Optionally, to get detailed lineage or lineage from stored procedures, in the **Data source connections** fields, enter the connection key and target.

    For instructions, expand the parameter description or see the "ODI Reader" section of the *MetaDex Scanner Reference*.

    The MetaDex server can process metadata without the connection information. For more information on the connectionless approach, see "Connection requirements" in the *MetaDex Scanner Administrator Guide*.

13. In the **Objects to be included during job** field, enter the objects to include in the job in the following format: `<project>/<folder path>/<object type>:<object name>`, for example, `ProjectName/**/Pack:*`

    An object include filter consists of the following elements:

    <project> is the project name in ODI Studio.

    <folder path> is the name of the folder in ODI Studio that contains the object that you want to include.

    <object type> is one of the following values: `Scen` for scenarios, `Pack` for packages, `Proc` for procedures, or `Map` for mappings.

    <object name> depends on the object type. If the object type is `Pack`, the object name is the name of the package.

    If you provide a project name in a filter, you must provide the package names of all the scenarios in that package in the following format: `<project>/<folder path>/Pack:<package name>/Scen:<scenario name>`

    If you omit the Scen:<scenario name> part, the filter enters `Scen:*` by default.

    You can substitute the elements with the following wildcard characters:

    - An asterisk (*) substitutes multiple characters in a project, a folder, or an object name.
    - A question mark (?) substitutes a single character in a project, a folder, or an object name.
    - A double asterisk (**) substitutes all possible folder depths.

14. In the **Objects to be excluded during job** field, enter the objects to exclude from the job in the following format: `<project>/<folder path>/<object type>:<object name>`, for example, `ProjectName/**/Pack:test*`

    An object exclude filter consists of the following elements:

<project> is the project name in ODI Studio.

<folder path> is the name of the folder in ODI Studio that contains the object that you want to include.

<object type> is one of the following values: `Scen` for scenarios, `Pack` for packages, `Proc` for procedures, or `Map` for mappings.

<object name> depends on the object type. For example, if the object type is `Pack`, the object name is the name of the package.

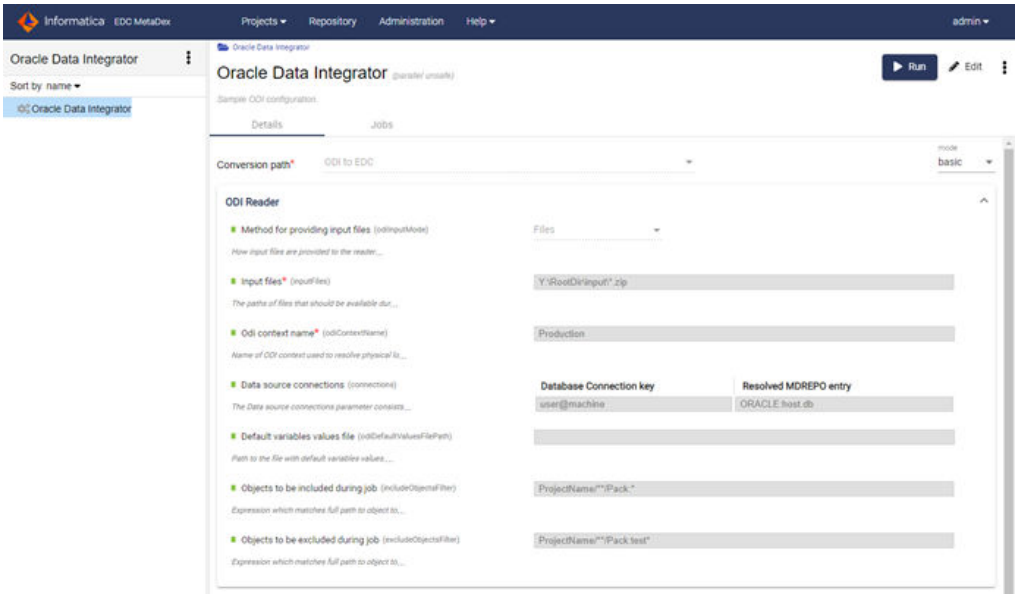
If you provide a project name in a filter, you must provide the package names of all the scenarios in that package in the following format: `<project>/<folder path>/Pack:<package name>/Scen:<scenario name>`

If you omit the Scen:<scenario name> part, the filter enters `Scen:*` by default.

You can substitute the elements with the following wildcard characters:

- An asterisk (*) substitutes multiple characters in a project, a folder, or an object name.
- A question mark (?) substitutes a single character in a project, a folder, or an object name.
- A double asterisk (**) substitutes all possible folder depths.

The following image shows the ODI Reader module parameters set to run a sample Oracle Data Integrator configuration:



15. Expand the **Enterprise Data Catalog Writer** module.

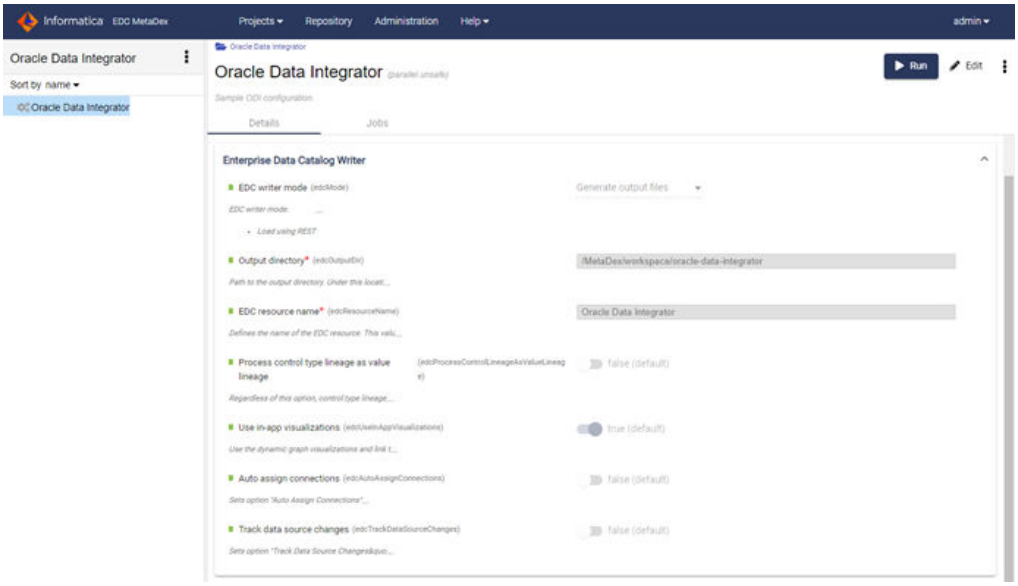
    The parameter fields appear.

16. In the **Output directory** field, enter the path to the output directory where the scanner stores extracted metadata or select the folder from the **File Manager**.

17. In the **EDC resource name** field, enter a unique name of the resource that the system creates in the Catalog tool.

    The resource contains metadata extracted by the configuration.

The following image shows the **Enterprise Data Catalog Writer** module parameters set to run a sample Oracle Data Integrator configuration:



18. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

19. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Extracting Oracle Data Integrator metadata with the ODI Direct Database Connection

Create a sample Oracle Data Integrator configuration to extract metadata with the Oracle Data Integrator MetaDex Direct Database Connection and load the metadata into the catalog.

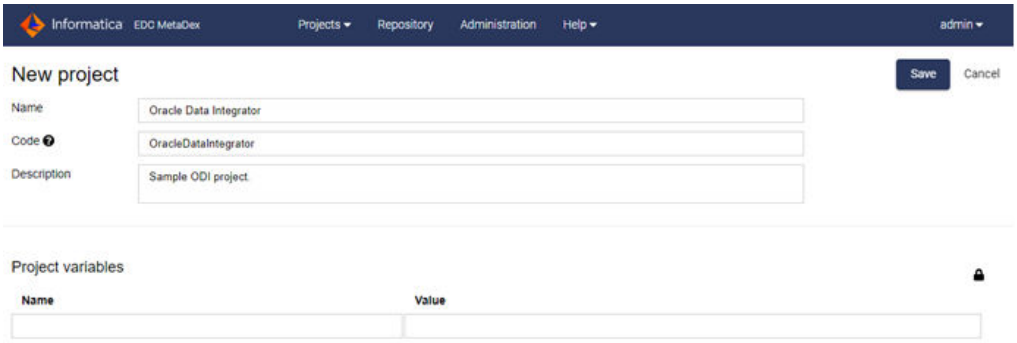
Before you begin, get access to the SNP_LOC_REP table for the user that connects to the ODI database.

1. Install the Oracle Data Integrator 12c client on the same machine as the MetaDex server.

   You can download the ODI client from the Oracle website.

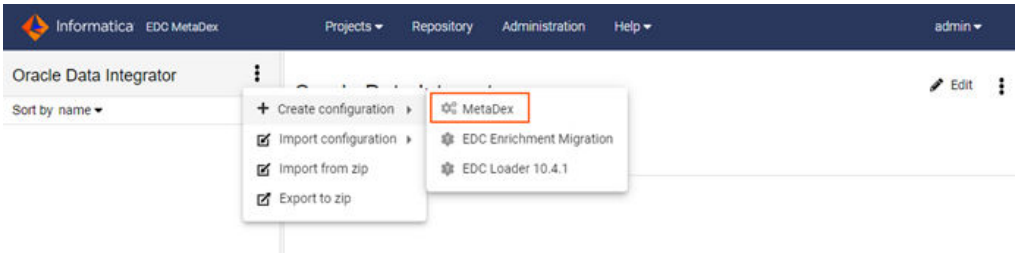
2. Create an Oracle Data Integrator project.

   The following image shows a **New project** page for a sample Oracle Data Integrator project:



3. In **Administration**, assign roles to users in the Oracle Data Integrator project.
4. In the project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The following image shows the **Create configuration** menu:

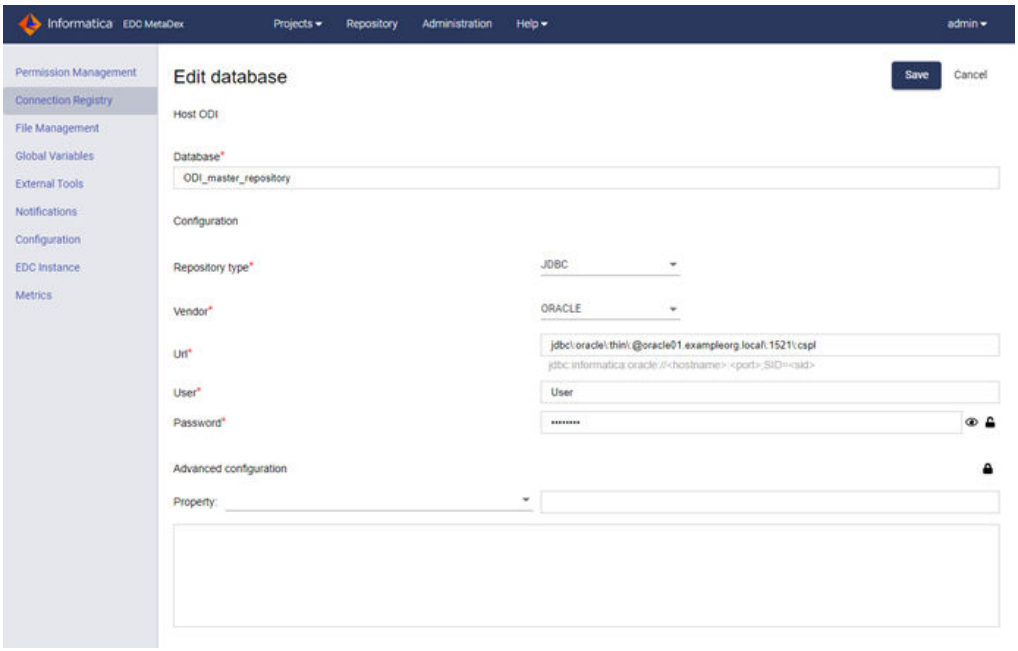


   The **New configuration** page appears.
5. Enter a name and an optional description for the configuration.
6. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
7. From the **Conversion path** list, select **ODI to EDC**.

   A list of expandable modules appears.
8. From the **mode** list, select the **basic** mode.
9. Expand the **ODI Reader** module.

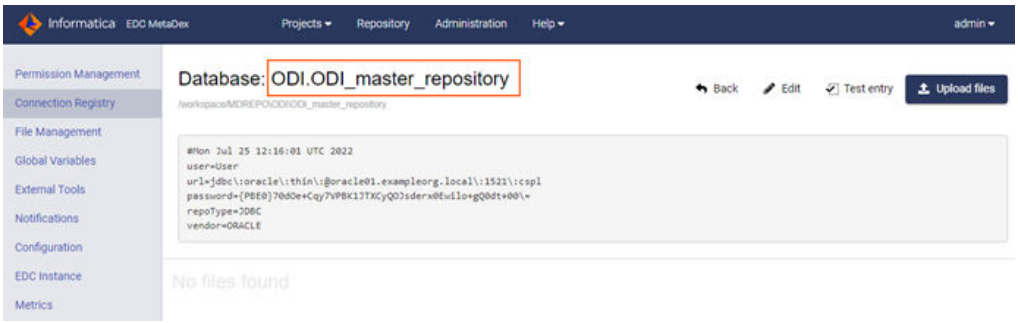
   The parameter fields appear.
10. From the **Method for providing input files** list, select **Exporter**.
11. In the connection registry, create an entry for a connection to the ODI master repository database.

    The following image shows a sample database entry in the connection registry for an ODI host:



    For more information, see "Creating a connection" in the MetaDex Scanner Administrator Guide.
12. Copy the connection name in the following format: <host name>.<database name>

The following image shows the connection name:



13. In the **ODI Master Repository Path** field, paste the connection name that you copied in Step 12. For example: `ODI.ODI_master_repository`
14. Optionally, if the Work Repository differs from Master Repository, repeat the same steps and paste the connection name in the **ODI Work Repository Path** field. If you want to override the work repository connection, select the Oracle connection that you created in Administrator.
15. In the **Odi context name** field, enter the name of a physical location, such as a database schema, for example, `Production`

    In Oracle Data Integrator, you use contexts to group physical resources together in the same environment.
16. Optionally, to get detailed lineage or lineage from stored procedures, in the **Data source connections** fields, enter the connection key and target.

    For instructions, expand the parameter description or see the "ODI Reader" section of the *MetaDex Scanner Reference*.

    The MetaDex server can process metadata without the connection information. For more information on the connectionless approach, see "Connection requirements" in the *MetaDex Scanner Administrator Guide*.
17. In the **Objects to be included during job** field, enter the objects to include in the job in the following format: `<project>/<folder path>/<object type>:<object name>`, for example, `ProjectName/**/Pack:*`

    An object include filter consists of the following elements:

    <project> is the project name in ODI Studio.

    <folder path> is the name of the folder in ODI Studio that contains the object that you want to include.

    <object type> is one of the following values: `Scen` for scenarios, `Pack` for packages, `Proc` for procedures, or `Map` for mappings.

    <object name> depends on the object type. If the object type is `Pack`, the object name is the name of the package.

    If you provide a project name in a filter, you must provide the package names of all the scenarios in that package in the following format: `<project>/<folder path>/Pack:<package name>/Scen:<scenario name>`

    If you omit the Scen:<scenario name> part, the filter enters `Scen:*` by default.

    You can substitute the elements with the following wildcard characters:

    - An asterisk (*) substitutes multiple characters in a project, a folder, or an object name.
    - A question mark (?) substitutes a single character in a project, a folder, or an object name.

- A double asterisk (**) substitutes all possible folder depths.

18. In the **Objects to be excluded during job** field, enter the objects to exclude from the job in the following format: `<project>/<folder path>/<object type>:<object name>`

    An object exclude filter, for example, `ProjectName/**/Pack:test*`, consists of the following elements:

    <project> is the project name in ODI Studio.

    <folder path> is the folder name in ODI Studio that contains the object that you want to include.

    <object type> is one of the following values: `Scen` for scenarios, `Pack` for packages, `Proc` for procedures, or `Map` for mappings.

    <object name> depends on the object type. For example, if the object type is `Pack`, the object name is the name of the package.

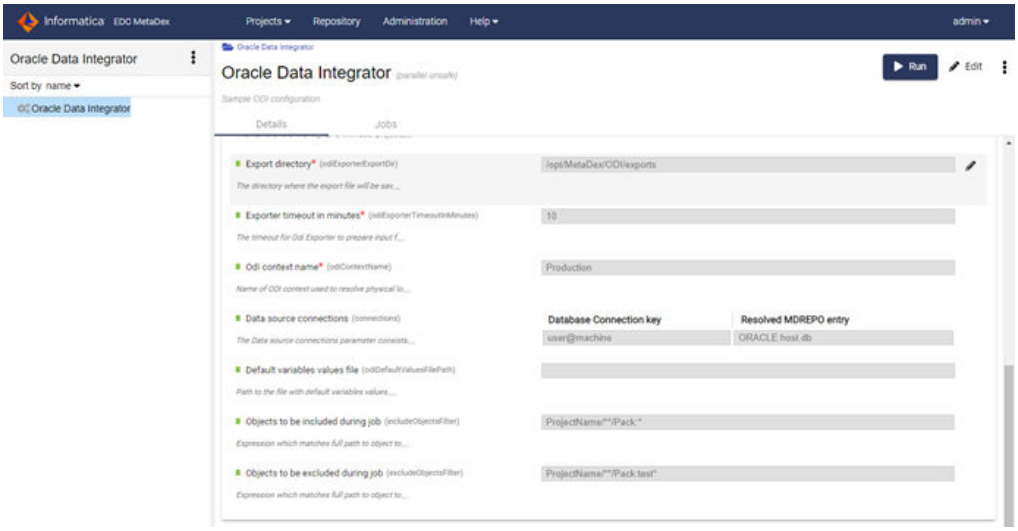
    If you provide a project name in a filter, you must provide the package names of all the scenarios in that package in the following format: `<project>/<folder path>/Pack:<package name>/Scen:<scenario name>`

    If you omit the Scen:<scenario name> part, the filter enters `Scen:*` by default.

    You can substitute the elements with the following wildcard characters:

    - An asterisk (*) substitutes multiple characters in a project, a folder, or an object name.
    - A question mark (?) substitutes a single character in a project, a folder, or an object name.
    - A double asterisk (**) substitutes all possible folder depths.

    The following image shows the second part of the ODI Reader module parameters set to run a sample Oracle Data Integrator configuration:



19. Expand the **Enterprise Data Catalog Writer** module.

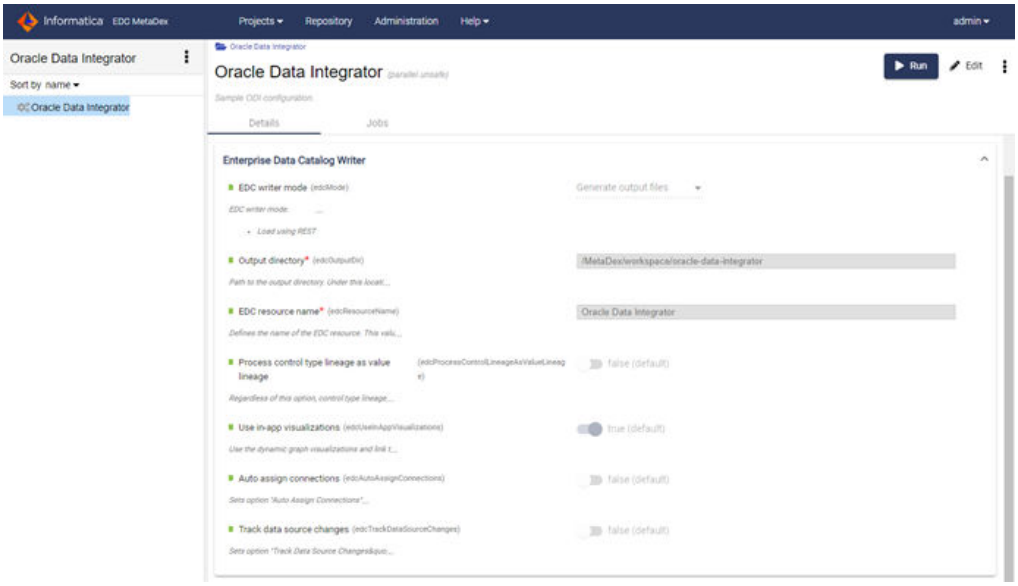
    The parameter fields appear.

20. In the **Output directory** field, enter the path to the output directory where the scanner stores extracted metadata or select the folder from the **File Manager**.

21. In the **EDC resource name** field, enter a unique name of the resource that the system creates in the Catalog tool.

    The resource contains metadata extracted by the configuration.

The following image shows the **Enterprise Data Catalog Writer** module parameters set to run a sample Oracle Data Integrator configuration:



22. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

23. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Note: When multiple objects create tables sharing the same name on the same connection, use the **Temporary Table** field to identify tables with duplicate names. In SQL Evaluator, in the **Temporary Table** field enter a pattern to identify duplicate table names.

When a table name matches the pattern, the MetaDex tool appends a unique prefix, such as #TMP_0_TABLENAME, #TMP_1_TABLENAME to its name.

The unique prefix lets the MetaDex scanner to display a separate lineage for each table.

SAP BODS

SAP Business Objects Data Services (BODS) is an ETL tool for extracting, transforming, and then loading data. It is designed to deliver enterprise-class solutions for data integration, data quality, data processing, and data profiling.

Prerequisites and permissions

To scan SAP BODS metadata, perform the following prerequisite tasks:

1. Split input into projects, and then create a configuration for each project. For example, if you split the input into 10 projects based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into projects.

2. To extract SAP BODS metadata, you need to export your entire repository to an XML file. For more information, see "Exporting a repository to a file" in SAP documentation.
3. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in SAP BODS. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.
4. Configure the SAP BODS MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

You don't need to download any drivers. Informatica provides the required drivers in the installation package provided by the installer.

Connection properties

Not applicable.

Types of input

In the MetaDex tool, you can extract SAP BODS metadata from XML files using the following input capabilities:

- Flow Control Component
  - Call Step
  - If Step
  - Parallel Step
  - Script
  - Try Step
  - While Step
- Dataflow Component
  - Excel Source
  - File Source
  - File Target
  - Query
  - Table Source
  - Table Target
  - Xml Source
  - Xml Target
- Dataflow Transform Component
  - Case Operation
  - Data Transfer
  - Date Generation
  - History Preserving
  - Key Generation
  - Map Operation

- Merge
- Pivot
- Reverse Pivot
- Row Generation
- SQL
- Table Comparison

For more information, see "Supported Components" in the "Business Objects Data Services Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

The MetaDex scanner creates the following types of objects in the catalog:

- Repository
- Project
- Job
- Workflow
- Dataflow
- Script

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- SQL script function.
- Stored procedure calls.

Relationships

Not applicable.

Rules and guidelines

When you use the SAP BODS MetaDex scanner, you can't extract metadata by exporting an entire repository of ATL files.

Talend

Talend is an ETL data integration platform that you can use to gain business insights from data.

Prerequisites

Before you configure a Talend MetaDex scanner, complete the following prerequisites:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. Export projects from the Talend workspace:
   a. In the Talend workspace, go to **File** > **Export** > **General** > **File System**.
   b. Select the applicable projects.
   c. Select **Select root directory** and click **Browse**.
   d. Choose the applicable directory.
   e. Click **Finish**.
3. Configure the Talend MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Connection properties

Not applicable.

Types of input

You can extract Talend metadata from Talend projects that contain the following components:

- Cloud. For example, S3, Snowflake, HDFS.
- Database. For example, relational databases.
- File. For example, XML, Flat, Excel.
- Processing. For example, map, join, filter.

For a detailed list of components, see "Supported Components" in the "Talend Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

The MetaDex scanner creates the following types of objects in the catalog:

- Project
- Folder
- Job
- Subjob

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.

- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- Stored procedure calls.

Relationships

Not applicable.

Rules and guidelines

Not applicable.

CHAPTER 4

Configuring reporting scanners

This chapter includes the following topics:

- Overview of reporting scanners, 108
- Microsoft Power BI, 108
- SAS, 115
- Microsoft SQL Server Analysis Services, 118
- Microsoft SQL Server Reporting Services, 120

Overview of reporting scanners

The reporting category of MetaDex scanners includes sources that allow you to build data visualizations in the form of reports or dashboards.

Before you create any configuration, validate the prerequisites and necessary permissions, and familiarize yourself with the parameters, authentication methods, types of input and assets that a given scanner generates. If you haven't created a configuration with the MetaDex tool before, familiarize yourself with the general MetaDex scanners configuration workflow before you proceed.

Microsoft Power BI

Microsoft Power BI is a business intelligence tool that allows you to connect data and create dashboards that can be shared. Note that Microsoft Power BI is not related to the Power BI Report Server.

Prerequisites and permissions

To scan Microsoft Power BI metadata, perform the following prerequisite tasks:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.

2. If you plan to connect Microsoft Power BI to Microsoft Azure through a REST API with the service principal authentication method:
   a. Enable the service principal.
   For more information, see the Knowledge Article:
   How to configure Service Principal for Microsoft Power BI.
   b. Enable Preview features for your organization in tenant settings.
   For more information, see the Knowledge Article: How to enable preview in Microsoft Power BI.
3. If you plan to connect Microsoft Power BI to Microsoft Azure through a REST API with the administrator user role:
   a. Assign one of the following Power BI administrator privileges to the administrator user role:
      - Microsoft 365 Global Administrator
      - Power BI Service Administrator
   b. Get administrator access at the workspace level.
4. Configure the connection. See "Connection properties" on page 109.
5. If you use an SSL-enabled proxy or a PowerBI instance that have non-standard SSL certificates, import the certificate CA using the following command: `keytool -importcert -alias proxyPBI / <certificate_location>/<certificate_file_name>.cer -keystore <metadex_jvm_keystore_location> -storepass <metadex_jvm_keystore_password>`

Note: You don't need to download any drivers. Informatica provides the required drivers in the installation package provided by the installer.

Connection properties

In the MetaDex tool, enable proxy if you cannot establish a direct connection between an Enterprise Data Catalog machine and the Microsoft Power BI cloud. For more information on Microsoft Power BI proxy parameters, see "Microsoft Power BI Reader" in the *MetaDex Scanner Reference*.

You can connect to Microsoft Power BI to Microsoft Azure through a REST API with the following types of authentication:

Service principal authentication

If you use service principal authentication method, configure the following parameters:

- REST API tenant ID
- REST API client secret
- REST API client ID

For more information on Microsoft Power BI REST API parameters, see "Microsoft Power BI Reader" in the *MetaDex Scanner Reference*.

Optionally, you can scan Microsoft Power BI metadata offline, directly from previously extracted JSON files.

Use the service principal authentication method for national clouds in the United States. The following table describes the additional parameters that you configure in the MetaDex tool:

Parameter	Description
authorityUrl	URL for user authentication. https://login.microsoftonline.com/organizations/
scope	Parameter that the endpoint uses to authenticate the user. https://analysis.usgovcloudapi.net/powerbi/api/.default
powerbiCloudRestApiUrl	URL for data and metadata download from the Power BI cloud. https://api.powerbigov.us

If you use other national clouds, see
Tutorial: Embed Power BI content into your application for national clouds.

Admin user role authentication

If you use admin user role authentication, configure the following parameters:

- REST API user
- REST API password
- REST API client ID

For more information on Microsoft Power BI REST API parameters, see "Microsoft Power BI Reader" in the *MetaDex Scanner Reference*.

For more information, see the Knowledge Article:
HOW TO: Extract a Microsoft Power BI JSON file with a REST API client in EDC.

Optionally, you can scan Microsoft Power BI metadata offline, directly from previously extracted JSON files.

You must provide the REST API URL in the MetaDex tool for both service principal and admin user role authentication methods.

The default REST API URL is https://api.powerbi.com. If you are using a different REST API URL, enter it in the **REST API url** field.

Types of input

In the MetaDex tool, you can extract Microsoft Power BI metadata from the following types of input:

- Workspaces:
  - Workspace of the group type
  - Workspace of the workspace type
- Reports that are based on the data set but do not contain column-level information.
  For the **Microsoft Power BI REST API extractor filters** parameter, use the value in the following format: `Include:<WorkspaceName>:<ReportName>`. For example, `Include:ClientWorkSpace:SalesReport`.
- Data sets:
  - WEB
  - AZURE
  - BLOBS

- SQL
- FILE
- AZURE_DATA_LAKE_STORAGE
- ODBC_S3
- FABRIC LAKEHOUSE
- FABRIC WAREHOUSE

Note: For column-level information for data sets, if available, see the Microsoft REST API results.

For more information, see "Supported Dataset and Report components" in the "Microsoft Power BI Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Workspace
- Dashboard
- Tile
- Report
- Data set

Lineage

You can view the following lineage:

- Reference data set or Actual Object to the PowerBI data set.
- Data set to Report.
- Report to Dashboard.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the Microsoft Power BI MetaDex scanner:

- You can't extract column-level information for reports.
- For more information on metadata scanning, considerations, and limitations, refer to the latest Microsoft documentation.

Creating a Microsoft Power BI configuration

When you create a Microsoft Power BI configuration you configure parameters for the Reader and Writer modules.

1. In your Microsoft Power BI project, from the **More** menu, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.

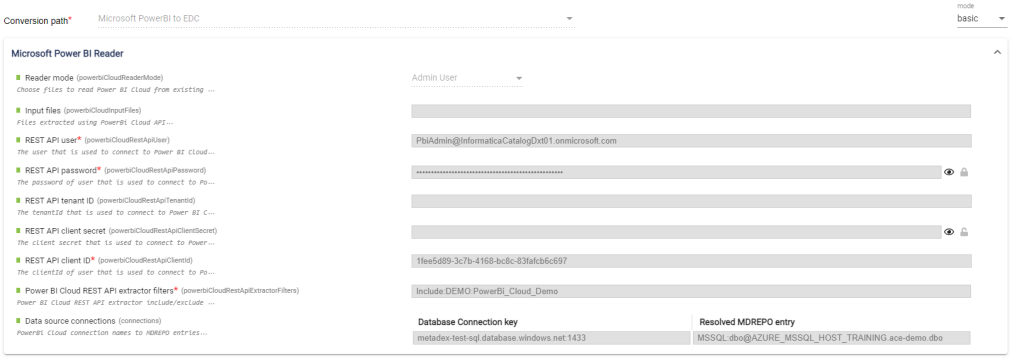
2. Enter a name and an optional description for the configuration.
3. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
4. From the mode list, based on your level of expertise, select the mode in which you want to create the configuration. You can select from the following modes:

Mode	Description
Basic mode	Default mode. Includes the minimum of data source connection properties required to scan metadata.
Advanced Mode	Includes additional properties. You can use this mode to adjust processing metadata to your requirements.
Expert mode	Includes all available properties. If you are an experienced user, or if Informatica Global Customer Support suggests it, use this mode to adjust processing metadata to your requirements.

   A list of possible configuration modules appears.

5. From the **Conversion path** list, select **Microsoft Power BI to EDC**.
6. Expand the **Microsoft Power BI** module.
7. In the **Reader mode** field, depending on the authentication method you want to choose, select one of the following options:
   - **Admin User**
   - **Service Principal**

   The following image shows example values:



8. Provide values for the following parameters:
   a. REST API user
   b. REST API password
   c. REST API client ID

   Note: Your Microsoft Power Bi administrator must provide you with the necessary data.
9. Provide the workspace and report names to configure the **Microsoft Power BI Cloud REST API extractor filters** parameter.

   The following example illustrates the correct syntax: <Include> or <Exclude>:<Name of Microsoft Power BI workspace>:<Name of report in the workspace>

   where: <Include> and <Exclude> determine if this is an "include" or "exclude" filter entry.

Use the following filter for the first configuration:



Use an asterisk to match any number of characters. For example, instead of separately entering the names of multiple reports that start with "Include:DEMO:PowerBi_Cloud_", enter `Include:DEMO:PowerBi_Cloud_*`

10. Expand the **Enterprise Data Catalog Writer** module and provide the following parameters:

Parameter	Description
EDC Writer mode	This parameter determines how the Enterprise Data Catalog Writer module operates. Set this parameter to **Generate output files**. After a successful parsing, you can change the parameter to **Load using REST API**. After the job is complete, the extracted metadata loads to your Enterprise Data Catalog instance.
Output directory	The path to your output directory. The scanner generates files containing extracted metadata and stores them in the output directory.
EDC address	The URL of the Enterprise Data Catalog tool. You must not change the global variables if the target of the metadata is a local Enterprise Data Catalog instance.
EDC user	The user name that you need to connect to the Enterprise Data Catalog tool.
EDC password	The password that you need to log in to the Enterprise Data Catalog tool.
EDC resource name	The name of the resource that the system creates in the Enterprise Data Catalog tool. It contains metadata extracted by the parsing configuration.

You can find descriptions of all the other parameters in the "Module Reference" chapter of the *MetaDex Scanner Reference*.

11. Click **Save** and then click **Run**.

    The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

12. Use the **WARN** filter field to filter and review the warning or error messages.

    When you select **WARN**, you can review the warnings and errors in one list.

13. Analyze the reported warning or error messages.
14. Adjust the configuration.
15. Run the job again.

    For more information, see "Running a scanner job" on page 20.

16. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

Troubleshooting Microsoft Power BI configuration issues

This section contains the most common configuration issues that a Microsoft Power BI MetaDex scanner can report after you run a job for the first time.

"Connectionless repository created for [host]" warning appears.

```
WARN: Connectionless repository created for [host].[metadex-test-
sql.database.windows.net:1433]
```

This warning appears when there is a connection missing to a database that the scanned Microsoft Power BI data set uses.

When the connection is missing, the scanner operates in an offline mode and tries to analyze SQL statements to establish source data structures. However, in some cases it requires a connection to the real source database, otherwise it may fail to scan some data. For more information, see "Connection requirements" in the *MetaDex Scanner Administrator Guide*.

Perform the following actions to solve the issue:

1. Add the correct database to the **Connection Registry** in the MetaDex tool. For more information on adding such connections, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*. .
2. Add the connection to the configuration.

   Note: The connection mapping entry consists of the **Database Connection key**, which you can find in the warning message, and a **Resolved MDREPO entry**. The following example presents the correct **Resolved MDREPO entry** syntax: <VENDOR>:<USER>@HOST.DATABASE.<SCHEMA>

   where: <HOST> and <DATABASE> are values that you configured when you added the connection to the connection registry.
3. Run the configuration again and verify the logs. For more information, see "Running a scanner job" on page 20.

"Could not get auth token..." warning or error appears.

```
WARN: Request failed. Reason: Could not get auth token due to Request https://
login.microsoftonline.com/com...
```
or
```
ERROR: Unexpected problem has occurred: Could not get scan
result from Power BI Cloud due to: Could not get auth token due to the Request https://
login.microsoftonline.com/commonoauth2/token return not accepted 400 code with body (...)
```

This warning and error message appear when there are Microsoft Power BI connection issues.

You might see the messages in the following situations:

- When you provide invalid credentials during the configuration.
- When you have insufficient privileges granted in Microsoft Power BI cloud.

  Note: For more information about the permissions, see "Microsoft Power BI" on page 108 and "Enable service principal authentication for read-only admin APIs" on the Microsoft website.

Perform the following actions to solve the issue:

1. Verify the credentials or permissions.
2. Run the configuration again and validate the logs.

"Could not create graph for dataset" error appears.

```
ERROR:(DatasetProcessor:85) - Could not create graph for dataset Asset_data due to Dataset
Asset_data does not contain information about t's columns. To make processing possible please
refresh dataset in Power BI Cloud and then try again.
```

This error appears when the Microsoft Power BI data set processor cannot generate a graph.

Perform the following actions to solve the issue:

1. Refresh the problematic data set in the Microsoft Power BI cloud.
2. Verify the privileges in the Microsoft Power BI cloud.

"Could not get scan result from Power BI Cloud" error appears

```
Could not get scan result from Power BI Cloud due to: Unable to check prerequisites status
```

This error appears if you don't have permission to use the getPrerequisitiesStatus API or if the following parameters are not set to "true":

- allowedToReturnDetailedMetadata
- allowedToReturnExpressions

Perform the following actions to solve the issue:

1. Verify that you have all the permissions described in "Microsoft Power BI" on page 108.
2. Send an HTTP GET request to verify that you enabled the required parameters.
   The following table lists the request parameters:

Parameter	Description
GET	The address of the resource that you request the data from. https://api.powerbi.com/v1.0/myorg/admin/workspaces/getPrerequisitesStatus
Authorization header	Provides credentials that authenticate a user agent with a server. Bearer {{BEARER}}

SAS

SAS is a data analytics and discovery software. You can use SAS solutions for data management, decision-making, business intelligence, risk management, marketing, and advanced data analysis.

Important: Configure the SAS MetaDex scanner only if you are an experienced user of SAS.

Prerequisites and permissions

Before you configure the SAS MetaDex scanner to extract metadata from the SAS data source, prepare the following prerequisites:

- Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

  Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.

- **The unloaded SAS catalog**
  To prepare the unloaded SAS catalog, perform the following steps:

  1. Download the SAS program from the following location: `${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app/utils/dbToCatalogRepo/sql/Sas/unloadSasCatalog.sas` along with the README file.

2. Provide the program and the README file to the SAS administrator.
   The SAS administrator runs the program.
3. Copy the results of the previous step to the following connection registry folder for the SAS system:
   `${INFA_HOME}/AdvScannersWorkspace/MDREPO/<host>/<database>`

   To create <host> and <database> entries, see "Creating a connection" in the *MetaDex Scanner Administrator Guide*.

- **The SAS code to be processed**
  To prepare the SAS code, select one of the following methods, depending on the type of your code:
  - **Original SAS programs and macros or EGP project files**
    Use this method when you do not have dynamic code driven by runtime values.
  - **SAS code extracted from execution logs**
    Use this method when you have a significant amount of dynamic code driven by runtime values.

  For more information on how to process SAS code, see "Types of input" on page 117.
- **CSV files referenced as a source in SAS programs**
  In the files, ensure that the headers contain only names of extracted columns.

Configure the SAS MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Permissions

Not applicable.

Connection properties

The SAS MetaDex scanner does not connect directly to the SAS application.

Types of input

In the MetaDex tool, you can extract SAS metadata from the following types of input:

Option	Description
Original SAS programs	1. In the MetaDex tool, use the **SAS to EDC** conversion path. 2. Copy the SAS programs from the SAS application instance. 3. During scanner configuration, point to the location of the SAS programs by using the following parameters: - Input files - Macro definitions files
EGP project files	1. In the MetaDex tool, use the **SAS EGP to EDC** conversion path. 2. Copy the SAS programs from the SAS application instance. 3. During scanner configuration, point to the location of the SAS programs by using the following parameter: Input files.
SAS code extracted from execution logs	Run your SAS programs with the following SAS options enabled: - MPRINT Use this option to display the SAS statements that are generated with macros. - MFILE Use this option to redirect logs to a specific file. - MLOGIC Use this option to add detailed macro execution logs. For more information about SAS macro system options, see SAS documentation. After execution, SAS generates process logs with one of the following programs, depending on your operating system: - `utils/extractSasFromLog.bat` - `utils/extractSasFromLog.sh` **Note:** For more information about processing logs, see the How to pre-process SAS log files to prepare them for the SAS advanced scanner Knowledge Base article. In the MetaDex tool, use the generated log output as input for the SAS MetaDex scanner.

For more information, see "Supported SAS Procedures" in the "Sas Reader" section of the *MetaDex Scanner Reference*.

Generated asset types

You can extract metadata directly from SAS scripts and indirectly from SAS execution logs.

Your MetaDex scanner creates the following types of objects in the catalog:

- Folder
- Program
- Step
- EGP project
- EGP flow
- Library
- Data set
- Work data set
- Variable

Lineage

You can view the following lineage:

- Summary lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- Stored procedure calls.

Relationships

Not applicable.

Rules and guidelines

Consider the following rules and guidelines when you use the SAS MetaDex scanner:

- If you process dynamic code using a method other than processing logs, you can't get lineage.
- When you process transpose procedures, you might need to perform additional configuration to get column-level lineage. For more information, see the HOW TO: Use special annotations in a PROC TRANSPOSE statement Knowledge Base (KB) article.
- Use the pattern replacer when you use a script that you do not want to edit manually. For more information, see the HOW TO: Use the pattern replacer in a script in EDC Knowledge Base (KB) article.

Microsoft SQL Server Analysis Services

Microsoft SQL Server Analysis Services (SSAS) is an online analytical processing and data mining tool in Microsoft SQL Server. You can use SSAS to analyze information from multiple databases or disparate tables or files.

Prerequisites

Before you configure the SSAS MetaDex scanner to extract metadata from the SSAS data source, prepare the following prerequisite tasks:

1. Split input into projects, and then create a configuration for each project. For example, if you split the input into 10 projects based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into projects.
2. Get the following SSAS project files:
   - SSAS Tabular files: BIM, SMPROJ, SETTINGS
   - SSAS Multidimensional files: DS, DSV, CUBE, DIM, DWPROJ

3. Get the database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in Microsoft SQL Server Analysis Services. To prepare the extract, use the Metadata Extractor.
4. Configure the following SSAS MetaDex scanners:
   - SSAS Tabular
   - SSAS Multidimensional

   For more information, see "Configuring MetaDex scanners" on page 16.

Permissions

Not applicable.

Connection properties

Not applicable.

Types of input

You can extract SSAS Tabular metadata from the following types of input:

- Tables
- Measurers and measure groups
- KPIs
- Perspectives and perspective tables

For more information, see "SSAS Tabular Reader Module" in the *MetaDex Scanner Reference*.

You can extract SSAS Multidimensional metadata from the following types of input:

- Cube
  - Cube dimension
    - Cube dimension hierarchies
  - Cube calculations
  - Cube perspectives
  - Cube KPIs
  - Cube partitions
  - Cube measure groups
- Dimensions
  - Dimensions hierarchies
- Data source views

For more information, see "SSAS Reader Module" in the *MetaDex Scanner Reference*.

Generated asset types

MetaDex scanners create assets that you can view in the Catalog tool as Enterprise Data Catalog objects.

You can see the hierarchy of the following SSAS Tabular MetaDex scanner:

- Folder
- Project

- Measure Group
- Table
- Perspective
- Perspective Table
- KPI

You can see the hierarchy of the following SSAS Multidimensional MetaDex scanner:

- Project
- Dimension
- Field
- Cube
- Data Source View

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.

Relationships

You can view the following usage associations for SSAS Tabular MetaDex scanner:

- Table
- MeasureGroup

Rules and guidelines

SSAS MetaDex scanners don't parse scripting languages, for example, VBA.

Microsoft SQL Server Reporting Services

Microsoft SQL Server Reporting Services (SSRS) is a server-based report generating software system. It is part of a suite of Microsoft SQL Server services that includes SSAS (SQL Server Analysis Services) and SSIS

(SQL Server Integration Services). You can use SSRS to prepare and deliver interactive and printed reports. SQL administrators can connect to SQL databases and use SSRS tools to format SQL reports.

Prerequisites

Before you configure the SSRS MetaDex scanner to extract metadata from the SSRS data source, complete the following prerequisites:

1. Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

   Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
2. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in Microsoft SQL Server Reporting Services. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.
3. You must process Microsoft SQL Server Analysis Services (SSAS) projects if you used Microsoft MDX queries in reports.

   Note: You use Microsoft MDX queries to gather data from deployed SSAS projects, such as cubes or dimensions.
4. Configure the SSRS MetaDex scanner. For more information, see "Configuring MetaDex scanners" on page 16.

Permissions

Get a Report Server user with permission to view and download reports and shared data sources from the SSRS administrator. Depending on the scenario, you need to get the Report Builder Role and the Publisher Role or the System User Role.

Connection properties

To connect the SSRS MetaDex scanner to the Reporting Server, enter the following parameters:

- Reporting Server user name
- Reporting Server user password
- Reporting Server security domain
- The following URL to the ReportService2010.asmx file: `https://<RS domain>/ReportServer/ReportService2010.asmx`

Types of input

You can extract SSRS metadata from the following types of input:

- Data sets
- Reports

Generated asset types

Your MetaDex scanner creates the following types of objects in the catalog:

- Folder
- Project

- DataSet
- Field
- Report
- PlSql DataSet Statement

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- Stored procedure calls.
- Embedded MDX queries.

Relationships

You can view the following usage associations for SSRS assets:

- DatabaseObjectsModel.SourceColumn
- SsrsModelV2.Field

CHAPTER 5

Configuring IBM mainframe scanners

This chapter includes the following topics:

- Overview of IBM mainframe scanners, 123
- COBOL and JCL, 123

Overview of IBM mainframe scanners

The IBM mainframe category of MetaDex scanners includes sources that correspond to procedural programming languages and database management technologies that occur within the mainframe operating system.

Before you create any configuration, validate the prerequisites and necessary permissions, and familiarize yourself with the parameters, authentication methods, types of input and assets that a given scanner generates. If you haven't created a configuration with the MetaDex tool before, familiarize yourself with the general MetaDex scanners configuration workflow before you proceed.

COBOL and JCL

Common business-oriented language (COBOL) is a computer programming language designed for business use. You can use COBOL in applications deployed on IBM mainframe computers, such as large-scale batch and transaction processing jobs.

For the COBOL MetaDex scanner, you can use one of the following conversion paths:

- JCL to EDC
- COBOL to EDC

Job control language (JCL) is an orchestration language that you use to instruct the system on how to run a batch job or start a subsystem. Since you usually call COBOL programs from JCL, use the JCL to EDC path for COBOL programs. When you use the JCL to EDC path, you know the values of parameters that JCL passes to COBOL programs.

In rare cases when you are unable to provide JCL files, use the COBOL to EDC path. JCL data sets can require a special configuration.

Important: Configure the IBM mainframe MetaDex scanners only if you understand IBM mainframe systems and JCL calls.

Common IBM mainframe prerequisites

Before you configure the COBOL or JCL MetaDex scanner to extract metadata from the respective data source, perform the following prerequisite tasks:

- Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

  Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
- Configure the scanner. See "Configuring MetaDex scanners" on page 16.

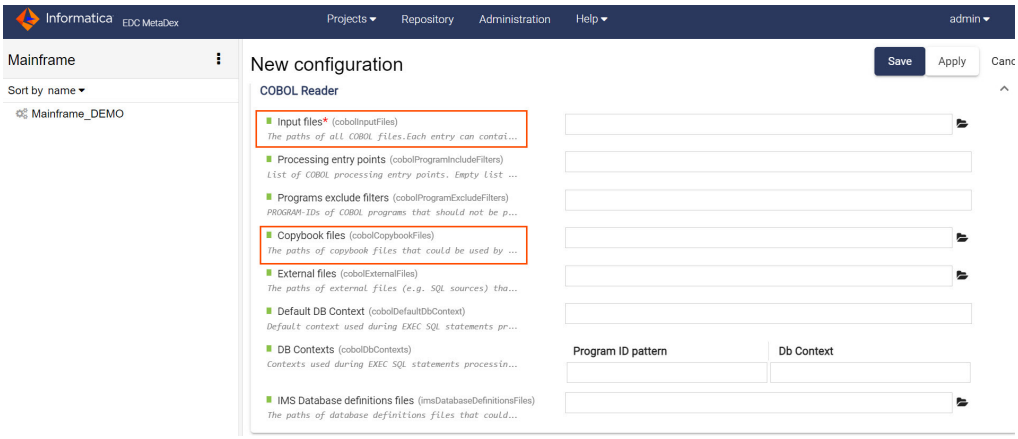
Prerequisites for the JCL to EDC conversion path

Before you configure the JCL MetaDex scanner to extract metadata from the JCL data source, perform the following prerequisite tasks:

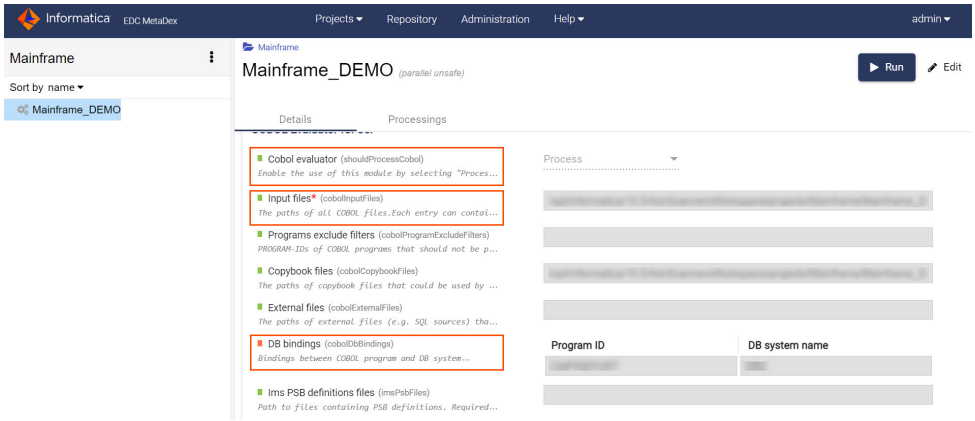
1. IBM mainframe MetaDex scanners read files from a local file system. Before you create a configuration, copy the following files from IBM mainframe into the location that your MetaDex scanner can access:
   - COBOL
   - COPYBOOK
   - JCL
   - Other files used by JCL, such as SQL scripts
2. Log in to the MetaDex tool.
3. On the **Details** tab of your configuration, expand the **JCL Reader** module.
4. Provide the following parameters for your configuration:
   a. In the **Input files** field, enter paths to input files that specify the location of the JCL jobs that you want to process. Input must be available and complete. For more information, see "Input files" in the "JCL Reader" section of the *MetaDex Scanner Reference*.
   b. In the **Jobs to be processed** field, enter the high-level jobs that you want to analyze.

      Note: If you leave the **Jobs to be processed** field empty, you will analyze all available jobs.
   c. In the **Jobs to be excluded** field, enter the jobs that you want to exclude in the section. For example, problematic jobs such as logs or cleanup.
   d. In the **JCL files host name** field, enter the name of the machine that hosts JCL files.
   e. To connect to a database, in the **Database connections** fields, enter values according to "Database connections" in the "JCL Reader" section of the *MetaDex Scanner Reference*.

The following image shows the **JCL Reader** module of the JCL to EDC conversion path:



5. Expand the **COBOL Evaluator for Jcl** module.
6. Provide the following parameters for your configuration:
   a. From the **Cobol evaluator** list, select **Process**.
   b. In the **Input files** field, enter paths to COBOL input files that specify the location of the COBOL jobs that you want to process. For more information, see "Input files" in the "Cobol Evaluator for Jcl" section of the *MetaDex Scanner Reference*.
   c. If you call any databases, in the **DB bindings** field, enter the ID of the program and the system name of the database.
   The following image shows the **Cobol Evaluator for JCl** module of the JCL to EDC conversion path:



7. Depending on the technology that you want to use, see the applicable prerequisites in this guide. For example, if the JCL MetaDex scanner processes SAS with the JCL to EDC conversion path, see "SAS" on page 115.

   Note: For a list of mainframe utilities that Informatica supports, see "List of Modules" in the "JCL to EDC" section of the *MetaDex Scanner Reference*.
8. If you encounter an unknown value error, provide the applicable values.

The following table describes the required parameters:

Parameter	Description
CYMD	Date in the following format: YYYMMDD
CHHMMSS	Time in the following format: HHMMSS
CDATE	Date in the following format: YYMMDD
CTIME	Time in the following format: _HHMM

The following text shows an example of an unknown value error:

```
ERROR (JclReader:109) - Exception thrown while converting JCL job: CVPTR200 to
graph. Message: Error occurred while processing FTP control statement from datasets
*. Parse error (NoViableAlt) on token: '.'.
```

9. Get database access or extract all necessary database data in one of the following ways:
   - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
   - Extract structures and definitions of all required database tables, views, and procedures referenced in COBOL. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.

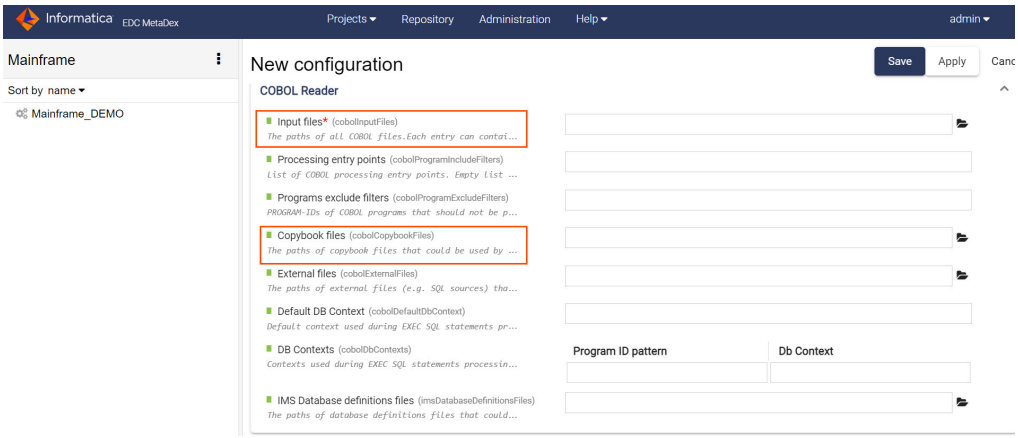
Prerequisites for the COBOL to EDC conversion path

Before you configure the COBOL MetaDex scanner to extract metadata from the COBOL data source, perform the following prerequisite tasks:

1. Log in to the MetaDex tool.
2. On the **Details** tab of your configuration, expand the **COBOL Reader** module.
3. Provide the following parameters for your configuration:
   a. In the **Input files** field, enter paths to all COBOL input files that specify the location of the COBOL jobs that you want to process. Input must be available and complete. For more information, see "COBOL Reader" in the *MetaDex Scanner Reference*.
   b. In the **Copybook files** field, enter the paths to copybook files that the COBOL programs use.

      Note: You can search recursively.

The following image shows the **COBOL Reader** module of the COBOL to EDC conversion path:



4. Get database access or extract all necessary database data in one of the following ways:
    - Get access to all required databases. For more information, see "Connections" in the *MetaDex Scanner Administrator Guide*.
    - Extract structures and definitions of all required database tables, views, and procedures referenced in COBOL. To prepare the extract, use the Metadata Extractor. For more information, see "Metadata Extractor" in the *MetaDex Scanner Administrator Guide*.

Permissions

Not applicable.

Connection properties

Not applicable.

Types of input

You can extract COBOL metadata from the following types of input:

- Files with JCL job definitions

  Note: Jobs allow you to run JCL programs. For a list of supported JCL programs, see "JCL Reader" in the *MetaDex Scanner Reference*.
- COBOL programs
- Copybooks

Generated asset types

MetaDex scanners create assets that you can view in the Catalog tool as Enterprise Data Catalog objects.

You can see the hierarchy of the following JCL and COBOL objects:

- Copybook
- JCL procedure definition > JCL procedure instance > JCL step
- IBM IMS database > IBM IMS segment
- SourceFile > COBOL program > COBOL program result set
- SourceFile > COBOL program > COBOL program parameter
- SourceFile > COBOL program > COBOL paragraph with error
- SourceFile > COBOL program > COBOL paragraph
- JCL job > JCL step

Note: You can't use other mainframe components, such as Easytrieve or Customer Information Control System (CICS).

To find the generated asset types in the Catalog Administrator tool, click: **Manage** > **Models** > **Advanced Scanners - Mainframe**.

Lineage

You can view the following lineage:

- Summary and transformation-level lineage.
- Data lineage at the data set level.
- Data lineage at the data element level.
- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.

- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.
- Embedded SQL.
- Stored procedure calls.

Relationships

Not applicable.

Rules and guidelines

Not applicable.

Creating a JCL configuration

When you create a JCL configuration you configure parameters for the reader and writer modules.

1. In the MetaDex tool, open your JCL project.
2. From the **More** menu, select **Create configuration** > **MetaDex**.

   The **New configuration** page appears.
3. Enter a name and an optional description for the configuration.
4. To run multiple instances of the configuration simultaneously, select **Parallel Safe**.
5. From the mode list, select the mode in which you want to create the configuration. You can select from the following modes:

Mode	Description
Basic mode	Default mode. Includes the minimum of data source connection properties required to scan metadata.
Advanced mode	Includes additional properties. You can use this mode to adjust processing metadata to your requirements.
Expert mode	Includes all available properties. If you are an experienced user or if Informatica Global Customer Support suggests it, use this mode to adjust processing metadata to your requirements.

   The list of possible configuration modules appears.
6. From the **Conversion path** list, select **JCL to EDC**.
7. In the **JCL Reader** module, provide the following parameters:

Parameter	Description
Connection registry path	The path to the connection registry (MDREPO) directory. For more information about the connection registry, see the *MetaDex Scanner Administrator Guide*.
Input files	Paths to JCL jobs you want to process. Entries can contain the following wildcard characters: "?" for single and "*" for multiple characters.

Parameter	Description
Jobs to be processed	The list of JCL jobs you want to process in the given order. If empty, the scanner processes all input jobs in alphabetic order.
JCL files host name	The host name for JCL files that the configuration processes.
Database connections	The list of connections that the JCL jobs use to connect to databases. All connection keys must be mapped to an MDREPO entry.

Default variables values file	The fully qualified path to the file that contains local variables, which are not set in the code. The following example shows a sample default variables file:
	
	`VARIABLE_1=VALUE_1`
	`VARIABLE_2=VALUE_2`
	`VARIABLE_3=VALUE_3`

8. To parse SAS programs that the JCL scripts trigger, enter the following parameters in the **SAS Evaluator for JCL** module:

Parameter	Description
Sas evaluator	Enables the analysis of SAS programs. Set to "Process."
Input files	Paths to SAS programs that you want to process. Entries can contain wildcard characters. Use a question mark (?) for single, and use an asterisk (*) for multiple characters.
Context	The context for SAS objects. Used to search and create database objects. The context contains dot-separated segments that specifiy default values for the host, database, and schema. For example HOST.DBNAME.LIBNAME. To override the default values, include explicit qualification of object references used in the parsed query.

Default variables values file	The fully qualified path to the file that contains local variables, which are not set in the code. You can set the default values globally or individually for SAS programs. The following example shows a sample default variables file:
	
	`[GLOBAL]`
	`VAR &start_date='YYYYMMDD'`
	`VAR &end_date='YYYYMMDD'`
	
	`[FILE:'PROGRAM1.sas']`
	`VAR &end_date='20171031'`
	
	`[FILE:'PROGRAM2.sas']`
	`VAR &end_date='20181031`

9. In the **COBOL evaluator for Jcl** module, provide the following parameters:

Parameter	Description
Cobol evaluator	Enables the analysis of COBOL programs. Set to "Process".
Connection registry path	The path to the connection registry (MDREPO) folder. When you provide the path for the JCL Reader module, it appears automatically in this module.
Input files	Paths to COBOL programs that you want to process. Entries can contain wildcard characters. Use a question mark (?) for single, and use an asterisk (*) for multiple characters.
Copybook files	Paths to Copybook files referenced in Cobol programs. Entries can contain wildcard characters. Use a question mark (?) for single, and use an asterisk (*) for multiple characters.
External files	Paths to external files, such as SQL sources, referenced in Cobol programs. Entries can contain wildcard characters. Use a question mark (?) for single, and use an asterisk (*) for multiple characters.
COBOL program context	COBOL programs are represented as stored procedures. Segment names are used as a procedure context.
DB bindings	Bindings between a COBOL program and a database connection. Used when the DSN SYSTEM information is missing in JCL code which triggers the COBOL program or when you use DSN SYSTEM BIND.

10. In the MetaDex tool, expand the **Enterprise Data Catalog Writer** module and provide values for the following parameters:

Parameter	Description
EDC writer mode	Set the value to **Generate output files**. After a successful parsing, you can change it to **Load using REST API**. After the job is complete, you load metadata to the catalog.
Output directory	The path to the output directory of the extracted metadata files.
EDC address	The host name and port number of the Catalog Service in the following format: <host>:<port>
EDC user	The user name configured for the catalog administrator user.
EDC password	The password for the Enterprise Data Catalog user.
EDC resource name	The name of the resource that the system creates in the catalog. It contains metadata extracted by the configuration.

For a complete list of parameter descriptions, see the "COBOL and JCL" chapter of the *MetaDex Scanner Reference*.

11. Click **Save** and then click **Run**.

The MetaDex scanner reads and parses the input. If the parser encounters any issues, it generates warning or error messages. For more information, see "Running a scanner job" on page 20.

12. Use the **WARN** filter field to filter and review the warning or error messages.

    When you select **WARN**, you can review the warnings and errors in one list.
13. Analyze the reported warning or error messages.
14. Adjust the configuration.
15. Run the job again.

    For more information, see “Running a scanner job” on page 20.
16. After you are ready to load the results into the catalog, change the **EDC writer mode** parameter value to **Load using REST API**.

CHAPTER 6

Configuring application scanners

Overview of application scanners

The applications category of MetaDex scanners includes software solutions that allow you to perform a variety of business and sales-related operations.

Before you create any configuration, validate the prerequisites and necessary permissions, and familiarize yourself with the parameters, authentication methods, types of input and assets that a given scanner generates. If you haven't created a configuration with the MetaDex tool before, familiarize yourself with the general MetaDex scanners configuration workflow before you proceed.

Salesforce

Salesforce is a Customer Relationship Management (CRM) software solution. It has many applications which support various features such as lead generation, lead acquisition, sales tracking, and deal closure. It is designed to manage the data of an organization around customers and sales.

Salesforce offers the following features:

- Contact management
- Opportunity management
- Customer engagement
- Lead management
- Partner management
- Email integration
- Sales forecasting

Salesforce provides version 53 of the REST API that the Salesforce MetaDex scanner uses.

Prerequisites

Before you configure the Salesforce MetaDex scanner to extract metadata from a Salesforce data source through REST API, complete the following prerequisite tasks:

- Learn the configuration workflow for MetaDex scanners. For more information, see "Configuring MetaDex scanners" on page 16.
- Split input into logical applications, and then create a configuration for each logical application. For example, if you split the input into 10 logical applications based on lines of business, create 10 configurations. If a line of business comprises different areas such as treasury or accounting, create separate configurations for those areas also.

  Warning: Jobs might run slowly or become unresponsive if you don't split input into logical applications.
- Verify that you have access to Salesforce REST API v53.

Permissions

In Salesforce, select the following permissions that the Salesforce MetaDex scanner requires to extract metadata:

- Modify All Data allows you to extract all objects. If you don't want Salesforce to modify all data, select View All Data.

  Note: If you select View All Data, you might not extract some objects. For more information, see the Knowledge Base article.
- API Enabled allows you to access the data of your organization through API requests to your Salesforce instance.

Connection properties

To connect to a live Salesforce instance through REST API, expand the **Salesforce Reader** module and, depending on the authorization method, enter the following information:

OAuth 2.0 Refresh Token Flow for Renewed Sessions

For OAuth 2.0 Refresh Token Flow for Renewed Sessions, request an updated Salesforce access token and enter the following parameters:

- REST API Consumer Key
- REST API Consumer Secret
- REST API Refresh Token

For more information about the parameters, see Salesforce documentation on OAuth 2.0 Refresh Token Flow for Renewed Sessions.

OAuth 2.0 Username-Password Flow for Special Scenarios

For OAuth 2.0 Username-Password Flow for Special Scenarios, enter the following parameters:

- REST API Consumer Key
- REST API Consumer Secret
- REST API user
- REST API password

Consider that OAuth 2.0 Username-Password Flow for Special Scenarios passes credentials back and forth. For more information about the restrictions for this method, and the parameters, see Salesforce documentation on OAuth 2.0 Username-Password Flow for Special Scenarios.

Standard Password Flow

For Standard Password Flow, enter the following parameters:

- REST API user
- REST API password
- REST API Security Token

For more information about the parameters, see Salesforce documentation on the API calls login.

Proxy server authentication

When you create a Salesforce MetaDex scanner in Advanced mode, you can enable the proxy server authentication. A proxy server is a system or a router that functions as a gateway between users and the internet. Proxy server authentication enables you to configure the authentication method for the proxy.

Configure the following required properties for proxy server authentication:

Property	Description
Proxy Host	Host name or IP address of the HTTPS proxy server.
Proxy Port	Port number of the HTTPS proxy server. For example, 30408.
Proxy Username	Authenticated user name to connect to the HTTPS proxy server.
Proxy Password	Password for the authenticated user.

Types of input

You can extract Salesforce metadata from the following types of input:

Organization

An organization asset is a root asset of Salesforce.

Object

An object asset contains fields and their properties.

List View

A list view asset contains list view columns and their properties.

Trigger

A trigger asset is a single object that contains code.

Generated asset types

The MetaDex scanner creates the following organization types of objects in the Catalog:

- List View and List View Column
- Trigger
- Object and Field

Lineage

You can view the following lineage:

- Data lineage at the data set level.
- Data lineage at the data element level.

- Control lineage at the data set level. You can view the lineage links only on the system lineage diagrams.
- Control lineage at the data element level. You can view the lineage links only on the system lineage diagrams.

Relationships

Not applicable.

Rules and guidelines

API Usage in a Salesforce account shows the number of API requests made for the account during the last 24 hours. The Salesforce MetaDex scanner runs until it exceeds the maximum capacity of requests.

APPENDIX A

Advanced Custom Metadata Loader

This appendix includes the following topics:

- Custom Metadata Loader overview, 136
- Advanced Custom Metadata Loader concepts, 137
- Adding Custom Metadata Loader to the MetaDex tool, 138
- Configuring Custom Metadata Loader, 139
- Troubleshooting Advanced Custom Metadata Loader, 145

Custom Metadata Loader overview

Use the Informatica Advanced Custom Metadata Loader to define custom models, upload them, and populate them with data.

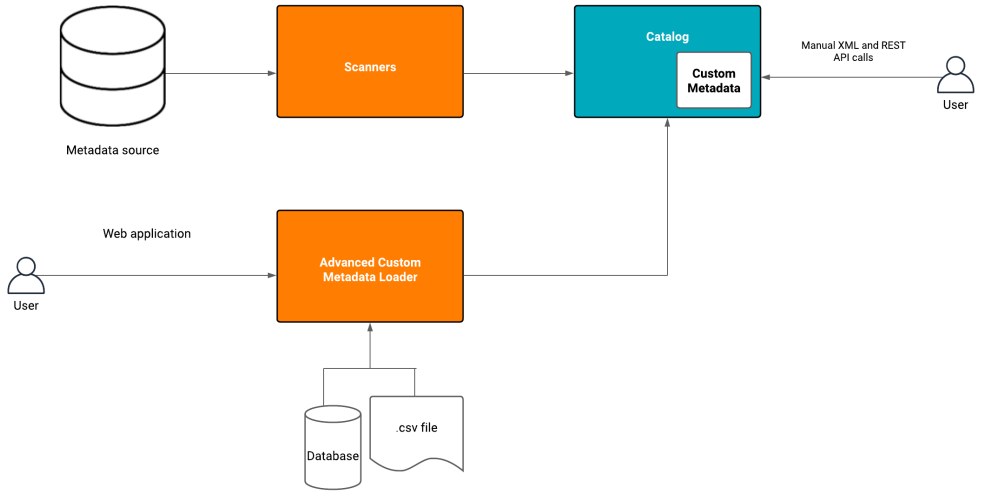
You can use scanners, both MetaDex and standard, to extract information from a metadata source and load it to the catalog. To ingest metadata from data sources for which Enterprise Data Catalog does not provide a resource, you can use the following methods:

- Custom metadata integration feature in Enterprise Data Catalog
- Custom Metadata Loader in the MetaDex tool. For more information about models, see “Advanced Custom Metadata Loader concepts” on page 137.

To load the custom metadata into the catalog, you can perform one of the following actions:

- Perform the custom metadata integration process manually. See "Custom Metadata Integration Overview" in the *Catalog Administrator Guide*.
- If you have an applicable license, add Advanced Custom Medatada Loader to the MetaDex tool.

The following image shows adding custom metadata to the Enterprise Data Catalog tool:



Advanced Custom Metadata Loader automates the process of integrating custom metadata and allows the scanner to ingest data without using any .xml files that you configure manually or REST APIs to create custom models.

Advanced Custom Metadata Loader concepts

The following key concepts of Informatica Advanced Custom Metadata Loader help you understand defining custom models, upload them, and populate them with data.

Models

Based on the model for the data source, the scanner extracts metadata from the data source. Enterprise Data Catalog, by default, provides models for multiple data sources from which you can extract metadata. These are system models.

To ingest custom metadata in the catalog, you must define a custom model. You use models and classes to classify assets and create a hierarchy to make the metadata clearer and more comprehensible. Each model contains classes connected in a parent-child type hierarchy. To improve the representation of custom metadata assets in the catalog, you can define additional class attributes. Optionally, you can define the attributes of the asset class.

The name of the model is visible in Enterprise Data Catalog. Each asset class in the model is associated with a name and an icon.

Data sets

You can define data sources to populate the model with data. There are various data source types with each type having its configuration parameters.

Data sources include a location that contains the record. See the following locations for each data source type:

- Excel file: the name of the Excel sheet
- CSV file: the name of the file
- Database: an SQL query

- XML and JSON file: an xquery that extracts data (JSON is transformed to XML)

Assets

Each entry defines one data stream arranged in tabular form (record set). Each row in a set represents a single asset.

Asset attributes come from data columns available in a record. By default, each attribute has an exact match in column names. Also, assets have the following columns:

- "Name"
- "Parent" for all assets below the root asset

Alternatively, you can define mappings between the attribute name and an expression that evaluates to the attribute value. Expressions consist of functions, operators, literals, and column references. For example:

```
toplevelasset + '/' + "parent asset" + '/' + Replace(asset, 'pre', '')
```

The following table describes string concatenation:

String concatenation	Description
toplevelasset	A reference to the column of the same name .
'/'	A literal. The forward slash concatenated with the string.
"parent asset"	A reference to the column. If it contains spaces, you must enclose it within double quotes.
Replace(asset, 'pre', '')	A function call. The function replaces occurrences of 'pre' in the value of the 'asset' column with an empty string.

Relationships

You must define relationships flow between assets. Each entry represents one kind of relationship: one pair of source and target asset types. You must define two attributes in a record set: 'source' and 'target' which are hierarchy paths to source and target assets.

Variables and settings

Any placeholder in the form of `VariableName` is replaced with the value of the variable.

Settings contain Enterprise Data Catalog connection details, database connection parameters, output directory, and other options.

You can find details of these settings in the configuration editor.

Adding Custom Metadata Loader to the MetaDex tool

To use the Advanced Custom Metadata Loader, you must add it as an external tool in the MetaDex tool.

1. In the header of the MetaDex tool, click **Administration**.
2. From the left pane, select **External Tools**.

   The **External Tools** page appears.

3. In the **External Tools** window, click **Add external tool**.

   The **New external tool** page appears.

4. Enter the tool details.

   The following table describes the tool properties:

Property	Description
Name	Required. The name of the tool. The recommended name is ACML.
Tool id	Required. The ID of the tool. By default, same as the name of the tool.
Description	The description of the tool.
Command	Required. The command that launches the tool. For ACML, run the following command: `${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app/custom-metadata-loader.sh`
Working directory	The path to the directory for storing temporary files.
Log Layout	Describes the format of logs created by the external tool.
Icon	Defines displayed image for the tool to differentiate configurations visually. If you do not select it, a default icon is used.
Config Schema	The JSON schema you copy and paste from the following location: `${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app/utils/acml/schemas/`

5. Add the command that you want to execute into the following configuration file: `${INFA_HOME}/AdvScannersWorkspace/etc/external-tools-allow-list.properties`

   For more information, see "External tools" in the *MetaDex Scanner Administrator Guide*.

6. On the **New external tool** page, click **Save**.

Configuring Custom Metadata Loader

You can use the Enterprise Data Catalog Advanced Custom Metadata Loader to define custom models and upload the models to Enterpirse Data Catalog.

1. In the MetaDex tool, on the **Projects** pane, click **ACML**.
2. From the menu, select **Create configuration** > **ACML**.

   The **New configuration** page appears.

3. In the **Name** field, enter the configuration name.
4. Optionally, enter the configuration description.
5. To scan multiple instances of the configuration simultaneously, select the **Parallel Safe** check box.
6. Configure settlings.

   See “Configuring Custom Metadata Loader settings” on page 140.

7. Configure data sets.

See "Configuring Custom Metadata Loader data sets" on page 141.

8. In the **Variables** pane, you can enter name-value pairs that are injected into configuration strings.
9. Define models.

   See "Defining Custom Metadata Loader models" on page 142.
10. Define assets.

    See "Defining Custom Metadata Loader assets" on page 143.
11. Define relationships.

    See "Defining Custom Metadata Loader relationships" on page 144.
12. Click **Save**.
13. Click **Apply**.

Configuring Custom Metadata Loader settings

Configure Enterprise Data Catalog connection and properties.

1. In the ACML project, on the **New configuration** page, on the **Settings** pane click **Expand**.
2. In the **EDC Configuration** pane, click **Expand**.
3. Enter the connection properties.

   The following table lists the connection configuration properties:

Property	Description
EDC Host	The Enterprise Data Catalog service URL.
EDC Login	The Enterpirse Data Catalog login.
EDC Password	The Enterpirse Data Catalog password.
Output dir	The directory where you want to save the output files.
Upload to EDC	Choose if you want to upload the generated files to Enterprise Data Catalog.
Resource name	The name under which the resources appear in Enterprise Data Catalog.
Upload timeout	The time for the Custom Metadata Loader to wait for the upload to complete. Default is 1200.

4. In the **Input options** pane, click **Expand**.
5. Enter the input options details.

   The following table lists the input options:

Property	Description
Working directory	A path to the directory that stores the auxiliary files.
Trim query results	Choose if you want to trim leading and trailing whitespaces in the column values.

Property	Description
Automatic object type inference	Choose if you want to automatically determine relations in model hierarchies.
Preserve temporary files	Choose if you want to keep or delete generated files after a finished job.
Save report as file	Choose if you want to generate a summary report of assets.
Report file path	A path to the report file that includes the file name.
Parent path separator	Enter a character that separates parents in the hierarchy path. Default is "/"
Name separator	Enter a character that separates a name in the hierarchy path. Default is ":"
Hierarchy path quoter	Enter a character to quote the hierarchy path. If you do not specify the quoter, the quotation is ignored.
Hierarchy path escape	Enter a character to escape other characters in the hierarchy path. Default is "\"

Configuring Custom Metadata Loader data sets

Data sets serve as an input for assets uploading process for the Custom Metadata Loader.

1. In the ACML project, on the **New configuration** page, on the **Datasets** pane, click **Expand**.
2. Click **Add Dataset**.
3. Define data sets by performing the following actions:
   a. From the **Dataset** list, select one of the data set sources:
      - Excel file
      - CSV file
      - JSON file
      - Database
      - Local database
   b. Depending on the chosen source, perform the following actions:

Option	Description
For the following sources: • Excel file • CSV file • XML file	Perform the following actions: 1. In the **Dataset name** field, enter the name of the data set. 2. In the **File path** field, enter the name of the data set.

Option	Description
For a JSON file,	Perform the following actions: 1. In the **Dataset name** field, enter the name of the data set. 2. In the **File path** field, enter the path to the file. 3. In the **Intermediate XML file path** field, enter the path to an XML file containing JSON data. 4. Expand **XML invalid characters replacement map**. 5. Click **Add Mapping**. 6. In the **From** field, enter a regular expression pattern to match with JSON keys. 7. In the **To** field, enter the expression that replaces matched part of JSON key in the XML tag. 8. If required, add another mapping.
For a database,	Perform the following actions: 1. In the **Dataset name** field, enter the name of the data set. 2. In the **JDBC driver path** field, enter the path to a JAR file with the JDBC driver. 3. In the **JDBC driver class** field, enter a fully qualified class name of the JDBC driver. 4. In the **JDBC connection string**, enter a JDBC connection URL. 5. In the **JDBC user name** field, enter the JDBC user name. 6. In the **JDBC user password** field, enter JDBC user password.
For a local database,	Perform the following actions: 1. In the **Dataset name** field, enter the name of the data set. 2. From the**Storage type** list, select where you want to keep the local database during processing. 3. Expand the **Source Locations** pane. 4. In the **Dataset name** field, enter the name of the source data set. 5. In the **Location name** field, enter the location in the source data set from which the data is read. 6. In the **Schema name** field, you can enter a chosen schema name. By default, the schema name generates after the source data set. 7. In the **Table name** field, you can enter a chosen table name. By default, the table name generates after the source location.

Defining Custom Metadata Loader models

Custom Metadata Loader generates and uploads package XML files for defined models.

1. In the ACML project, on the **New configuration** page, expand the **Models** pane.
2. Click **Add Model**.
3. Define a model by performing the following actions:
   a. From the list, choose the model type.

      You can choose one the following options:

      - **Generate and upload** to generate a model and upload it to Enterprise Data Catalog.
      - **Generate only** to generate a model.
      - **External** for models that exist in Enterprise Data Catalog. You create a new relationship with an asset of this model.

   b. In the **Model name** field, enter the name of the model.
   c. In the **Version** field, enter the version number.

d. If you want to add more models, click **Add Model**.

4. Expand the **Model classes** pane.
5. Click **Add Model class**.
6. Depending on the model type, add a model class by performing the following actions:

Option	Description
For the external model type,	Perform the following actions: 1. In the **Class name** field, enter the name of the asset class from the Catalog. 2. In the **Class name plural** entry field, enter the plural form of the asset class name.
For the following model types: • Generate and upload • Upload only	Perform the following actions: 1. In the **Class name** field, enter the name of the asset class from the Catalog. 2. In the **Class name plural** entry field, enter the plural form of the asset class name. 3. In **Super classes**, select the applicable check box. **Note:** Super classes are Enterprise Data Catalog built-in types that the class inherits from. You can choose from the following options: • core.DataElement. If the class takes part in relationships of the type core.DirectionalDataFlow, Detailed Lineage, or core.DirectionalControlFlow. • core.DataSet. If the class takes part in relationships of the type core.DataSetDataFlow, coreDataSetControlFlow or is the parent of the asset that is a part of Detailed Lineage. If applicable, you can select both classes.

7. Expand the **Parent classes** pane.
8. Enter the name of the parent class.
9. Choose if you want to enable the parent classes of the configured model class in the detailed lineage.
10. For generate and upload and upload only types, click **Add Adttribute**.
11. In the **Attribute name** entry field, enter the name of the attribute.
12. From the **Attribute type** list, select the appropriate type.

    You can choose from the following types:

    - **core.String**
    - **core.RichText**
    - **core.Integrer**
    - **core.Decimal**
    - **core.Date**
    - **core.Boolean**
13. In the **Attribute allowed values**, you can enter the allowed values of the attribute.
14. If the attribute is multivalued, select **true**.

Defining Custom Metadata Loader assets

Custom Metadata Loader retrieves asset records form data sets and uploades them to Enterpise Data Catalog.

1. In the ACML project, on the **New configuration** page, ond the **Assets** pane, click **Expand**.

2. Click **Add Asset** and perform the following actions:
    a. In the **Asset type** field, enter the asset type defined in the **Models** section.
    b. If the model cannot be defined automatically, in the **Asset model** field, enter the model that the asset class comes from.
    c. In the **Dataset** field, enter the name of the data set from the **Datasets** section.
    d. In the **Location** field, enter the location of the record set inside the data set.

       For an Excel file, it is the name of the sheet. For a CSV file it is the name of the file in the directory, and for the database it is an SQL query.
    e. If you want to import assets with replace or merge policy, from the **Import as complete** list, select **true**.
    f. If there are any multivalued attributes, in the **Multivalued attribute separator (default ",")** field, you can enter a character to separate the attributes.
    g. If there are any multivalued attributes, in the **Multivalued attribute quote character (default - ignore quotation)** field, you can enter a character to quote multivalued attributes.
    h. If there are any multivalued attributes, in the **Multivalued attribute escape character (default "\")** field, you can enter a character to escape other characters in multivalued attributes.
3. If the attribute you enter does not match with a single column in the data set, in the **Attribute to column name mappings** pane, click **Add Mapping**.
4. Add mapping by performing the following actions:
    a. In the **Mapping name** entry, enter the name of the attribute.
    b. In the **Expressions** entry, enter the expression from the data source that corresponds to the value of the attribute.
    c. From the **Attribute type** list, select one of the following types:
        - Built-in
        - Custom single-valued
        - Custom multiple-valued

Defining Custom Metadata Loader relationships

Define relationships between assets.

1. In the ACML project, on the **New configuration** page, on the **Relationships** pane, click **Expand**.
2. Add a relationship by performing the following actions:
    a. Click **Add Relationship**.
    b. From the **Relationship 1** list, select the relationship type.

       You can choose one of the following options:
        - **Internal relationship** for relationships between objects within one resource.
        - **External relationship** for relationships between objects that are in Enterprise Data Catalog.
3. In the **Relationship name** field, enter the name of the relationship.
4. From the **Association type** list, select the appropriate type.

   You can choose one of the following types:
    - **core.DirectionalDataFlow**

- **core.DataSetDataFlow**
- **core.DataSetControlFlow**
- **core.DirectionalControlFlow**

5. From the **Association type** list, select the appropriate type.
6. In the **Source object type** field, enter the class name that defines the type of source asset of the relationship defined in **Models**.
7. If the source model does not appear, in the **Source object model** field, enter the model.
8. If the target model does not appear, in the**Target object model** field, enter the model.
9. In the **Dataset** field, enter the name of the data set defined in the **Dataset** section.
10. In the **Location** field, enter the location of the record set inside the data set.

    For an Excel file, it is the sheet name. For a CSV file it is the file name in the directory, for a database it is an SQL query.
11. In the **Location** field, enter the location of the record set inside the data set.
12. If the reference you use is not defined in a single column in the data set, click **Add Mapping**.
13. In the **Mapping name** field, enter the name of the attribute.
14. In the **Expressions** field, enter the expression from the data source that corresponds to the value of the attribute.
15. For an external relationship, get the fully qualified IDs defined for both source and target objects:
    - You can add applicable **Source/Target Mapping**.
    - In the record set, you can provide proper values in the **Source** and **Target** columns.

Troubleshooting Advanced Custom Metadata Loader

You can encounter errors while using the Enterprise Data Catalog Advanced Custom Metadata Loader.

The Enterprise Data Catalog Advanced Custom Metadata Loader prints basic logging information during execution to the standard output. Errors, if any, appear in the standard output. For more details, see the log file created in the working directory.